An Unofficial Compendium of Worked Solutions to the Problems in *Algebra*

by I.M. Gelfand & A. Shen

Deepak Venkatesh

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Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *Algebra* by I. M. Gelfand and A. Shen, first published in 1993. The original text presents a wide range of problems in elementary algebra, some accompanied by solutions and others left as exercises for the reader.

The aim of this book is to provide clear, logically consistent solutions to all problems addressed, written in a style accessible to motivated students and self learners. Proofs have been verified primarily by hand, and in some cases supported by small computational checks written in Scheme (a dialect of Lisp) or in Python. Large language models were used only as auxiliary tools, primarily for generating figures in LaTeX and for reviewing exposition.

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Project repository:
https://github.com/deepak-venkatesh/gelfand-shen-algebra

For

My Daughter and Son

Contents

1 Introduction

No problems in this chapter.

2 Exchange of terms in addition

No problems in this chapter.

3 Exchange of terms in multiplication

No problems in this chapter.

4 Addition in the decimal number system

Problem 1.

Stack 8s knowing that 8×5 ends with a 0 (that is 40). This gives a carry over of 4. So we need $4 + 8 + 8$ to get a number that ends with 0 for the tens place. This gives a carry over of 2. This 2 can get added to 8 in the hundreds place. The tens place structure shown below.

$$\begin{array}{r} \cdots 8 \\ \cdots 8 \\ \cdots 8 \\ + \cdots 8 \\ \hline \cdots 0 \end{array}$$

$$\begin{array}{r} 888 \\ 088 \\ 008 \\ +\ 008 \\ \hline 1000 \end{array}$$

Answer is self verifiable.

Problem 2.

$$\begin{array}{r} AAA \\ +\ BBB \\ \hline AAAC \end{array}$$

The solution lies in the fact that in the answer the thousandth's place A has to be 1. This is so because whenever there is a carry over the tens digit will be 1 in addition (in this structure). At the maximum level it would be 9 + 9 = 18 for instance.
So we have A as 1.

$$\begin{array}{r} 111 \\ +\ BBB \\ \hline 111C \end{array}$$

Now for B it has to be 9 because if it was any other number then the answer could not have 1s in the places it has now. If B was 0 then the thousandth place 1 in the answer would not materialize. So we have now.

$$\begin{array}{r} 111 \\ +\ 999 \\ \hline 111C \end{array}$$

We can easily see that C is 0 now. So we have

$$A = 1$$

$$B = 9$$

$$C = 0$$

Answer is self verifiable.

5 The multiplication table and the multiplication algorithm

Problem 3.

This looks tricky but is fairly easy to understand the pattern once written down. 1001 multiplied by any 3 digit number will be that 3 digit number repeating twice. This is so because the '001' in 1001 when multiplied by the 3 digit number gives itself and then the '1' in the thousandth's place in 1001 and gives the 3 digit number. It is like concatenation of a 3 digit number to itself when multiplied by 1001.

$$\begin{array}{r} 715 \\ \times\ 001 \\ \hline 715 \end{array}$$

$$\begin{array}{r} 715 \\ \times\ 1001 \\ \hline 715 \\ +\ 715000 \\ \hline 715715 \end{array}$$

Answer is self verifiable.

Answer is 715715.

Problem 4.

This is similar to the previous problem except that we have a 2 digit number getting multiplied by '01'. It will still result in a concatenation.

Verified in Scheme

```
> (* 101010101 57)
5757575757
```

Answer is 5757575757

Problem 5.

This is on the same lines as previous two problems.

$$\begin{array}{r} 1020304050 \\ \times\ 10001 \\ \hline 1020304050 \\ +\ 10203040500000 \\ \hline 10204060804050 \end{array}$$

Verified in Scheme

```
> (* 10001 1020304050)
10204060804050
```

Answer is 10204060804050

Problem 6.

This is a trick I have been teaching all kids.

To look at an easier version of the problem say we have to $11 * 11$. This is 121. Two 1s is getting multiplied by two 1s (eleven in this case). So we have the mnemonic 1..2...*then..reverse*. When we make one of the numbers as three 1s that is $111 * 11$ then we repeat the center digit 1..2..2...*then..reverse*, the answer being 1221.

In this example we have 11111 multiplied by 1111. This should give us 12344321. Two 4s in center.

Let us look at a pattern in Scheme to verify.

```
> (* 1111 1111)
1234321

> (* 11111 1111)
12344321

> (* 111111 1111)
123444321

> (* 1111111 1111)
1234444321
```

Answer is 12344321

Problem 7.

The solution is provided in the book. Its an easy problem where we use the last digits of the 3s multiplication table.

$$\begin{array}{r} 1ABCDE \\ \times\ 3 \\ \hline ABCDE1 \end{array}$$

The only was to get 1 as the answer when 3 is multiplied by E is $3 * 7$. The carryover is 2. So now we have $(3 * D) + 2$ which should end with E which we know is 7 now. So we get 3 times 5 plus 2 which ends in 7. Therefore D is 5. We keep going till we reach the end at the final number.

This number is actually important since this is the number which repeats when we divide a number by 7. This number is starting with 1 and with only 0s thereafter. Next few problems have this trick involved.

Answer is verified.

Answer is 142857.

6 The division algorithm

Problem 8.

This is inverse of the previous chapter.

$$
\begin{array}{r}
1001001 \\
123\,\overline{)\,123123123} \\
\underline{123} \\
0123 \\
\underline{123} \\
0123 \\
\underline{123} \\
0
\end{array}
$$

We can see the pattern of 001..001..001 in the answer. Suppose we had 1234123412341234 divided by 1234. What would we get? It would be 0001..0001..0001..0001

Verified in Scheme

```
> (/ 1234123412341234 1234)
1000100010001
> (/ 123123123 123)
1001001
```

Answer is 1001001

Problem 9.

We can simplify the problem here. We have 1111111 (seven 1s) which will divide a long series of 1s. That means for every group of seven 1s the quotient will be 1. Since there are 100 1s we will have 14 groups of seven 1s that makes it 98 1s. The last two 1s will be the remainder. So the remainder is 11.

A smaller pattern here

$$
\begin{array}{r|l}
 & 100.\overline{0000099} \\
\hline
1111111 & 111111111.0000000 \\
 & \underline{1111111} \\
 & \qquad 011.000000 \\
 & \qquad \underline{9.999999} \\
 & \qquad 1.0000010 \\
 & \qquad \underline{9999999} \\
 & \qquad\qquad 11
\end{array}
$$

Answer is 11

Problem 10.

We get here the cyclical nature of the quotient when we divide by 7. example

$$
\begin{array}{r|l}
 & 142857.\overline{142857} \\
\hline
7 & 1000000.000000 \\
 & \underline{7} \\
 & 30 \\
 & \underline{28} \\
 & \ 20 \\
 & \ \underline{14} \\
 & \ \ 60 \\
 & \ \ \underline{56} \\
 & \ \ \ 40 \\
 & \ \ \ \underline{35} \\
 & \ \ \ \ 50 \\
 & \ \ \ \ \underline{49} \\
 & \ \ \ \ \ 1.0 \\
 & \ \ \ \ \ \ \underline{7} \\
 & \ \ \ \ \ \ 30 \\
 & \ \ \ \ \ \ \underline{28} \\
 & \ \ \ \ \ \ \ 20 \\
 & \ \ \ \ \ \ \ \underline{14} \\
 & \ \ \ \ \ \ \ \ 60 \\
 & \ \ \ \ \ \ \ \ \underline{56} \\
 & \ \ \ \ \ \ \ \ \ 40 \\
 & \ \ \ \ \ \ \ \ \ \underline{35} \\
 & \ \ \ \ \ \ \ \ \ \ 50 \\
 & \ \ \ \ \ \ \ \ \ \ \underline{49} \\
 & \ \ \ \ \ \ \ \ \ \ \ 1
\end{array}
$$

So 1000000 (7 digit number) when divided by 7 will give a recurring quotient with 142857. Therefore when we divide 1000...0 (20 zeroes) we have 18 zeroes consumed with 3 times 142857 appearing in the quotient. Then the last 2 zeroes will give the quotient of 14 and a remainder of 2. Thus the quotient should be 14285714285714285714 and a remainder of 2.

Verified in Scheme

```
> (/ 100000000000000000000 7)
14285714285714285714 2/7
```

Problem 11.

As shown in previous problem number 10 this pattern of 142857 will repeat. This the cyclical number we get in this instance.

Problem 12.

This is fairly similar to the previous two problems. The only difference is that when we start with a different number the cyclical pattern of division by 7 starts with a different digit, but the pattern holds. Let us take the example of 2000000 divided by 7.

$$
\begin{array}{r@{}l}
 & \,285714.\overline{285714} \\
7\,\big) & \overline{2000000.000000} \\
 & \underline{14} \\
 & \;\;60 \\
 & \;\;\underline{56} \\
 & \;\;\;\;40 \\
 & \;\;\;\;\underline{35} \\
 & \;\;\;\;\;\;50 \\
 & \;\;\;\;\;\;\underline{49} \\
 & \;\;\;\;\;\;\;\;10 \\
 & \;\;\;\;\;\;\;\;\;\;\underline{7} \\
 & \;\;\;\;\;\;\;\;\;\;30 \\
 & \;\;\;\;\;\;\;\;\;\;\underline{28} \\
 & \;\;\;\;\;\;\;\;\;\;\;\;2.0 \\
 & \;\;\;\;\;\;\;\;\;\;\;\;\underline{1.4} \\
 & \;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;60 \\
 & \;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\underline{56} \\
 & \;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;40 \\
 & \;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\;\underline{35} \\
 & \;50 \\
 & \;\underline{49} \\
 & \;10 \\
 & \;\underline{7} \\
 & \;30 \\
 & \;\underline{28} \\
 & \;2
\end{array}
$$

We see the same pattern but it starts at 2. So the pattern is 285714.

So the answers are
2000...00(20 0s): Quotient is 8571428571428571428. Remainder is 4.
3000...00(20 0s): Quotient is 4285714285714285714 2. Remainder is 6.
4000...00(20 0s): Quotient is 5714285714285714285 7. Remainder is 1.
5000...00(20 0s): Quotient is 7142857142857142857 1. Remainder is 3.
6000...00(20 0s): Quotient is 8571428571428571428 5. Remainder is 5.

Verified in Scheme

```
> (/ 20000000000000000000 7)
2857142857142857142 4/7
> (/ 30000000000000000000 7)
4285714285714285714 6/7
> (/ 40000000000000000000 7)
5714285714285714285 1/7
> (/ 50000000000000000000 7)
7142857142857142857 3/7
```

```
> (/ 6000000000000000000000 7)
857142857142857142857 5/7
```

Problem 13.

The guess here should be that each of the answers will be in some permutation of 142857 barring when multiplied by 7. Let us check.

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 1 \\ \hline 1\,4\,2\,8\,5\,7 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 2 \\ \hline 2\,8\,5\,7\,1\,4 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 3 \\ \hline 4\,2\,8\,5\,7\,1 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 4 \\ \hline 5\,7\,1\,4\,2\,8 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 5 \\ \hline 7\,1\,4\,2\,8\,5 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 6 \\ \hline 8\,5\,7\,1\,4\,2 \end{array}$$

$$\begin{array}{r} 1\,4\,2\,8\,5\,7 \\ \times \qquad\qquad 7 \\ \hline 9\,9\,9\,9\,9\,9 \end{array}$$

The way to answer this is to start at the one's place digit and work backwards.

Answer is verified above.

Problem 14.

Let us go for the first 10 natural numbers from 1 to 10.

Case for 1:

Anything divided by 1 is the same thing. So the dividend and quotient is same and there is no remainder.

$$\begin{array}{r} 1000000 \\ 1\overline{)1000000} \\ \underline{1} \\ 0000000 \end{array}$$

Case for 2:

Since the dividend ends with 0 it is an even number. So half of dividend is the quotient and remainder is 0.

$$\begin{array}{r} 500000 \\ 2\overline{)1000000} \\ \underline{10} \\ 000000 \end{array}$$

Case for 3:

In this case we will always get a remainder of 1 since 1 less than 10 or 100 or 1000 is divisible by 3.

$$\begin{array}{r} 333333.\overline{3} \\ 3\overline{)1000000.0} \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 1.0 \\ \underline{9} \\ 1 \end{array}$$

Case for 4:

Except for 10 as the dividend where we will get a remainder of 2 and a quotient fo 2 also, the rest of the dividends will always be one fourths of the dividend since the dividend ends with 2 zeroes. The remainder will be 0.

$$\begin{array}{r} 2.5 \\ 4\overline{)10.0} \\ \underline{8} \\ 2.0 \\ \underline{2.0} \\ 0 \end{array} \qquad \begin{array}{l} \,250000 \\ 4\overline{)1000000} \\ \,\underline{8} \\ \,20 \\ \,\underline{20} \\ \,00000 \end{array}$$

Case for 5:

Here every number from 10 onwards will be divisible by 5. There will be no remainder.

$$\begin{array}{l} \,200000 \\ 5\overline{)1000000} \\ \,\underline{10} \\ \,000000 \end{array}$$

Case for 6:

In this case the remainder will always be 4. We will be stuck in an infinite loop of dividing 40 by 6 once we finish our first division of 10 by 6. The quotient therefore will be 1 followed by all 6s. Example below.

$$\begin{array}{r} 166666.\overline{6} \\ 6\overline{)1000000.0} \\ \underline{6} \\ 40 \\ \underline{36} \\ 40 \\ \underline{36} \\ 40 \\ \underline{36} \\ 40 \\ \underline{36} \\ 40 \\ \underline{36} \\ 4.0 \\ \underline{3.6} \\ 4 \end{array}$$

Case for 7 done earlier.

Case for 8:

The pattern in this case is that we have 1, 2, 5 as the quotient. And once there are no remainders left we keep appending 0s to the quotient of 125.

$$\begin{array}{r} 1.25 \\ 8\,\overline{)10.00} \\ \underline{8} \\ 2.0 \\ \underline{1.6} \\ 40 \\ \underline{40} \\ 0 \end{array} \qquad \begin{array}{r} 12.5 \\ 8\,\overline{)100.0} \\ \underline{8} \\ 20 \\ \underline{16} \\ 4.0 \\ \underline{4.0} \\ 0 \end{array} \qquad \begin{array}{r} 125 \\ 8\,\overline{)1000} \\ \underline{8} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 0 \end{array} \qquad \begin{array}{r} 1250 \\ 8\,\overline{)10000} \\ \underline{8} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 00 \end{array} \qquad \begin{array}{r} 12500 \\ 8\,\overline{)100000} \\ \underline{8} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 000 \end{array}$$

$$\begin{array}{r} 125000 \\ 8\,\overline{)1000000} \\ \underline{8} \\ 20 \\ \underline{16} \\ 40 \\ \underline{40} \\ 0000 \end{array}$$

Case for 9:

The first division by 9 gives a remainder of 1 and then an endless loop of 10 divided by 9. Remainder will always be 1 and quotient will be 1111....

$$\begin{array}{r} 111111.\bar{1} \\ 9\,\overline{)1000000.0} \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 10 \\ \underline{9} \\ 1.0 \\ \underline{9} \\ 1 \end{array}$$

Case for 10:

Fairly simple. Just remove the last zero from the dividend to get the quotient and there is no remainder.

$$\begin{array}{r} 100000 \\ 10\overline{\smash{)}1000000} \\ \underline{10} \\ 000000 \end{array}$$

7 The binary system

Problem 15.

Binary to Decimal conversion can be written by:

$[(0 \text{ or } 1) \times 2^0] + [(0 \text{ or } 1) \times 2^1] + [(0 \text{ or } 1) \times 2^2]$...

The numbers in the given list are basically the set of whole numbers.

Binary	**Decimal**	**Expanded form**
0	0	0
1	1	2^0
10	2	2^1
11	3	$2^1 + 2^0$
100	4	2^2
101	5	$2^2 + 2^0$
110	6	$2^2 + 2^1$
111	7	$2^2 + 2^1 + 2^0$
1000	8	2^3
1001	9	$2^3 + 2^0$
1010	10	$2^3 + 2^1$
1011	11	$2^3 + 2^1 + 2^0$
1100	12	$2^3 + 2^2$
1101	13	$2^3 + 2^2 + 2^0$
1110	14	$2^3 + 2^2 + 2^1$
1111	15	$2^3 + 2^2 + 2^1 + 2^0$
10000	16	2^4
10001	17	$2^4 + 2^0$
10010	18	$2^4 + 2^1$
10011	19	$2^4 + 2^1 + 2^0$
10100	20	$2^4 + 2^2$
10101	21	$2^4 + 2^2 + 2^0$
10110	22	$2^4 + 2^2 + 2^1$

Answer is verified

Problem 16.

This problem is basically binary representation of a natural number.

Let $S = \{2^0, 2^1, \ldots, 2^{n-1}\}$. Every integer m with $0 \leq m \leq (2^n - 1)$ can be written uniquely as a sum of distinct elements of S.

The proof for this can be demonstrated using induction but we will skip that here. The 3$^{\text{rd}}$ column in the previous problem (problem number 15) already shows the solution for this problem.

Problem 17.

We refer back to the table in problem 15. For 14 in decimal the equivalent binary representation is 1110. 10000 binary is $(1{*}2^4)+(0{*}2^3)+(0{*}2^2)+(0{*}2^1)+(0{*}2^0)$ and that is 16.

Problem 18.

From the binary representation theorem given in problem 16 let us look at a number of the form $2^n \leq 45$. The biggest n here is 5 where we get $2^5 = 32$. So we have 32 as 100000. We need 13 more. We apply the same logic and arrive at 1101 for 13. Thus adding the binary representations we get 101101 as the binary form of 45.

Answer is 101101

Problem 19.

10101101 in binary can be converted to decimal easily.

$10101101 = (1{*}2^7)+(0{*}2^6)+(1{*}2^5)+(0{*}2^4)+(1{*}2^3)+(1{*}2^2)+(0{*}2^1)+(1{*}2^0)$
$10101101 = 128 + 32 + 8 + 4 + 1 = 173$

Answer is 173

Problem 20.

Binary Addition

$0 + 0 = 0$ this is so because $(0 * 2^0) + (0 * 2^0)$
$0 + 1 = 1$ this is so because $(0 * 2^0) + (1 * 2^0)$
$1 + 1 = 0$ this is so because $(1{*}2^0)+(1{*}2^0)$ the answer is 2 which is $(1{*}2^1)+(0{*}2^0)$ thus carry 1

$$\begin{array}{r} 1010 \\ +\,101 \\ \hline 1111 \end{array}$$

This is $10 + 5 = 15$ in base 10.

$$\begin{array}{r} 1111 \\ +1 \\ \hline 10000 \end{array}$$

This is 15 + 1 = 16 in base 10.

$$\begin{array}{r} 1011 \\ +1 \\ \hline 1100 \end{array}$$

This is 11 + 1 = 12 in base 10.

$$\begin{array}{r} 1111 \\ +1111 \\ \hline 11110 \end{array}$$

This is 15 + 15 = 30 in base 10.

Problem 21.

Binary Subtraction

0 - 0 = 0 this is so because $(0 * 2^0) - (0 * 2^0)$
1 - 0 = 1 this is so because $(1 * 2^0) - (0 * 2^0)$
0 - 1 = 1 this is so because $(0 * 2^0) - (1 * 2^0)$ results in a borrow of 10_2. So now we have a 2 in decimal subtracted with 1. Thus there is a borrow of 1 in this case
1 - 1 = 0 this is so because $(1 * 2^0) - (1 * 2^0)$

$$\begin{array}{r} 1101 \\ -101 \\ \hline 1000 \end{array}$$

This is 13 - 5 = 8 in base 10.

$$\begin{array}{r} 110 \\ -1 \\ \hline 101 \end{array}$$

This is 6 - 1 = 5 in base 10.

$$\begin{array}{r} 1000 \\ -1 \\ \hline 111 \end{array}$$

This is 8 - 1 = 7 in base 10.

Problem 22.

Binary multiplication

0 multiplied by anything is 0 and 1 multiplied by 1 is 1. In the binary case we have
0 * 0 = 0 this is so because $(0 * 2^0) * (0 * 2^0)$
0 * 1 = 0 this is so because $(0 * 2^0) * (1 * 2^0)$
1 * 1 = 1 this is so because $(1 * 2^0) * (1 * 2^0)$

$$\begin{array}{r} 1101 \\ \times\, 1010 \\ \hline 0000 \\ 11010 \\ 000000 \\ +\, 1101000 \\ \hline 10000010 \end{array}$$

In base 10 this is $13 * 10 = 130$.

Problem 23.

Binary Division

This is same as long division in base 10. Please note that the hint given has a different dividend than that given in the question. This is perhaps a typographical error.

For the problem in the hint:

$$11001_2 \div 101_2 = 101_2 \text{ remainder } 0_2$$

This is $\frac{25}{5}$ in base 10

For the actual problem:

$$11011_2 \div 101_2 = 101_2 \text{ remainder } 10_2$$

This is $\frac{27}{5}$ in base 10

Problem 24.

Binary fractions

For fractions in binary, the key difference from base 10 is the following:

A fraction has a terminating binary expansion if and only if its denominator, when written in lowest terms, is a power of 2.

For example,

$$\frac{1}{4} = 0.01_2$$

However,

$$\frac{1}{3}$$

does not have a terminating binary representation. Performing binary division, we obtain

$$\frac{1}{3} = 0.01010101\ldots_2$$

Thus, just as $\frac{1}{3}$ is a non-terminating decimal in base 10, it is also a non-terminating fraction in base 2.

8 The commutative law

No problems

9 The associative law

Problem 25.

Tried it. Beg to differ from Gelfand and Shen on this one. The flavor and aroma are different in the two processes described in the equation.

Problem 26.

First do the addition of 17999 + 1 to get 18000 then add 357 since the last 3 digits of 18000 are 0s. The answer is 18357.

Problem 27.

In such cases add 1 and subtract 1 at the end. So we get 18357 from the same steps as problem 26 then we subtract 1. The answer is 18356.

Problem 28.

Here we add 899 + 101 first. It is 900 + 100 which is a thousand. 1000 + 1343 is 2343.

Problem 29.

25.4 is done first to give 100. Then then answer is 3700.

Problem 30.

In this we do 125.8 first which again gives us 1000. The final answer is 37000.

10 The use of parentheses

Problem 31.

This is not a useful problem from an algebra perspective. This is a combinatorics problem and a good one. Let us try and build a reasoning around how to solve for the simplest cases.

For every case we need to partition the numbers into 2 groups. Each group will have to stand on its own. And in each of these sub groups we have a situation for which we would have already done the count prior.

Case 0: No number - We do not need to put any parentheses.

$$0 \rightarrow 0$$

Case 1: 2 - We do not need any parentheses but can put one like (2).

$$1 \rightarrow 1$$

Case 2: 2.3 - Just one like (2.3)

$$2 \rightarrow 1$$

Case 3: 2.3.4 - Partitioning into smaller groups gives a group of 2 and 1. It is (2.3).4 and 2.(3.4) Thus we get 1 + 1 = 2.
Case 3 $\rightarrow 1 + 2$ - 2.(3.4) $\rightarrow 1 + 1$

$$3 \rightarrow 2$$

Case 4: 2.3.4.5 - The book solves this. Let us make partitions 2.(3.4.5). We notice that we have simplified it to case 3 here which will repeat twice. The other partition is (2.3).(4.5) which is two cases before i.e. case 2. So we get the following: $2 + 2$ from Case 3 + 1 from Case 1.
Case 4 $\rightarrow 1 + 3$ - 2.(3.4.5) $\rightarrow 2 + 2$
Case 4 $\rightarrow 2 + 2$ - (2.3).(4.5) $\rightarrow 1$

$$4 \to 5$$

Case 5: 2.3.4.5.6 - This is the question we are asked. The algorithm requires us to partition. 2.(3.4.5.6) is one way to partition and we have reduced the problem to Case 4 above which repeats twice. The other partition is (2.3).(4.5.6). In this case we go two steps back. So we get $5 + 5 + 2 + 2$.
Case 5 $\to 1 + 4$ - 2.(3.4.5.6) $\to 5 + 5$
Case 5 $\to 2 + 3$ - (2.3).(4.5.6) $\to 2 + 2$

$$5 \to 14$$

Case 6: 2.3.4.5.6.7 - Let us take it a notch higher. The sub cases which can be built are
Case 6 $\to 1 + 5$ - 2.(3.4.5.6.7) $\to 14 + 14$
Case 6 $\to 2 + 4$ - (2.3).(4.5.6.7) $\to 5 + 5$
Case 6 $\to 3 + 3$ - (2.3.4).(5.6.7) $\to 2 \times 2$

$$6 \to 42$$

In fact we can even make it generic. Essentially what we are doing is partitioning numbers. Then for each partition we are looking back at previous permutations and adding. Note that in some cases we need to multiply too (for instance last sub case in Case 6). Can we generalize this? Yes, these are basically Catalan numbers! The nth Catalan number is given by the expression for all $n \geq 0$

$$\frac{(2n)!}{(n+1)!\, n!}$$

Problem 32.

This too is a combinatorics problem.

Let us take a simple trivial case of the problem.
2 $\to$ we do not need any parentheses
2.3 $\to$ we still do not need any parentheses
2.3.4 $\to$ now we need to put the first pair so that it becomes (2.3).4. So for $n = 3$ digits we have 2 i.e. $2(n - 2)$ parentheses.
2.3.4.5 $\to$ we need to put one additional pair ((2.3).4).5. So for $n = 4$ digits we have 4 i.e. $2(n - 2)$ parentheses.

Generalizing, for $n > 2$ number of parentheses is

$$2(n - 2)$$

In the given question we have the question as 2.3.4.5.....97.98.99.100. These are $(100 - 2) + 1$ digits. Therefore $n = 99$ and total number of parentheses are $2 \times (99 - 2)$ which is 194.

Answer is 194.

Problem 33.

This is easily doable like the way the child Gauss did. Basically there are 100 elements in this series

$$\begin{array}{rcl} S & = & 1 + 2 + 3 + \ldots + 99 + 100 \\ +S & = & 100 + 99 + 98 + \ldots + 2 + 1 \\ \hline 2S & = & 101 + 101 + 101 + \ldots + 101 \end{array}$$

$$2S = 100 \times 101$$

$$S = \frac{100 \times 101}{2} = 5050$$

11 The distributive law

Problem 34.

$$\begin{aligned} &= 1001 \times 20 \\ &= (1000 + 1) \times 20 \\ &= 20000 + 20 \\ &= 20020 \end{aligned}$$

In this we could simply multiply 1001 by 2 to get 2002 and then append a 0 behind it too.

Problem 35.

$$
\begin{aligned}
&= 1001 \times 102 \\
&= (1000 + 1) \times 102 \\
&= 102000 + 102 \\
&= 102102
\end{aligned}
$$

Problem 36.

$$(a + b + c + d + e)(x + y + z)$$

For each number from a to e we will get 3 terms one for each x, y and z. Thus total of 3 * 5 terms. 15 terms.

12 Letters in algebra

Problem 37.

Let small vessel volume be x and big vessel volume be y, then

$$x + y = 5$$

$$2x + 3y = 13$$

Solving these two linear equations we get $y = 3$ and $x = 2$. To solve such equations make the coefficients of one of the unknown same and subtract one equation from the other.

Problem 38.

The simple explanation is given in the book.

$$
\begin{gathered}
x \\
x + 3 \\
2.(x + 3) \\
(2x + 6) - x \\
(x + 6) - 4 = x + 2 \\
(x + 2) - x \\
2
\end{gathered}
$$

13 The addition of negative numbers

No problems

14 The multiplication of negative numbers

No problems

15 Dealing with fractions

Problem 39.

The explanation of this problem is humorous! But such a nice way to put it.

If only vodka bottles were used in the explanation everything would have fallen in place with this soviet era content.

Anyways.

$\frac{1}{3} \times \frac{7}{7}$ and $\frac{2}{7} \times \frac{3}{3}$

$\frac{7}{21}$ and $\frac{6}{21}$

We conclude that $\frac{1}{3}$ is bigger.

Problem 40.

This is a problem also found in the initial assessment for the Gelfand Correspondence Program.

$$\frac{10001}{10002} = 1 - \frac{1}{10002}$$

Similarly,

$$\frac{100001}{100002} = 1 - \frac{1}{100002}$$

As we can see the $\frac{1}{100002}$ much more smaller than $\frac{1}{10002}$. Thus when we subtract a smaller number from 1 we are left with a bigger number.

Answer is $\frac{100001}{100002}$

Problem 41.

Good problem.

$$\frac{12345}{54321} \times \frac{54322}{54322}$$

$$\frac{12346}{54322} \times \frac{54321}{54321}$$

Now lets only look at the numerators since the denominators are equal.

$$(12345) \times (54321 + 1)$$
$$(12345 + 1) \times (54321)$$

Make them both have common terms

$$(12345 \times 54321) + 12345$$
$$(12345 \times 54321) + 54321$$

It is clear now that the second fraction is bigger because it has 54321 in the numerator vs 12345 in the first fraction when all other terms are same in numerator and denominator.
Answer is $\frac{12346}{54322}$

Just to verify this in Scheme. We get a positive fraction when we subtract the first fraction from the second one.

```
> (- (/ 12346 54322) (/ 12345 54321))
6996/491804227
```

Problem 42.

These 3 problems are pretty difficult actually for middle school kids.

(a) We will use proof by contradiction to prove this.

Assume the greatest common divisor of a and b is m.

$$gcd(a, b) = m, m > 1$$

So $m|a$ and $m|b$ ($x|y$ reads x *divides* y)

Therefore m should also divide $ad - bc$ i.e. $m|(ad - bc)$

But we know $ad - bc = \pm 1$. So in $\dfrac{(ad - bc)}{m}$ denominator has to be $+1$

Thus m is $+1$ and $gcd(a, b) = 1$

Hence $\dfrac{a}{b}$ cannot be further simplified. The same proof can be used for $\frac{c}{d}$.

(b) Now to second part of this problem. I struggled with this one quite a bit. A little bit background on Farey Sequence. Niven and Zuckerman (1972) defined Farey Sequence as

> *The sequence of all reduced fractions with denominators not exceeding n, listed in order of their size, is called the Farey sequence of order n.*

Sometimes the definition is restricted to the interval 0 to 1. In this interval say we look at Farey sequence of order 3 which is given by

$$F_3 = \left\{ \frac{0}{1}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{1}{1} \right\}$$

The third term minus the second term here is

$$\frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

The numerator is 1. In Farey Sequences the numerator on subtraction of two consecutive elements is always ± 1

The problem says that in a Farey sequence of $\dfrac{a}{b}$ and $\dfrac{c}{d}$ the fraction $\dfrac{a+b}{c+d}$ is always between $\dfrac{a}{b}$ and $\dfrac{c}{d}$. Let us plug in the numbers from the F_3 sequence quoted above.

$$\frac{a+b}{c+d} = \frac{1+3}{1+2} = \frac{4}{3}$$

So the inequality is now

$$\frac{1}{3} < \frac{4}{3} < \frac{1}{2}$$

But $\frac{4}{3}$ is greater than 1 and does not lie there! The point is that there is a typographical mistake in this problem. The actual fraction between $\frac{a}{b}$ and $\frac{c}{d}$ is called the Mediant fraction and is given as

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$$

To verify

$$\frac{1}{3} < \frac{2}{5} < \frac{1}{2}$$

And this is correct.

Back to part (b) of the problem now. We need to prove the below.

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$$

Since $\frac{a}{b}$ and $\frac{c}{d}$ are consecutive terms in a Farey sequence, we have

$$bc - ad = 1,$$

and in particular $ad < bc$, with $b > 0$ and $d > 0$.

We now prove the two inequalities using cross-multiplication.

First,

$$\frac{a}{b} < \frac{a+c}{b+d} \iff a(b+d) < b(a+c) \iff ab + ad < ab + bc \iff ad < bc,$$

which is true.

Similarly,

$$\frac{a+c}{b+d} < \frac{c}{d} \iff d(a+c) < c(b+d) \iff ad + cd < bc + cd \iff ad < bc,$$

which is also true.

Hence,

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d},$$

and therefore the mediant fraction $\frac{a+c}{b+d}$ always lies strictly between $\frac{a}{b}$ and $\frac{c}{d}$.

(c) Last part of this challenging problem. We have say

$$\frac{a}{b} < \frac{e}{f} < \frac{c}{d}$$

$$\frac{e}{f} - \frac{a}{b} = \frac{be - af}{bf}$$

Since $be - af$ will be a positive integer and therefore at least 1, we can say

$$\frac{e}{f} - \frac{a}{b} \geq \frac{1}{bf}$$

Similarly we get

$$\frac{c}{d} - \frac{e}{f} \geq \frac{1}{df}$$

Now we have to visualize. Moving from $\frac{a}{b}$ to $\frac{e}{f}$ and from that finally to $\frac{c}{d}$. Total Distance is $\frac{1}{bd}$ (from the denominator between $\frac{a}{b}$ and $\frac{c}{d}$). So we get

$$\frac{1}{bd} = \left(\frac{e}{f} - \frac{a}{b}\right) + \left(\frac{c}{d} - \frac{e}{f}\right)$$

But we know from the inequalities above.

$$\frac{1}{bd} \geq \frac{1}{bf} + \frac{1}{df} = \frac{b+d}{bdf}$$

$$\frac{1}{bd} \geq \frac{b+d}{bdf}$$

$$1 \geq \frac{b+d}{f}$$

$$f \geq b + d$$

Hence proved that f cannot be less than $b + d$

Difficult problems for middle schoolers!

Problem 43.

This is application of mediant formula we derived in problem 42.

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$$

The end 2 pieces of $\frac{1}{20}$ can be omitted so we are left with 18 pieces. There are 6 red marks (7 equal segments will require 6 marks) and 12 green marks (13 equal segments). We can visualize from the left of the stick the first cut would be at $\frac{1}{20}$ which we have omitted then the next cut will be at $\frac{1}{13}$ and then at $\frac{1}{7}$. Let us find mediant between $\frac{1}{13}$ and $\frac{1}{7}$.

$$\frac{1}{13} < \frac{1+1}{13+7} < \frac{1}{7}$$
$$\frac{1}{13} < \frac{2}{20} < \frac{1}{7}$$

Note that the mediant will always lie at $\frac{k}{20}$. Thus the cut will be there and each of the 18 pieces will have only color either green or red.

The problem is solved in the book.

Problem 44.

First expression is

$$5\% \times 7 \times 10^9$$
$$35 \times 10^7$$

Second expression is

$$7\% \times 5 \times 10^9$$
$$35 \times 10^7$$

Thus they are equal.

Problem 45.

The more systematic way to reason is to say what number k when multiplied by $\frac{2}{3}$ gives $\frac{1}{2}$.

$$k \times \frac{2}{3} = \frac{1}{2}$$
$$k = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4}$$

So we fold the string in half once, then re fold it. We get quarter of the original two thirds. Now we cut off one of the one fourths and then we are left with three fourths of the original two thirds which is now half.

The problem is solved in the book.

16 Powers

Problem 46.

(a) 1024 is the answer. It is good to have the powers of 2 memorized for quick computation - 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024 (for folks interested in computers this will be second nature).

(b) 1000 - a thousand.

(c) 10000000 - 10 million - In India this is also called a Crore.

Problem 47.

Assuming by *decimal digits* the authors mean digits. Then the answer is 10001. 1000 zeroes and a one 1.

Problem 48.

Total number of seconds in a year

$$60 \times 60 \times 24 \times 365 = 31536000$$

Distance traveled in 4 light years will be

$$4 \times 3 \times 10^5 \times 31536000 = 37843200000000$$

That is a whopping 37.8432 Trillion Kilometers

17 Big numbers around us

Problem 49.

(a) 2^{10} is 1024 so $2^{20} = 2^{10} \times 2^{10} = 1024 \times 1024$
This will 1 followed by 6 other digits. So total digits will be 7.

Verified in Scheme

```
> (expt 2 20)
1048576
```

(b) $2^{100} = 2^{10} \times 2^{10}$ ....ten times
$1024 \times 1024 \times 1024$..... ten times
$(10^3 + 24)^2 \times (10^3 + 24)^2$....five times
$(1000000 + 576 + 48000) \times (1000000 + 576 + 48000)$ five times
$1048576 \times 1048576 \times 1048576 \times 1048576 \times 1048576$
So we get 31 digits. Why? Because the 1000000 will grow much bigger than the 48576. This is intuitively speaking. Ideally we should do it properly. Let us try that.

A natural number N is given. How many digits does this have?
If it is between 1 (included) to less than 10 it has 1 digit. If it is from 10 (included) to less than 100 then it has 2 digits. If it is between 100 (included) to less than 1000 it has 3 digits and so on. Representing it in inequality form.

$$
\begin{gathered}
10^0 \leq N < 10^1 \ \rightarrow 1 \ digit \\
10^1 \leq N < 10^2 \ \rightarrow 2 \ digits \\
10^2 \leq N < 10^3 \ \rightarrow 3 \ digits \\
10^3 \leq N < 10^4 \ \rightarrow 4 \ digits \\
..... \\
10^{k-1} \leq N < 10^k \ \rightarrow k \ digits
\end{gathered}
$$

So we have an inequality for the number of digits k for a number N.

$$10^{k-1} \leq N < 10^k$$

Let us take logarithms on both sides for the number 2^n. I understand students would not have been taught this yet. But I am sure if kids are working through this book they are smart enough to pick this up.

$$
\begin{gathered}
\log_{10} 10^{k-1} \leq \log_{10} 2^n < \log_{10} 10^k \\
k - 1 \leq n \log 2 < k
\end{gathered}
$$

rearranging the inequality above we get

$$n \log 2 < k \leq n \log 2 + 1$$

To reiterate the number 2^n will have k digits and this k is certainly bigger than $n \log 2$ and an integer and also lesser than or equal to $n \log 2 + 1$.
We can write k as

$$k = \lfloor n \log 2 \rfloor + 1$$

So we conclude that the number 2^n has $\lfloor n \log 2 \rfloor + 1$ digits.

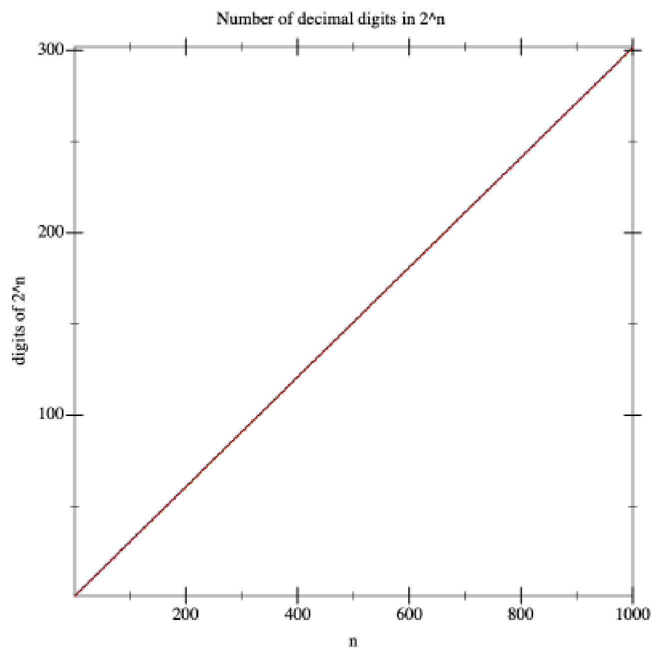
Let us try out some examples.

$2^{10} = \lfloor 10 \log 2 \rfloor + 1 = \lfloor 10 \times 0.30102999 \rfloor + 1 = 3 + 1 = 4$
Indeed $2^{10} = 1024$ which has 4 digits.
Note value of $\log 2$ is 0.30102999.

$2^{100} = \lfloor 100 \log 2 \rfloor + 1 = \lfloor 100 \times 0.30102999 \rfloor + 1 = 30 + 1 = 31$
We can verify with certainty that now the answer to this question is 31.

(c) For this part of the problem I will have to use programming. I am unsure how else to construct this graph.
Using Racket (a Scheme)

```
#lang racket
(require plot)

;; Function for the actual computation
(define (digits-of-2^n n)
  (add1 (floor (/ (* n (log 2)) (log 10)))))

;; Plot digits for n from 1 to 1000
(plot
 (function digits-of-2^n 1 1000)
 #:x-label "n"
 #:y-label "digits of 2^n"
 #:title "Number of decimal digits in 2^n")
```



18 Negative powers

Problem 50.

(a) $\frac{1}{10}$ or 0.1

(b) $\frac{1}{100}$ or 0.01

(c) $\frac{1}{1000}$ or 0.001

19 Small numbers around us

Problem 51.

As per notation yes both are true.

$$a^{-n} = \frac{1}{a^n}$$
$$a^{-(-k)} = \frac{1}{a^{-k}}$$
$$a^k = \frac{1}{a^{-k}}$$

a^0 is 1 so $\dfrac{1}{1}$ is 1 again.

Problem 52.

(a) $a^{10}b^4$

(b) $2.a^3b^{-2}$

Problem 53.

(a) $\dfrac{a^3}{b^5}$

(b) $\dfrac{1}{a^2b^7}$

20 How to multiply a^m by a^n, or why our definition is convenient

Problem 54.

Not sure of the ask of this question. Probably the answer is a^{m-n}.

21 The rule of multiplication for powers

Problem 55.

(a) $n = 2000 - 1001 = 999$

(b) $1001 + n = -2$ thus $n = -1003$

(c) $\dfrac{1}{1000}$ vs $\dfrac{1}{1024}$. Thus 10^{-3} is bigger.

(d) $1000 - n = 501$, $n = 499$

(e) $1000 - n = -4$, $n = 1004$

(f) $2 \times 100 = n$, $n = 200$

(g) $(2 \times 3)^{100} = a^{100}$, $a = 6$

(h) $10 \times 15 = n$, $n = 150$

Problem 56.

It does not matter what sign m and n have as such. Specifically if $m > 0$ and $n < 0$ then $(a^m)^{-n} = \dfrac{1}{(a^m)^n}$.

For either of them to be zero the answer would be 1 since one of the powers is zero.

Problem 57.

Again signs do not make any difference to the formula $(ab)^n = a^n \cdot b^n$

Problem 58.

If $a = 0$ then it will 0.
If $a > 0$ then it will be $-a^{775}$
If $a < 0$ then it will be a^{775}

Problem 59.

Ideally $b \neq 0$ is the first call out.
Otherwise it is easy to put any number whether integer or fraction as n here.
So not sure of the intent of the problem in this case.

Problem 60.

We can manipulate the base $4^{\frac{1}{2}}$ to $(2^2)^{\frac{1}{2}}$. Thus this can be simplified to $2^{(2\times\frac{1}{2})}$ giving 2^1. But we need to be careful here since $-2 \times -2 = (-2)^2$ also. Therefore the answer will ± 2.

Similarly for $27^{\frac{1}{3}}$ should give the third root because of $3 \times 3 \times 3$ giving 3^3. Here we will not get -3 else that would make the answer negative and incorrect.

22 Formula for short multiplication: The square of a sum

Problem 61.

Application of $(a+b)^2 = a^2 + b^2 + 2ab$

(a)

$$\begin{gathered} 101^2 \\ = (100+1)^2 \\ = 10000 + 1 + 200 \\ = 10201 \end{gathered}$$

(b)

$$\begin{gathered} 1002^2 \\ = (1000+2)^2 \\ = 1000000 + 4 + 4000 \\ = 1004004 \end{gathered}$$

Problem 62.

Let product p be

$$p = m \times n$$

Now m and n the factors become 10% bigger

$$
\begin{gathered}
(m + 10\%m) \times (n + 10\%n) \\
1.1m \times 1.1n \\
1.21m \times n \\
1.21p \\
(p + 21\%p)
\end{gathered}
$$

Thus the product becomes 21% bigger.

Problem 63.

This question is to drive home the point made in the text that "The square of the sum of two terms is the sum of their squares plus two times the product of the terms."

The core message is that *square of the sum* and *sum of the squares* are two different things. Rightly so. Students need to be careful, that is all.

The answer to the problem is 'No'. Why?

Case 1: NN is me a man. I, the father, have a son. The father of the son is me. So this refers to me. Now my father has a son but he could have more than one son. In my case we are actually two brothers. So not always true.

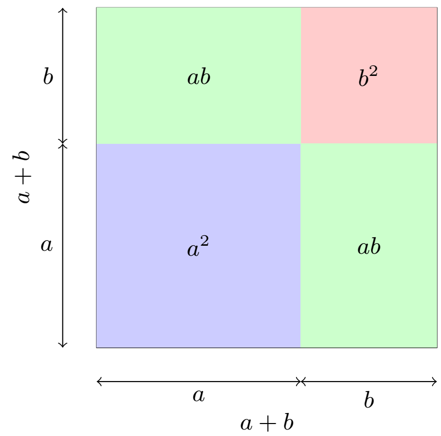
Case 2: NN is my wife. My wife's son has a father which is me. But I am not my wife. So this is incorrect already. My wife's father does have a son who is my wife's brother. But my brother in law and wife are not the same person.

Luckily my real family is good to answer this question!

23 How to explain the square of the sum formula to your younger brother or sister

Problem 64.

This is a simple representation of the formula $(a + b)^2 = a^2 + b^2 + 2ab$.



Problem 65.

(a) $99^2 = (100-1)^2 = 10000 + 1 - 200 = 9801$

(b) $998^2 = (1000-2)^2 = 1000000 + 4 - 4000 = 996004$

Problem 66.

(a) When $a = b$ then
The square of the sums gives $4a^2$ or $4b^2$
The square of the difference gives 0

(b) When $a = 2b$ then
The square of the sums gives $\frac{9}{4}a^2$ or $9b^2$
The square of the difference gives $\frac{a^2}{4}$ or b^2

24 The difference of squares

Problem 67.

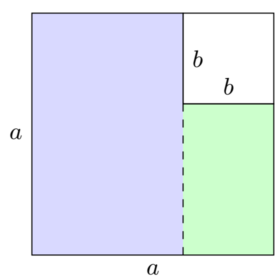
$$(a+b)(a-b) = a^2 \cancel{- ab} \cancel{+ ba} - b^2 = a^2 - b^2$$

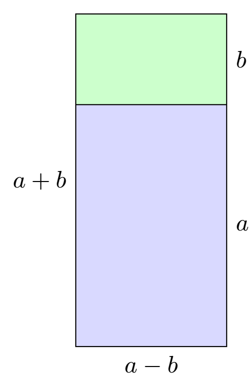
Problem 68.

$$101 \times 99 = (100+1)(100-1) = 100^2 - 1^2 = 9999$$

Problem 69.

We just cut it vertically as shown with the dotted line and then stack the two rectangles with $(a-b)$ side matching.





Problem 70.

Let the larger number be n then the other number will be $(n-2)$. We can write:

$$n(n-2) + 1 = n^2 - 2n + 1 = (n-1)^2$$

$(n-1)^2$ is a perfect square and the number $(n-1)$ is between n and $(n-2)$.

Problem 71.

The difference between the squares of two consecutive numbers n and $(n+1)$ is

$$(n+1)^2 - n^2 = n^2 + 1 + 2n - n^2 = 2n + 1$$

Now difference between the squares of the next two consecutive numbers $(n+1)$ and $(n+2)$ is

$$(n+2)^2 - (n+1)^2 = n^2 + 4 + 4n - n^2 - 1 - 2n = 2n + 3$$

So the difference between the two differences is

$$(2n+3) - (2n+1) = 3 - 1 = 2$$

2 is the constant difference. This is called an arithmetic progression.

Problem 72.

This is a nice trick. Let a number be of the form $n5$. This is a two digit number but we can extend the logic for higher digit numbers.

$n5$ can be written as $10n + 5$.

So the square will be $(10n + 5)^2$. This can be rearranged as.

$$(10n + 5)^2 = 100n^2 + 25 + 100n = 100n(n + 1) + 25$$

The $100n(n+1)$ is a number n times $(n+1)$ i.e. two consecutive numbers. Multiplying by 100 gives it the correct place in decimal value system as thousandth for this two digit square. We already have the left over 25 for the ending two digits. So we get

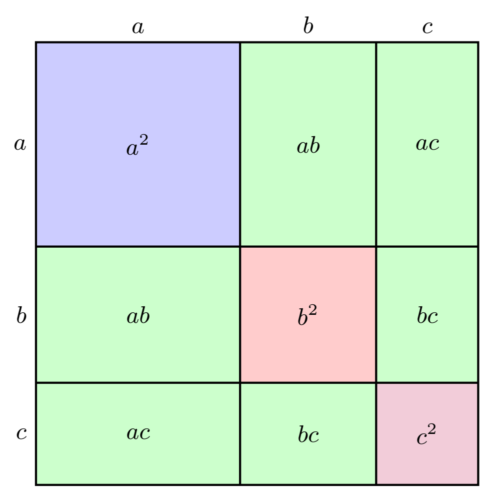
$$(n5)^2 = (n \times (n + 1))25$$

Thus we can prove this trick.

Problem 73.

$$\begin{gathered}(a + b + c)^2 = (a + b + c) \times (a + b + c) \\ = a^2 + ab + ac + ab + b^2 + bc + ac + bc + c^2 \\ = a^2 + b^2 + c^2 + 2(ab + bc + ca)\end{gathered}$$

Visually we see it as below.



Problem 74.

$$(a+b-c)^2 = a^2 + b^2 + c^2 + 2(ab - bc - ac)$$

Problem 75.

Consider $(a+b)$ as say A so we have $(A+c)(A-c)$. We get

$$(A+c)(A-c) = A^2 - c^2$$
$$(a+b)^2 - c^2$$
$$a^2 + b^2 - c^2 + 2ab$$

Problem 76.

Consider $(b+c)$ as say B so we have $(a+B)(a-B)$. We get

$$(a+B)(a-B) = a^2 - B^2$$
$$a^2 - (b+c)^2$$
$$a^2 - b^2 - c^2 - 2bc$$

Problem 77.

We can change it to $(a+(b-c))(a-(b-c))$, this is of the form $(a+B)(a-B)$.

$$a^2 - B^2$$
$$a^2 - b^2 - c^2 + 2bc$$

Problem 78.

We see a pattern here and can avoid long multiplications

$$(a^2 - 2ab + b^2)(a^2 + 2ab + b^2)$$
$$(a-b)^2(a+b)^2$$
$$((a-b)(a+b))^2$$
$$(a^2 - b^2)^2$$
$$a^4 - 2a^2b^2 + b^4$$

25 The cube of the sum formula

Problem 79.

Essentially the use of the formula $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

$$\begin{aligned}11^3 &= (10+1)^3\\ &= 10^3 + 1^3 + 3.10.1(10+1)\\ &= 1000 + 1 + 330\\ &= 1331\end{aligned}$$

Problem 80.

Same as the previous problem.

$$\begin{aligned}101^3 &= 100^3 + 1^3 + 3.100.1(100+1)\\ &= 1000000 + 1 + 30300\\ &= 1030301\end{aligned}$$

Problem 81.

This is again failry simple

$$\begin{aligned}(a-b)^3 &= (a-b)^2 \times (a-b)\\ &= (a^2 + b^2 - 2ab) \times (a-b)\\ &= a^3 + ab^2 - 2a^2b - a^2b - b^3 + 2ab^2\\ &= a^3 - b^3 + 3ab^2 - 3a^2b\\ &= a^3 - b^3 - 3ab(a-b)\end{aligned}$$

26 The formula for $(a+b)^4$

No problems.

27 Formulas for $(a+b)^5$, $(a+b)^6$,... and Pascal's triangle

Problem 82.

For this problem we will use the Pascal's triangle for each of the questions asked.

$$11^3 = (10+1)^3 = 10^3 + 3 \cdot 10^2 \cdot 1 + 3 \cdot 10 \cdot 1^2 + 1^3 = 1000 + 1 + 300 + 30 = 1331$$

$$\begin{aligned} 11^4 = (10+1)^4 &= 10^4 + 4 \cdot 10^3 \cdot 1 + 6 \cdot 10^2 \cdot 1^2 + 4 \cdot 10 \cdot 1^3 + 1^4 \\ &= 10000 + 4000 + 600 + 40 + 1 = 14641 \end{aligned}$$

$$\begin{aligned} 11^5 = (10+1)^5 &= 10^5 + 5 \cdot 10^4 \cdot 1 + 10 \cdot 10^3 \cdot 1^2 + 10 \cdot 10^2 \cdot 1^3 + 5 \cdot 10 \cdot 1^4 + 1^5 \\ &= 100000 + 50000 + 10000 + 1000 + 50 + 1 = 161051 \end{aligned}$$

$$\begin{aligned} 11^6 = (10+1)^6 &= 10^6 + 6 \cdot 10^5 \cdot 1 + 15 \cdot 10^4 \cdot 1^2 + 20 \cdot 10^3 \cdot 1^3 + 15 \cdot 10^2 \cdot 1^4 + 6 \cdot 10 \cdot 1^5 + 1^6 \\ &= 1000000 + 600000 + 150000 + 20000 + 1500 + 60 + 1 = 1771561 \end{aligned}$$

Problem 83.

$$(a+b)^7 = a^7 + 7a^6b + 21a^5b^2 + 35a^4b^3 + 35a^3b^4 + 21a^2b^5 + 7ab^6 + b^7$$

Problem 84.

In this set of 3 problems the b is replaced by $-b$ and for odd powers we can take that into consideration.

$$\begin{aligned} (a-b)^4 &= a^4 + 4a^3(-b)^1 + 6a^2(-b)^2 + 4a(-b)^3 + (-b)^4 \\ &= a^4 - 4a^3b + 6a^2b^2 - 4ab^3 + b^4 \end{aligned}$$

$$\begin{aligned} (a-b)^5 &= a^5 + 5a^4(-b)^1 + 10a^3(-b)^2 + 10a^2(-b)^3 + 5a(-b)^4 + (-b)^5 \\ &= a^5 - 5a^4b + 10a^3b^2 - 10a^2b^3 + 5ab^4 - b^5 \end{aligned}$$

$$\begin{aligned} (a-b)^6 &= a^6 + 6a^5(-b)^1 + 15a^4(-b)^2 + 20a^3(-b)^3 + 15a^2(-b)^4 + 6a(-b)^5 + (-b)^6 \\ &= a^6 - 6a^5b + 15a^4b^2 - 20a^3b^3 + 15a^2b^4 - 6ab^5 + b^6 \end{aligned}$$

Problem 85.

The sum of the coefficients in Pascal's triangle will be a power of 2 as shown in the table below.

Row	Coefficients	Sum of Coefficients
1	1	1
2	1 + 1	2
3	1 + 2 + 1	4
4	1 + 3 + 3 + 1	8
5	1 + 4 + 6 + 4 + 1	16
6	1 + 5 + 10 + 10 + 5 + 1	32
7	1 + 6 + 15 + 20 + 15 + 6 + 1	64
n	...	2^{n-1}

Problem 86.

In this scenario the variable a or b has a coefficient which is the sum of the coefficients in that specific row of the Pascal's triangle.

$$(a+b)^2 = 4a^2$$
$$(a+b)^3 = 8a^3$$
$$(a+b)^4 = 16a^4$$

Problem 87.

Yes, each of the expressions collapses to the sum of the coefficients pertaining to it as per the Pascal's triangle.

Problem 88.

Every expression turns to 0 since $a - b = 0$

28 Polynomials

Problem 89.

$$\begin{aligned}
&(1+x-y)(12-zx-y) \\
&= 12 - xz - y + 12x - x^2z - xy - 12y + xyz + y^2 \\
&= 12 - xz - 13y + 12x - x^2z - xy + xyz + y^2
\end{aligned}$$

Problem 90.

(a)

$$(1+x)(1+x^2) = 1 + x^2 + x + x^3 = 1 + x + x^2 + x^3$$

(b)

$$\begin{aligned}
&(1+x)(1+x^2)(1+x^3)(1+x^4) = (1+x+x^2+x^3)(1+x^3+x^4+x^7)\\
&= 1 + 0x + 0x^2 + 1x^3 + 1x^4 + +0x^5 + 0x^6 + 1x^7 + 0x^8 + 0x^9 + 0x^{10}\\
&+0 + 1x + 0x^2 + 0x^3 + 1x^4 + +1x^5 + 0x^6 + 0x^7 + 1x^8 + 0x^9 + 0x^{10}\\
&+0 + 0x + 1x^2 + 0x^3 + 0x^4 + +1x^5 + 1x^6 + 0x^7 + 0x^8 + 1x^9 + 0x^{10}\\
&+0 + 0x + 0x^2 + 1x^3 + 0x^4 + +0x^5 + 1x^6 + 1x^7 + 0x^8 + 0x^9 + 1x^{10}\\
&= 1 + x + x^2 + 2x^3 + 2x^4 + 2x^5 + 2x^6 + 2x^7 + x^8 + x^9 + x^{10}
\end{aligned}$$

(c)

$$\begin{aligned}
&(1+x+x^2+x^3)^2\\
&= 1 + 1x + 1x^2 + 1x^3 + 0x^4 + 0x^5 + 0x^6\\
&+0 + 1x + 1x^2 + 1x^3 + 1x^4 + 0x^5 + 0x^6\\
&+0 + 0x + 1x^2 + 1x^3 + 1x^4 + 1x^5 + 0x^6\\
&+0 + 0x + 0x^2 + 1x^3 + 1x^4 + 1x^5 + 1x^6\\
&= 1 + 2x + 3x^2 + 4x^3 + 3x^4 + 2x^5 + x^6
\end{aligned}$$

(d) This is essentially the previous problem. There is a pattern in this - the coefficients go from 1 to n where n is the number of terms. In this case $n = 11$ (1 and all the 10 x^y terms). Then the n goes back in reverse counting to 1. Meanwhile the x^y keep increasing. We can simply write the answer as.

$$1 + 2x + 3x^2 + 4x^3 + 5x^4 + 6x^5 + 7x^6 + 8x^7 + 9x^8 + 10x^9 + 11x^{10} + 10x^{11} + 9x^{12} + 8x^{13}$$
$$+7x^{14} + 6x^{15} + 5x^{16} + 4x^{17} + 3x^{18} + 2x^{19} + x^{20}$$

(e) We need not multiply these long polynomials. To get x^{30} there is only one way possible where the x^{10} of the three expressions multiply with itself. Thus coefficient of x^{30} is 1.

For x^{29} too we can reason out. The only way to get 29 is a sum of $(9, 10, 10)$ in various permutations. That is only 3 ways $(9, 10, 10)$, $(10, 9, 10)$, and $(10, 10, 9)$. Anything lower that 8 is not possible (we will need one more since 8, 10, 10 will give 28).

(f) This one too we do not need to multiply at all. When multiplying by 1 we would get the same term $(1 + x + x^2 ... + x^{10})$ but when we multiply by $-x$ we shift the expression all with minus signs like $(-x - x^2 - x^3 - ... - x^{10} - x^{11})$. Adding them all up would just leave the extreme terms intact rest will cancel out each other. The answer will be $(1 - x^{11})$.

(g) This is a quick multiplication and cancelling out of terms

$$= a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$$
$$= a^3 + b^3$$

(h) This one we cannot do any quick multiplication. But we can reason it out.

$= 1 + x + x^2 + x^3 + ... + x^{10}$
$+0 - x - x^2 - x^3 - ... - x^{10} - x^{11}$
$+0 + 0 + x^2 + x^3 + ... + x^{10} + x^{11} + x^{12}$
$-$...
$+$...
.... $- x^{18} - x^{19}$
.... $+ x^{18} + x^{19} + x^{20}$

All the odd terms will cancel out when added together. Thus the answer is
$1 + x^2 + x^4 + x^6 + x^8 + ... + x^{18} + x^{20}$

For some of the problems in this section application of sum of a geometric series makes the problem much easier. But the book has not introduced it yet. So we must skip it.

29 A digression: When are polynomials equal?

Problem 91.

Put $x = -1$, this makes the left hand as 0 but the right hand side of the equation is non zero. Thus, these two are not equal polynomials.

Problem 92.

In this instance $x = 1$ or $x = -1$ does not cut it. But $x = -3$ makes the right hand side zero and left hand side as non zero. Thus, these two are not equal polynomials.

Problem 93.

$(x + 1)^2 - (x - 1)^2$
Not a good idea. $x = 2$ shows that both sides when evaluated give different values. It is better to expand the left hand side since it is simple.

30 How many monomials do we get?

Problem 94.

A polynomial with 4 monomials when multiplied by another polynomial with 4 terms will yield 16 monomial terms in total.

Problem 95.

Yes, they can yield lower than 16 monomials if the monomial are similar terms.

Problem 96.

Not at all possible. I like the recommendation by the author 'If you think so, please reconsider the problem several years from now.'

Problem 97.

This is not a trivial problem for a middle school student. This proof requires a little bit of effort. The student needs to know proof by contradiction. Let us try it.

Assume, for the sake of contradiction, that there exist two polynomials $P(x)$ and $Q(x)$, each having at least two nonzero terms, such that their product has only one term. That is,

$$P(x)\,Q(x) = cx^k$$

for some constant $c \neq 0$.

Let $P(x)$ have its lowest degree term of degree i and highest degree term of degree j, with $i < j$, and let $Q(x)$ have its lowest degree term of degree m and highest degree term of degree n, with $m < n$. Such indices exist since each polynomial has at least two nonzero terms.

Then, in the product $P(x)Q(x)$:

- the term of lowest degree has degree $i + m$,
- the term of highest degree has degree $j + n$.

Since $i < j$ and $m < n$, we have

$$i + m < j + n.$$

Thus the product $P(x)Q(x)$ must contain at least two terms of different degrees, namely x^{i+m} and x^{j+n}.

This contradicts the assumption that $P(x)Q(x)$ has only one term. Hence, there is no possible way in which the product of two polynomials, each with at least two nonzero terms, can result in a single term polynomial.

Problem 98.

Yes, it is possible. A good example is given in the book. But how can we prove it? If we can show that even a single case exists then we can say that this assertion is true.

$(x^2 + 2xy + 2y^2)(x^2 - 2xy + 2y^2)$ is the example shown.

$$(x^2 + 2y^2 + 2xy)(x^2 + 2y^2 - 2xy)$$
$$(x^2 + 2y^2)^2 - (2xy)^2$$
$$x^4 + 4y^4 + 4x^2y^2 - 4x^2y^2$$
$$x^4 + 4y^4$$

31 Coefficients and values

Problem 99.

For $a = 1$ and $b = 1$ we simply get the sum of the coefficients for each row in the Pascal's triangle. That is
1, 2, 4, 8, 16, 32, 64 ... $2^n - 1$ for the n^{th} row.

Problem 100.

Adding the numbers with alternating signs will make the sum of coefficients 0 as they will cancel each other out. This is happening because the odd powers end up as a minus sign for one of the terms.

Problem 101.

The hint is good to solve this problem.

At $x = 1$ the polynomial will be of the form $(1 + 2 \cdot 1)^{200}$. This gives us the answer 3^{200}.

The polynomial can be written as below for $x = 1$

$$P(x) = a_0 + a_1x + a_2x^2 + ...$$
$$P(x) = a_0 + a_1 + a_2 + ...$$

Problem 102.

Similar to the last problem we put $x = 1$ and get $(1 - 2 \cdot 1)^{200}$. The answer is 1.

Problem 103.

Same logic, we substitute $x, y = 1$. We get $(1 + 1 - 1)^3$ which is 1.

Problem 104.

For terms not containing y we can simply put $y = 0$ and work it out at $x = 1$. This gives us $(1 + 1 - 0)^3$ which is 8.

Problem 105.

This one is slightly tricky and we should use concepts of sets. There are essentially 4 types of sets of monomials which is possible:

- Constant term that is 1
- Only x terms
- Only y terms
- xy terms

We know certain sums already

Terms	Sums of Coefficients
1	1
x	?
y	?
xy	?
all terms	1

We know that 1 and the x terms only have a sum of 8, therefore only x will be $8 - 1$ which is 7. So let us update the table.

Terms	Sums of Coefficients
1	1
x	7
y	?
xy	?
all terms	1

Now we can figure out the sum of the coefficients of terms not containing x. We put $x = 0$. We get $(1 + 0 - 1)^3$ since $y = 1$ and that is 0. Next we remove the constant 1 from this 0 to get only the sum from y terms, the answer is -1. Updating the table.

Terms	Sums of Coefficients
1	1
x	7
y	-1
xy	?
all terms	1

Finally to get the coefficients for the xy terms we can simply equate the individual terms to the total sum as below.

$$\begin{aligned} 1 + 7 - 1 + coeff_{xy} &= 1 \\ coeff_{xy} &= -6 \end{aligned}$$

So finally to get the sum of coefficients of terms containing x we add the x only sums and xy sums. This gives $7 - 6$ and the answer is 1.

32 Factoring

Problem 106.

Fairly easy

$$\begin{aligned} (ac + ad + bc + bd) &= a(c + d) + b(c + d) \\ &= (c + d)(a + b) \end{aligned}$$

Problem 107.

(a)

$$\begin{aligned} ac + bc - ad - bd &= c(a + b) - d(a + b) \\ (a + b)(c - d) \end{aligned}$$

(b)

$$\begin{aligned}1 + a + a^2 + a^3 &= 1 + a^2 + a + a^3 \\ &= (1 + a^2) + a(1 + a^2) \\ &= (1 + a^2)(1 + a)\end{aligned}$$

(c) This is not an easy one to get it. One of the ways we can solve is to work backwards from a geometric series. First let us derive the sum of a geometric series.

$1.S = 1 + a + a^2 + a^3 + a^4 + ... + a^{13} + a^{14}$
$aS = 0 + a + a^2 + a^3 + a^4 + ... + a^{13} + a^{14} + a^{15}$

Subtract the above two.

$(S - aS) = 1 - a^{15}$

$S = \dfrac{(1 - a^{15})}{(1 - a)}$

Now we can do the following manipulations

$S = \dfrac{(1^3 - (a^5)^3)}{(1 - a)}$

$S = \dfrac{(1 - a^5)(1^2 + (a^5)^2 + 1.a^5)}{(1 - a)}$

$S = \dfrac{(1 - a^5)(1 + a^{10} + a^5)}{(1 - a)}$

Look at the term $\dfrac{(1 - a^5)}{(1 - a)}$. This itself is a sum of geometric series up to a^4. So we substitute that geometric series.

$S = (1 + a + a^2 + a^3 + a^4)(1 + a^5 + a^{10})$
Thus the factorization is $(1 + a + a^2 + a^3 + a^4)(1 + a^5 + a^{10})$
(d)

$$\begin{gathered}x^4 - x^3 + 2x - 2 \\ x^3(x - 1) + 2(x - 1) \\ (x^3 + 2)(x - 1)\end{gathered}$$

Problem 108.

$$\begin{gathered}a^2+3ab+2b^2\\a^2+2ab+b^2+ab+b^2\\(a+b)^2+b(a+b)\\(a+b)(a+2b)\end{gathered}$$

Problem 109.

(a)

$$\begin{gathered}a^2-3ab+2b^2\\a^2-2ab+b^2-ab+b^2\\(a-b)^2-b(a-b)\\(a-b)(a-2b)\end{gathered}$$

(b)

$$\begin{gathered}a^2+3a+2\\a^2+a+2a+2\\a(a+1)+2(a+1)\\(a+1)(a+2)\end{gathered}$$

Problem 110.

(a)

$$\begin{gathered}a^2+4ab+4b^2\\(a+2b)^2\end{gathered}$$

(b)

$$\begin{gathered}a^4+2a^2b^2+b^4\\(a^2+b^2)^2\end{gathered}$$

(c)

$$\begin{gathered}a^2-2a+1\\(a-1)^2\end{gathered}$$

Problem 111.

One thing to know for kids/students is that when they see a solution in a book they might wonder and be impressed that in one attempt the book came up with an elegant solution. This inductive thinking is not born solely from intelligence but from pattern recognition and pattern recognition in turn comes from practice. And when anyone practices they make mistakes, hit roadblocks, turn around and try again till they find a good correct proof. That's all on this at this time.

Now the problem.

$$
\begin{gathered}
x^5 + x + 1 \\
x^5 + x^4 + x^3 + x^2 + x + 1 - x^4 - x^3 - x^2 \\
x^3(x^2 + x + 1) + (x^2 + x + 1) - x^2(x^2 + x + 1) \\
(x^3 + 1 - x^2)(x^2 + x + 1)
\end{gathered}
$$

Problem 112.

We can simply substitute b instead of a and then $-b$ instead of a in a^2. In both cases we will get b^2.

Problem 113.

In such problems we look at what could be a factor and here the authors point out that when $a = b$ the answer is 0, thus $(a - b)$ should be a factor. We should divide the given polynomial by $(a - b)$. This gives us $(a^2 + ab + b^2)$. Thus the factor is simply $(a - b)(a^2 + ab + b^2)$.

Problem 114.

Similar logic as the last problem. In this case both a and b should be 0 to be a factor. They need to be of opposite signs, thus dividing $(a^3 + b^3)$ by $(a + b)$ we get $(a^2 - ab + b^2)$. Thus the factor of $(a^3 + b^3)$ is $(a + b)(a^2 - ab + b^2)$.

Problem 115.

The book has this solved but this is probably not what should be the answer, there is an additional factorization possible.

$$a^4 - b^4$$
$$(a^2)^2 - (b^2)^2$$
$$(a^2 + b^2)(a^2 - b^2)$$
$$(a^2 + b^2)(a + b)(a - b)$$

Problem 116.

(a) Again in this case if $a = b$ we get a factor. So we divide $(a^5 - b^5)$ with $(a - b)$. We get the factors as $(a - b)(a^4 + a^3b + a^2b^2 + ab^3 + b^4)$.

A general formula can be derived for the factorization of $(a^n - b^n)$ seeing the last few questions. This is true for all n.

$(a^n - b^n) = (a - b)(a^{(n-1)} + a^{(n-2)}b + a^{(n-3)}b^2 + ... + ab^{(n-2)} + b^{(n-1)})$

(b) We can use the general formula derived earlier, but first

$(a^{10} - b^{10}) = ((a^5)^2 - (b^5)^2) = (a^5 + b^5)(a^5 - b^5)$

We can also derive the factorization formula for $(a^n + b^n)$ as below. But this is true only when n is odd.

$(a^n + b^n) = (a + b)(a^{(n-1)} - a^{(n-2)}b + a^{(n-3)}b^2 - ... - ab^{(n-2)} + b^{(n-1)})$

Applying these two to the given problem we get

$(a^{10} - b^{10}) = (a+b)(a-b)(a^4 - a^3b + a^2b^2 - ab^3 + b^4)(a^4 + a^3b + a^2b^2 + ab^3 + b^4)$

(c)

This can be stated as $(a^7 - 1^7)$, thus we could use the same logic as above

$(a^7 - 1) = (a - 1)(a^6 + a^5 + a^4 + a^3 + a^2 + a + 1)$

We can now use the sum of the geometric series derived earlier to check our factorization too.

$(a^7 - 1) = (a - 1)(\dfrac{1 - a^{6+1}}{(1 - a)})$

An important takeaway from this set of problems are these two factorizations.

$$(a^n - b^n) = (a - b)(a^{(n-1)} + a^{(n-2)}b + a^{(n-3)}b^2 + ... + ab^{(n-2)} + b^{(n-1)})$$

true for all n

$$(a^n + b^n) = (a + b)(a^{(n-1)} - a^{(n-2)}b + a^{(n-3)}b^2 - ... - ab^{(n-2)} + b^{(n-1)})$$

true only when n is odd

Problem 117.

$$a^2 - 4b^2 = a^2 - (2b)^2 = (a + 2b)(a - 2b)$$

Problem 118.

(a) $a^2 - 2 = (a + \sqrt{2})(a - \sqrt{2})$

(b) $a^2 - 3b^2 = (a + \sqrt{3}b)(a - \sqrt{3}b)$

(c) $a^2 + 2ab + b^2 - c^2 = (a + b)^2 - c^2 = (a + b + c)(a + b - c)$

(d) $a^2 + 4ab + 3b^2 = a^2 + 4ab + 4b^2 - b^2 = (a + 2b)^2 - b^2$
$= (a + 2b + b)(a + 2b - b) = (a + 3b)(a + b)$

Problem 119.

In this case we could use our earlier derived formula but we are going to use square roots as explained in the solution to this problem.

$$a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 = (a^2 + b^2)^2 - (\sqrt{2}ab)^2$$
$$(a^2 + b^2 + \sqrt{2}ab)(a^2 + b^2 - \sqrt{2}ab)$$

Problem 120.

Polynomials of the form $a^{2n} + b^{2n}$ factor only when n has an odd divisor as shown by the authors. If n is not an odd number then the factorization can be only done over real numbers such as square roots and/or complex numbers. The authors introduce complex numbers immediately after this problem.

Problem 121.

$$(2+3\sqrt{-1})(2-3\sqrt{-1})$$
$$(2^2-(3\sqrt{-1})^2)$$
$$(4-(9\times(-1)))$$
$$(4+9)$$
$$13$$

Problem 122.

The authors say these sets of problems are more difficult.

(a)

$$x^4+1=(x^2+1)^2-2x^2$$
$$(x^2+1)^2-(\sqrt{2}x)^2$$
$$(x^2+1-\sqrt{2}x)(x^2+1+\sqrt{2}x)$$
$$((x+1)^2-2x-\sqrt{2}x)((x+1)^2-2x+\sqrt{2}x)$$
$$((x+1)^2-(2+\sqrt{2})x)((x+1)^2-(2-\sqrt{2})x)$$

(b)

$$x(y^2-z^2)+y(z^2-x^2)+z(x^2-y^2)$$
$$xy^2-xz^2+yz^2-yx^2+zx^2-zy^2$$
$$zx^2-yx^2+xy^2-xz^2+yz^2-zy^2$$
$$x^2(z-y)-x(z^2-y^2)+zy(z-y)$$
$$x^2(z-y)-x(z-y)(z+y)+zy(z-y)$$
$$(z-y)(x^2-x(z+y)+zy)$$
$$(z-y)(x-y)(x-z)$$

(c) This is similar to an earlier problem (number 111)
Let us do some addition of terms to make it consistent across so that we can derive a common term. It is the same as division in a way. Arranging the terms basis their powers we can see a clean pattern.

$a^{10}+a^9+a^8$
..... $-a^9-a^8-a^7$
.... $+a^7+a^6+a^5$
.... $-a^6-a^5-a^4$

.... $+a^5+a^4+a^3$
.... $-a^3-a^2-a$
.... $+a^2+a+1$

We can take the term (a^2+a+1) out from each row above.
This leaves us with $(a^8-a^7+a^5-a^4+a^3-a+1)$

Therefore the factors are $(a^2+a+1)(a^8-a^7+a^5-a^4+a^3-a+1)$.

The actual method to solve such problems requires cyclotomic factorization which is a part of under graduate abstract algebra so we should skip that.

(d) We can see that at $a+b+c=0$ the given polynomial should yield no remainder if divided. Thus $a+b+c$ should be a factor. This is similar to the reasoning in the problem 113 and 114.

Now this $a+b+c$ has to at least multiply by $a^2+b^2+c^2$ and other terms so that we end up with $-3abc$ while other terms such as ab^2+ac^2 cancel out. This is also fairly visible if we have the term $(-ab-bc-ca)$. Finally, we get

$$a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-(ab+bc+ca))$$

(e) This is an easy problem actually after a series of difficult problems.

Let us expand $(a+b+c)^3$ in a way where we initially treat $(b+c)$ as one term. So we have to basically do $(a+k)^3$ which is $(a^3+k^3+3ak(a+k))$. Now put back $(b+c)$ instead of k. $(a^3+(b+c)^3+3a(b+c)(a+b+c))$ is what we get. Further expanding

$$(a+b+c)^3=a^3+b^3+c^3+3bc(b+c)+3a(b+c)(a+b+c)$$
$$(a+b+c)^3=a^3+b^3+c^3+3(b+c)(bc+a(a+b+c))$$
$$(a+b+c)^3=a^3+b^3+c^3+3(b+c)(bc+a^2+ab+ac)$$
$$(a+b+c)^3=a^3+b^3+c^3+3(b+c)(a^2+a(b+c)+bc)$$
$$(a+b+c)^3=a^3+b^3+c^3+3(b+c)(a+b)(a+c)$$
$$(a+b+c)^3-a^3-b^3-c^3=3(b+c)(a+b)(a+c)$$

(f) This is another easy problem and we can see an easy substitution will collapse this problem's solution into a few lines.

Substitute the following
$P=a-b$
$Q=b-c$

$R = c - a$

We can transform the given polynomial now:

$$\begin{aligned} P^3 + Q^3 + R^3 &= (P+Q+R)^3 - 3(P+Q)(Q+R)(R+P) \\ &= (a-b+b-c+c-a)^3 - 3(a-b+b-c)(b-c+c-a)(c-a+a-b) \\ &= 0^3 - 3(a-c)(b-a)(c-b) \\ &= 3(a-b)(b-c)(c-a) \end{aligned}$$

Problem 123.

A hint is given which leads us to in an easy factorization.

$$\begin{gathered} (a+b) < 1 + ab \\ 1 + ab - (a+b) > 0 \\ (1-a)(1-b) > 0 \end{gathered}$$

Now both a and b are greater than 1 then both $(1-a)$ and $(1-b)$ when multiplied will always be positive. Hence proved.

Problem 124.

We know the following factorization:

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

If $a^2 + ab + b^2 = 0$, then the right-hand side is zero, hence

$$a^3 - b^3 = 0.$$

From $a^3 - b^3 = 0$ we obtain $a^3 = b^3$, which implies

$$a = b.$$

Substituting $a = b$ into $a^2 + ab + b^2$ gives

$$3a^2 = 0,$$

hence $a = b = 0$.

Problem 125.

This is an easy problem. We have already derived it earlier.

$$(a+b+c)^3 = a^3 + b^3 + c^3 + 3(a+b)(b+c)(c+a)$$

If $a+b+c=0$ then we can also write:
$a+b=-c$
$b+c=-a$
$c+a=-b$

Putting these three into the above equation we get

$$0^3 = a^3 + b^3 + c^3 + 3(-c)(-a)(-b)$$

Thus,

$$a^3 + b^3 + c^3 = 3abc$$

Problem 126.

Simplifying both the sides

$$abc = (a+b+c)(ab+bc+ca)$$

We observe a pattern on the right hand side

$$abc = 3abc + (a^2b + a^2c + b^2a + b^2c + c^2a + c^2b)$$

The term $a^2b + a^2c + b^2a + b^2c + c^2a + c^2b$ should equate to $-2abc$.
Given that $a = -banda = -c$ then substituting $b = -a$ and $c = -a$
$= a^2(-a-a) + (-a)^2(a-a) + (-a)^2(a-a)$
$= -2a^3$
This is same as $-2abc$ because $-2a(-a)(-a)$ i.e. $-2a^3$. Same holds true for other two combinations.

33 Rational expressions

No problems

34 Converting a rational expression into the quotient of two polynomials

Problem 127.

The objective is to get a fraction in which both numerator and denominator are polynomials.

(b) $\dfrac{ac}{b^2}$

(c) $\dfrac{x}{(1+x)}$

(d)

$$\frac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{x}}}}$$

$$\frac{1}{1+\cfrac{1}{1+\cfrac{x}{x+1}}}$$

$$\frac{1}{1+\cfrac{x+1}{2x+1}}$$

$$\frac{2x+1}{3x+2}$$

(e)

$$\frac{\frac{x}{y}+\frac{y}{z}+\frac{z}{x}}{\frac{y}{x}+\frac{z}{y}+\frac{x}{z}}+1$$

$$\frac{x^2z+y^2x+z^2y}{y^2z+z^2x+x^2y}+1$$

$$\frac{x^2z+x^2y+y^2x+y^2z+z^2x+z^2y}{y^2z+z^2x+x^2y}$$

(g)

$$\frac{1}{\left(\frac{\frac{1}{a}+\frac{1}{b}}{2}\right)}$$

$$\frac{2ab}{a+b}$$

Problem 128.

$$\frac{(x-a)(x-b)}{(c-a)(c-b)}+\frac{(x-a)(x-c)}{(b-a)(b-c)}+\frac{(x-b)(x-c)}{(a-b)(a-c)}$$

In the book the authors expand right at the start. We should defer it to the end as far as possible. A lot of terms cancel out in basic manipulations itself.

Numerator when cancels out with denominator:

$$\begin{gathered}(x-a)(x-b)(a-b)\cancel{(b-c)}\cancel{(a-b)}\cancel{(a-c)}\\ +(x-a)(x-c)\cancel{(c-a)}\cancel{(c-b)}\cancel{(a-b)}(c-a)\\ +(x-b)(x-c)\cancel{(c-a)}(b-c)\cancel{(b-a)}\cancel{(b-c)}\end{gathered}$$

The denominator becomes:

$$\cancel{(c-a)}(c-b)(b-a)\cancel{(b-c)}\cancel{(a-b)}(a-c)$$

Working on the numerator now:

$$(x-a)(x-b)(a-b)+(x-a)(x-c)(c-a)+(x-b)(x-c)(b-c)$$

There is a certain symmetry we observe here. The a and b logically grouped with x, so with the other combinations.
We notice that x^2 on the first addition term yields $ax^2 - bx^2$, this will cancel out with the x^2 terms in the next two operands between the two addition signs. Thus all x^2s cancel out. Similarly we also notice all the x terms also cancel out through, we are left with a compact expression in the numerator now where we can easily factorize.

$$a^2b-ab^2+ac^2-a^2c+b^2c-bc^2$$

Assume a^2 to be the unknown for the quadratic factorization then

$$a^2(b-c)+a(c^2-b^2)+(cb^2-bc^2)$$

Take $(c-b)$ out.

$$(c-b)[-a^2+a(c+b)-bc]$$

$$(c-b)(b-a)(a-c)$$

Both the numerator and denominator are same. Thus, the answer is 1.

Problem 129.

Case $x = a$: The first two terms in the addition becomes 0. The last term evaluates to 1.

Case $x = b$: Same logic as above. The expression evaluates to 1.

Case $x = c$: Same logic as above. The expression evaluates to 1.

Problem 130.

These problems should be solved in general terms to avoid confusion.

Let the volume of each half of the pool be X. Let the rate of flow in the first half be p and in the second half be q.

If the first half is filled in a units of time and the second half in b units of time, then

$$X = ap$$
$$X = bq$$

To fill the entire pool (volume $2X$), both pipes work together for time t:

$$2X = t(p + q).$$

Substituting $p = \frac{X}{a}$ and $q = \frac{X}{b}$, we get

$$2X = t\left(\frac{X}{a} + \frac{X}{b}\right).$$

Cancelling X from both sides,

$$2 = t\left(\frac{1}{a} + \frac{1}{b}\right).$$

Hence,

$$t = \frac{2ab}{a + b}.$$

Problem 131.

This is similar to the previous problem.

Assume the length from point A to B be D and the speed of the motor boat be v and river be r. Then we can formulate the following equations.

$$D = (v + r)a$$
$$D = (v - r)b$$

Students should note that while going with the stream the boat's speed is added to the river's speed. But while going against the current of the river the river's speed needs to be deducted from the boat's speed. A side question is what if the speed of the boat was lower than that of the river and what if in another case it was equal?

Now when we need to find the time taken for the boat to travel D when the speed of the river is 0 then we have to find t in the below equation.

$$D = t \times v$$

From the earlier two equations equating r we get

$$\frac{D}{a} - v = v - \frac{D}{b}$$
$$D = 2\left(\frac{1}{a} + \frac{1}{b}\right)V$$

Thus we now have

$$t = 2\left(\frac{1}{a} + \frac{1}{b}\right)$$

Problem 132.

This is again similar to earlier problems. Assume half the trip is D distance and it takes t_1 time while the other half takes t_2 time. We can write

$$2D = \left(\frac{D}{v_1} + \frac{D}{v_2}\right)v$$
$$v = \frac{2v_1v_2}{v_1 + v_2}$$

Problem 133.

(a)

$$(x+\frac{1}{x})^2 = x^2 + \frac{1}{x^2} + 2$$
$$7^2 = x^2 + \frac{1}{x^2} + 2$$
$$x^2 + \frac{1}{x^2} = 49 - 2 = 47$$

(b)

$$(x+\frac{1}{x})^3 = x^3 + \frac{1}{x^3} + 3(x+\frac{1}{x})$$
$$x^3 + \frac{1}{x^3} = (x+\frac{1}{x})^3 - 3(x+\frac{1}{x})$$
$$x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7$$
$$x^3 + \frac{1}{x^3} = 322$$

Problem 134.

We can use the general expansion of the expression $(x+y)^n$ and look at the coefficients from the Pascal's triangle.

$$(x+y)^n = x^n + C_1 x^{n-1} y + C_2 x^{n-2} y^2 + ... + C_{n-1} x y^{n-1} + y^n$$

One important point to note is that as per Pascal's triangle if n is odd then all the coefficients exist in pairs, for instance C_1 will be same as C_{n-1}. But if n is even then the middle term's coefficient will not have a pair. But this does not matter since that coefficient is an integer itself. Going back to the above equation and substituting $\frac{1}{x}$ instead of y.

$$(x+\frac{1}{x})^n = x^n + C_1 x^{n-1}\frac{1}{x} + C_2 x^{n-2}\frac{1}{x^2} + ... + C_{n-1} x \frac{1}{x^{n-1}} + \frac{1}{x^n}$$

rearrange the terms

$$(x+\frac{1}{x})^n = x^n + \frac{1}{x^n} + C_1 x^{n-1}\frac{1}{x} + C_{n-1} x \frac{1}{x^{n-1}} + C_2 x^{n-2}\frac{1}{x^2} + C_{n-2} x^2 \frac{1}{x^{n-2}} ...$$

$$(x+\frac{1}{x})^n = x^n + \frac{1}{x^n} + C_1 x^{n-2} + C_{n-1}\frac{1}{x^{n-2}} + C_2 x^{n-4} + C_{n-2}\frac{1}{x^{n-4}} ...$$

We also know that the corresponding coefficients (from the earlier statement) exists in parirs/middle one for even n is alone. Thus $C_1 = C_{n-1}$ and so on. We reduce the equation to the following.

$$(x + \frac{1}{x})^n = x^n + \frac{1}{x^n} + C_1 \left(x^{n-2} + \frac{1}{x^{n-2}} \right) + C_2 \left(x^{n-4} + \frac{1}{x^{n-4}} \right) ...$$

The smallest term in this equation will be $x^2 + \frac{1}{x^2}$. This we can show is an integer already.

$$(x + \frac{1}{x})^2 = x^2 + \frac{1}{x^2} + 2$$

$$x^2 + \frac{1}{x^2} = (x + \frac{1}{x})^2 - 2$$

The right hand side is an integer minus 2. Now we can look at $x^4 + \frac{1}{x^4}$.

$$(x^2 + \frac{1}{x^2})^2 = x^4 + \frac{1}{x^4} + 2$$

$$x^4 + \frac{1}{x^4} = (x^2 + \frac{1}{x^2})^2 - 2$$

The right hand side here too is an integer. Thus, we can go down the rabbit hole and say as we go up in the higher powers of even numbers the term $x^{2k} + \frac{1}{x^{2k}}$ will always be an integer for a given integer $k > 0$.

Thus the original expression collapses to the following:

$$(x + \frac{1}{x})^n = x^n + \frac{1}{x^n} + C_1 \left(x^{n-2} + \frac{1}{x^{n-2}} \right) + C_2 \left(x^{n-4} + \frac{1}{x^{n-4}} \right) ...$$

$$integer_1 = x^n + \frac{1}{x^n} + integer_2$$

Hence we have proved that $x^n + \frac{1}{x^n}$ is always an integer.

Please note the proper way of proving this is via mathematical induction but the authors have not introduced this technique yet.

Problem 135.

The pattern is that of a Fibonacci sequence. For the two instances we simply subsititute the given $\frac{2x+1}{3x+2}$ first and get the first answer then substitute that answer once more to get the second answer.

The first answer is $\frac{3x+2}{5x+3}$

The second answer is $\dfrac{5x+3}{8x+5}$

The general term can be got as the k_{th} term in a Fibonacci sequence.

$$\frac{k_{n-1}x + k_{n-2}}{k_n x + k_{n-1}}$$

35 Polynomial and rational fractions in one variable

Problem 136.

We need to only worry about the highest degree term and that will be $(2x)^5$, that is $32x^5$.

Problem 137.

Polynomials P has a degree m, thus the highest degree term will be something like C_1x^m. Similarly the other polynomial Q will have its highest degree term as C_2x^n. The product of the polynomials $P \times Q$ will have its highest degree term as $C_1 \cdot C_2 x^{m+n}$.

Problem 138.

(a) Since it is the sum of the two polynomials and one of them is higher than the other we are sure that the highest degree of the resultant polynomial will be 9.

(b) Since both the polynomials have the same degree there is one possibility that they cancel each other out if the coefficients are equal but of opposite signs. Thus the degree of the resultant polynomial will be 7 or less.

Problem 139.

In this case what we are essentially doing is $(y^7)^{10}$. Thus the degree of the resultant polynomial will be 70.

36 Division of polynomials in one variable; the remainder

Problem 140.

The quotient will have degree 4, since it is obtained from $\dfrac{x^7}{x^3}$.

The remainder must be a polynomial whose degree is strictly less than the degree of the divisor.

Since the divisor has degree 3, the remainder polynomial can have degree 0, 1, or 2.

Problem 141.

The solution is given in the book. P is divided by S which gives the following.

$$P = Q_1S + R_1$$
$$P = Q_2S + R_2$$

R_1 and R_2 will have a degree smaller than S. Then

$$Q_1S + R_1 = Q_2S + R_2$$
$$R_1 - R_2 = (Q_2 - Q_1)S$$

$R_1 - R_2$ has a smaller degree than S since individually too they have a smaller degree. They cannot be a multiple of S either. Therefore, $Q_1 - Q_2 = 0$, that is $Q_1 = Q_2$ and thus $R_1 = R_2$.

Problem 142.

In this case the fraction cannot be further divided as such. The quotient is 0 and the remainder is the dividend itself.

Problem 143.

(a)

$$\begin{array}{r|l} & \quad\quad\quad x^2 + x + 1 \\ \hline x - 1 & x^3 \quad\quad\quad\quad - 1 \\ & -x^3 + x^2 \\ \hline & \quad\quad x^2 \\ & \quad -x^2 + x \\ \hline & \quad\quad\quad x - 1 \\ & \quad\quad -x + 1 \\ \hline & \quad\quad\quad\quad 0 \end{array}$$

(b)

$$
\begin{array}{rl}
 & x^3 + x^2 + x + 1 \\
x - 1 \big) & \overline{x^4 \qquad\qquad\qquad - 1} \\
 & \underline{-x^4 + x^3} \\
 & \quad x^3 \\
 & \quad \underline{-x^3 + x^2} \\
 & \qquad x^2 \\
 & \qquad \underline{-x^2 + x} \\
 & \qquad\quad x - 1 \\
 & \qquad\quad \underline{-x + 1} \\
 & \qquad\qquad 0
\end{array}
$$

(c)

$$
\begin{array}{rl}
 & x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1 \\
x - 1 \big) & \overline{x^{10} \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad - 1} \\
 & \underline{-x^{10} + x^9} \\
 & \quad x^9 \\
 & \quad \underline{-x^9 + x^8} \\
 & \qquad x^8 \\
 & \qquad \underline{-x^8 + x^7} \\
 & \qquad\quad x^7 \\
 & \qquad\quad \underline{-x^7 + x^6} \\
 & \qquad\qquad x^6 \\
 & \qquad\qquad \underline{-x^6 + x^5} \\
 & \qquad\qquad\quad x^5 \\
 & \qquad\qquad\quad \underline{-x^5 + x^4} \\
 & \qquad\qquad\qquad x^4 \\
 & \qquad\qquad\qquad \underline{-x^4 + x^3} \\
 & \qquad\qquad\qquad\quad x^3 \\
 & \qquad\qquad\qquad\quad \underline{-x^3 + x^2} \\
 & \qquad\qquad\qquad\qquad x^2 \\
 & \qquad\qquad\qquad\qquad \underline{-x^2 + x} \\
 & \qquad\qquad\qquad\qquad\quad x - 1 \\
 & \qquad\qquad\qquad\qquad\quad \underline{-x + 1} \\
 & \qquad\qquad\qquad\qquad\qquad 0
\end{array}
$$

(d)

$$
\begin{array}{rl}
 & \phantom{-x^3 - {}} x^2 - x + 1 \\
\cline{2-2}
x + 1 \big) & x^3 \phantom{{}-x^2 - x} + 1 \\
 & \underline{-x^3 - x^2} \\
 & - x^2 \\
 & \phantom{-x^3 -{}} \underline{x^2 + x} \\
 & \phantom{-x^3 - x^2 +{}} x + 1 \\
 & \phantom{-x^3 - x^2 {}} \underline{- x - 1} \\
 & \phantom{-x^3 - x^2 - x - {}} 0
\end{array}
$$

(e)

$$
\begin{array}{rl}
 & \phantom{-x^4 - {}} x^3 - x^2 + x - 1 \\
\cline{2-2}
x + 1 \big) & x^4 \phantom{{}-x^3 - x^2 - x} + 1 \\
 & \underline{-x^4 - x^3} \\
 & - x^3 \\
 & \phantom{-x^4 -{}} \underline{x^3 + x^2} \\
 & \phantom{-x^4 - x^3 +{}} x^2 \\
 & \phantom{-x^4 - x^3 {}} \underline{- x^2 - x} \\
 & - x + 1 \\
 & \phantom{-x^4 - x^3 - x^2 -{}} \underline{x + 1} \\
 & \phantom{-x^4 - x^3 - x^2 - x + {}} 2
\end{array}
$$

Problem 144.

This one is just a specific case of the formula, a geometric progression where $b = 1$.

$$(a^n - b^n) = (a - b)(a^{(n-1)} + a^{(n-2)}b + a^{(n-3)}b^2 + ... + ab^{(n-2)} + b^{(n-1)})$$
$$(a^n - 1) = (a - 1)(a^{(n-1)} + a^{(n-2)} + a^{(n-3)} + ... + a + 1)$$

Let us rearrange the equation.

$$(1 + a + a^2 + a^3 + a^4 + ... + a^{n-2} + a^{n-1}) = \frac{(a^n - 1)}{(a - 1)}$$

Let us substitute $a = 2$ in the above equation.

$$(1 + 2 + 2^2 + 2^3 + 2^4 + ... + 2^{n-2} + 2^{n-1}) = \frac{(2^n - 1)}{(2 - 1)} = (2^n - 1)$$

We can see the left hand side is the summation of all the powers of 2 up to the number $(n - 1)$ and the right hand side is one less than 2^n.

37 The remainder when dividing by $x - a$

Problem 145.

(a) For all powers of x we will get $x^n = 1$ when we substitute $x = 1$.

(b) For only odd powers of x we will get $x^n = -1$ when we substitute $x = -1$.

Problem 146.

(a) We can easily observe that at $x = 1$ and $x = -2$ the polynomial will yield 0. So we know $(x-1)$ and $(x+2)$ will be factors. Then dividing the polynomial with these two factors.

$$
\begin{array}{r@{}rrrrr}
 & & & x^2 & -x & +3 \\ \cline{2-6}
x^2+x-2\big) & x^4 & & & +5x & -6 \\
 & -x^4 & -x^3 & +2x^2 & & \\ \cline{2-4}
 & & -x^3 & +2x^2 & +5x & \\
 & & x^3 & +x^2 & -2x & \\ \cline{3-5}
 & & & 3x^2 & +3x & -6 \\
 & & & -3x^2 & -3x & +6 \\ \cline{4-6}
 & & & & & 0
\end{array}
$$

Thus the factorization is

$$x^4 + 5x - 6 = (x-1)(x+2)(x^2 - x + 3)$$

If we try to factorize $(x^2 - x + 3)$ we will end up with complex roots to this quadratic so we will skip that.

(b) Here we see all terms are positive so we should look for negative numbers which would turn the polynomial into 0. The first factor we can see is $(x+1)$. Now we will divide the polynomial by this factor.

$$
\begin{array}{r@{}rrrrr}
 & & x^3 & -x^2 & +4x & +1 \\ \cline{2-6}
x+1\big) & x^4 & & +3x^2 & +5x & +1 \\
 & -x^4 & -x^3 & & & \\ \cline{2-3}
 & & -x^3 & +3x^2 & & \\
 & & x^3 & +x^2 & & \\ \cline{3-4}
 & & & 4x^2 & +5x & \\
 & & & -4x^2 & -4x & \\ \cline{4-5}
 & & & & x & +1 \\
 & & & & -x & -1 \\ \cline{5-6}
 & & & & & 0
\end{array}
$$

Now we have to figure out whether the polynomial $x^3 - x^2 + 4x + 1$ can be further factorized or not. By visual inspection we can deduce that any integer from -5 to $+5$ will not turn the polynomial 0.

So the factorization is

$$x^4 + 3x^2 + 5x + 1 = (x + 1)(x^3 - x^2 + 4x + 1)$$

(c) The visual inspection immediately gives is the following two factors $(x + 1)$ and $(x - 2)$. Dividing now.

$$\begin{array}{r l}
 & \qquad\qquad\quad x+1 \\
\cline{2-2}
x^2 - x - 2 \big) & x^3 \qquad\quad - 3x - 2 \\
 & -x^3 + x^2 + 2x \\
\cline{2-2}
 & \qquad\quad x^2 \;\; - x - 2 \\
 & \qquad -x^2 \;\; + x + 2 \\
\cline{2-2}
 & \qquad\qquad\qquad\quad 0
\end{array}$$

Thus the factorization is

$$x^3 - 3x - 2 = (x + 1)^2(x - 2)$$

Problem 147.

The proof is given in the book with the common error made when highlighted. We have used this logic in the earlier problem already when we divided the polynomial by the product of all its known factors. The authors state that $P = (x - 1)Q$. Then they substitute 2 and find that 2 is a root of Q. Finally arriving at the expression $P = (x - 1)(x - 2)R$.

Problem 148.

This is simply the highest degree of the polynomial. The authors show this with an example. But we can simply state the following.

$$P(x) = (x - n_1)(x - n_2)(x - n_3)(x - n_4)...1$$

Here the R equivalent polynomial is of 0 degree and the constant is 1.

Problem 149.

We simply check if the polynomial gives a remainder of 0 when divided by $(x-1)$ and $(x + 1)$. The other easier way is to substitute 1 and -1 in the polynomial and see if they are roots.

Problem 150.

This is only possible when n is even. We can do a short proof of the same but it is fairly visible when we look at the problem.

If n was even we see the proof below and it is correct.

$$x^n - 1 = x^{2k} - 1 = (x^k - 1)(x^k + 1)$$

If n was odd then.

$$x^n - 1 = x^{2m+1} - 1$$

For $(x-1)$ the right hand side will yield 0 but $(x+1)$ will yield -2. Hence for n as an odd number this will not hold true.

Problem 151.

We can substitute $x = 1$ and $x = -1$ in the equation $P(x) = (x^2 - 1) + (ax + b)$. This will give us two linear equations in x. We can solve them simultaneously and arrive at the values of a and b.

Problem 152.

We can write the given division as

$$P(x) = Q(x)(x^2 - 1) + (5x - 7) = Q(x)(x-1)(x+1) + (5x - 7).$$

Remainder upon division by $(x-1)$

Since $(x^2 - 1)$ is divisible by $(x-1)$, the remainder of $P(x)$ upon division by $(x-1)$ is the same as the remainder of $(5x - 7)$ upon division by $(x-1)$.

By the Remainder Theorem,

$$\operatorname{rem}_{x-1} P(x) = P(1) = 5.1 - 7 = -2.$$

Equivalently, we can write

$$5x - 7 = 5(x-1) - 2,$$

hence

$$P(x) = (x-1)\big(Q(x)(x+1) + 5\big) - 2,$$

so the remainder is -2.

Remainder upon division by $(x+1)$

Similarly,

$$\text{rem}_{x+1}\, P(x) = P(-1) = 5(-1) - 7 = -12,$$

since $(x^2 - 1)$ is also divisible by $(x + 1)$.

Problem 153.

(a)

We know as the authors assure us that when we substitute x_1 in the polynomial we will get 0 (a root). Thus let us make the a polynomial with the new roots from the old one.

$$(x - x_1)(x - x_2)(x - x_3) = x^3 + x^2 - 10x + 1$$

In this let us put the new roots. But in doing so we want our right hand side to also adjust so that when we substitute the root the resulting polynomial is 0. So for the root $(x_1 + 1)$ we will have to subtract 1 from the xs in the polynomial.

$$(x - (x_1 + 1))(x - (x_2 + 1))(x - (x_3 + 1)) = (x - 1)^3 + (x - 1)^2 - 10(x - 1) + 1$$

The right hand side can be simplified to the following.

$$x^3 - 2x^2 - 9x + 11$$

(b)

Similar logic as the previous one but in this case we will need to divide by 2 in the polynomial for all the$x$ terms.

$$\left(\frac{x}{2}\right)^3 + \left(\frac{x}{2}\right)^2 - 10\left(\frac{x}{2}\right) + 1$$

$$\frac{x^3}{8} + \frac{x^2}{4} - 5x + 1$$

Getting integer coefficients

$$x^3 + 2x^2 - 40x + 8$$

(c)

Same logic here but we take reciprocal in the polynomial.

$$\frac{1}{x^3} + \frac{1}{x^2} - \frac{10}{x} + 1$$

Getting integer coefficients

$$x^3 - 10x^2 + x + 1$$

Problem 154.

The immediate thing to visually see is that $(x^2 - 3x + 2)$ is $(x-1)(x-2)$. Thus if we substitute 1 or 2 in the cubic polynomial we should get 0. Thus we will have our two equations in two unknowns and we can solve for a and b.

$$1 + a + 1 + b = 0$$
$$8 + 4a + 2 + b = 0$$

These two give us $a = -\frac{8}{3}$ and $b = \frac{2}{3}$.

38 Values of polynomials, and interpolation

Problem 155.

The table is below

x	$x^3 - 2$	value
0	$0^3 - 2$	-2
1	$1^3 - 2$	-1
2	$2^3 - 2$	6
3	$3^3 - 2$	25
4	$4^3 - 2$	62
5	$5^3 - 2$	123
6	$6^3 - 2$	214

Problem 156.

This is just a substitution problem. We can either plug in numbers and show that successive differences will result in a constant difference in this case 2. Or we can consider a generic number k and show a generic answer. Let us work with k.

$P(x)$	x^2-x-4	Expansion	Value
$k-1$	$(k-1)^2-(k-1)-4$	$k^2+1-2k-k+1-4$	k^2-3k-2
k	k^2-k-4	k^2-k-4	k^2-k-4
$k+1$	$(k+1)^2-(k+1)-4$	$k^2+1+2k-k-1-4$	k^2+k-4
$k+2$	$(k+2)^2-(k+2)-4$	$k^2+4+4k-k-2-4$	k^2+3k-2

Now we just take first successive differences.

First Difference	Value
$k^2-k-4-k^2+3k+2$	$2k-2$
$k^2+k-4-k^2+k+4$	$2k$
$k^2+3k-2-k^2-k+4$	$2k+2$

The value columns already shows that the difference in these terms is 2. So if we take another set of differences in these values we will get the series as a constant 2.

Problem 157.

This is a generalization of the above solution. Assume the polynomial is ax^2+bx+c. Then we can build a similar table.

$P(x)$	ax^2bx+c	Value
$k-1$	$a(k-1)^2+b(k-1)+c$	$ak^2-2ak+bk+a-b+c$
k	ak^2+bk+c	ak^2+bk+c
$k+1$	$a(k+1)^2+b(k+1)+c$	$ak^2+2ak+bk+a+b+c$
$k+2$	$a(k+2)^2+b(k+2)+c$	$ak^2+4ak+bk+4a+2b+c$

Now we just take first successive differences.

First Difference	Value
$ak^2+bk+c-ak^2+2ak-bk-a+b-c$	$2ak-a+b$
$ak^2+2ak+bk+a+b+c-ak^2-bk-c$	$2ak+a+b$
$ak^2+4ak+bk+4a+2b+c-ak^2-2ak-bk-a-b-c$	$2ak+3a+b$

The value column can again be observed for successive differences which is

$$2ak+a+b-(2ak-a+b)$$

and

$$2ak+3a+b-(2ak+a+b)$$

The above two differences yield a constant difference of $2a$. Thus we have proved all differences will be $2a$.

Problem 158.

For a polynomial of degree 3, the third finite difference is constant.

We verify this using the cubic polynomial

$$P(x) = (x+1)^3 = x^3 + 3x^2 + 3x + 1.$$

Evaluate $P(x)$ at successive integers:

$$\begin{aligned} P(x) &= x^3 + 3x^2 + 3x + 1, \\ P(x+1) &= x^3 + 6x^2 + 12x + 8, \\ P(x+2) &= x^3 + 9x^2 + 27x + 27, \\ P(x+3) &= x^3 + 12x^2 + 48x + 64. \end{aligned}$$

First differences:

$$\begin{aligned} \Delta_1 &= P(x+1) - P(x) = 3x^2 + 9x + 7, \\ \Delta_2 &= P(x+2) - P(x+1) = 3x^2 + 15x + 19, \\ \Delta_3 &= P(x+3) - P(x+2) = 3x^2 + 21x + 37. \end{aligned}$$

Second differences:

$$\begin{aligned} \Delta_2 - \Delta_1 &= 6x + 12, \\ \Delta_3 - \Delta_2 &= 6x + 18. \end{aligned}$$

Third difference:

$$(6x + 18) - (6x + 12) = 6,$$

which is a constant.

Hence, the third finite difference of a cubic polynomial is constant.

The same result holds for any cubic polynomial

$$ax^3 + bx^2 + cx + d,$$

where the constant third difference equals $6a$.

Problem 159.

This is a classic number theory problem and I will show it with Lisp code too. Pleasantly surprised to see this problem in this book. Would love it when my daughter and son come across this for the first time in their lives. I hope they admire the mathematics of Euler as much as I do.

> *"Euler, the master of us all."*
>
> — *Carl Friedrich Gauss*

We can test whether a number is prime if we can check whether the greatest divisor for that number is the number itself and no other divisor greater than qual to 2 exists. This algorithm has probably been used for centuries.

So we will check for every value of n starting from 1 and go on. Manually it is very easy to check up to $n = 10$ since that results in only 151 and we notice that all numbers $43, 47, 53, 61, 71, 83, 97, 113, 131, 151$ are prime. But a general solution is pretty difficult. In fact this was Problem number #6 in the International Math Olympiad in 1987! Stating the problem below:

Let n be an integer greater than or equal to 2. Prove that if $k^2 + k + n$ is prime for all integers k such that $0 \leq k \leq \sqrt{n/3}$, then $k^2 + k + n$ is prime for all integers k such that $0 \leq k \leq n - 2$.

We know the answer but now let us make our computer wizard do the job of computation. Here is the Scheme/Lisp code and the output generated.

```
#lang sicp

;; check whether a value of a given polynomial is prime or not


;; polynomial given x^2 + x + 41
(define (poly x)
  (+ (* x x) x 41))

;; print value of polynomial and whether it's prime or not
(define (print-polyval-prime from to)
  (define (iter n)
    (if (> n to)
        (quote done)
        (let ((value (poly n)))
          (display "n = ") (display n)
          (display "   f(n) = ") (display value)
          (display "   prime? -> ") (display (if (prime? value) "yes" "no"))
          (newline)
          (iter (+ n 1)))))
  (iter from))

;; Prime testing procedure
(define (prime? n)
  (= n (smallest-divisor n)))

(define (smallest-divisor n)
  (find-divisor n 2))

(define (find-divisor n test-divisor)
  (cond ((> (square test-divisor) n) n)
        ((divides? test-divisor n) test-divisor)
```

```
      (else (find-divisor n (+ test-divisor 1)))))

(define (square x)
  (* x x))

(define (divides? x y)
  (= (remainder y x) 0))
```

Below is the output printed in the REPL in Dr.Racket. We can see that at $n = 40$ we get an output value of 1681 which is not a prime. 1681 is the square of 41.

```
> (print-polyval-prime 1 41)
n = 1   f(n) = 43   prime? -> yes
n = 2   f(n) = 47   prime? -> yes
n = 3   f(n) = 53   prime? -> yes
n = 4   f(n) = 61   prime? -> yes
n = 5   f(n) = 71   prime? -> yes
n = 6   f(n) = 83   prime? -> yes
n = 7   f(n) = 97   prime? -> yes
n = 8   f(n) = 113   prime? -> yes
n = 9   f(n) = 131   prime? -> yes
n = 10   f(n) = 151   prime? -> yes
n = 11   f(n) = 173   prime? -> yes
n = 12   f(n) = 197   prime? -> yes
n = 13   f(n) = 223   prime? -> yes
n = 14   f(n) = 251   prime? -> yes
n = 15   f(n) = 281   prime? -> yes
n = 16   f(n) = 313   prime? -> yes
n = 17   f(n) = 347   prime? -> yes
n = 18   f(n) = 383   prime? -> yes
n = 19   f(n) = 421   prime? -> yes
n = 20   f(n) = 461   prime? -> yes
n = 21   f(n) = 503   prime? -> yes
n = 22   f(n) = 547   prime? -> yes
n = 23   f(n) = 593   prime? -> yes
n = 24   f(n) = 641   prime? -> yes
n = 25   f(n) = 691   prime? -> yes
n = 26   f(n) = 743   prime? -> yes
n = 27   f(n) = 797   prime? -> yes
n = 28   f(n) = 853   prime? -> yes
n = 29   f(n) = 911   prime? -> yes
n = 30   f(n) = 971   prime? -> yes
n = 31   f(n) = 1033   prime? -> yes
n = 32   f(n) = 1097   prime? -> yes
n = 33   f(n) = 1163   prime? -> yes
n = 34   f(n) = 1231   prime? -> yes
n = 35   f(n) = 1301   prime? -> yes
n = 36   f(n) = 1373   prime? -> yes
n = 37   f(n) = 1447   prime? -> yes
```

```
n = 38   f(n) = 1523   prime? -> yes
n = 39   f(n) = 1601   prime? -> yes
n = 40   f(n) = 1681   prime? -> no
n = 41   f(n) = 1763   prime? -> no
done
```

Problem 160.

This problem is solved but it is essentially solving linear equations.

$$P(x) = ax + b$$

Substituting 1 and 2 as given we get

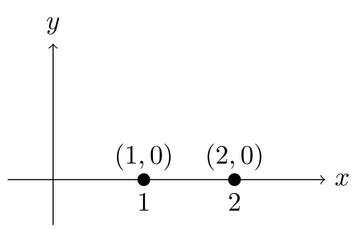
$$\begin{aligned} a + b &= 7 \\ 2a + b &= 5 \end{aligned}$$

We compute from the above two equations that $a = -2$ and $b = 9$. Thus $P(x) = -2x + 9$.

Problem 161.

We can think of this as a straight line on a co-ordinate plane. This line at $x = 1$ gives $y = 0$ so the point $(1, 0)$ lies on it. The other point is $(2, 0)$. We notice that both the points lie on the X axis which means the line is the X axis itself.

Thus $P(x) = 0$.



Problem 162.

The polynomial can be either linear or quadratic. Since the graph of the polynomial crosses the x-axis at $x = 1$ and $x = 2$, these values are roots of the polynomial. Hence the polynomial must be divisible by $(x - 1)$ and $(x - 2)$.

We may therefore write

$$P(x) = (x - 1)(x - 2) \cdot Q(x),$$

where $Q(x)$ is another polynomial.

We are given that the degree of $P(x)$ does not exceed 2. Since $(x-1)(x-2)$ already has degree 2, it follows that $Q(x)$ must be a constant.

Let $Q(x) = k$. Then

$$P(x) = k(x-1)(x-2) = k(x^2 - 3x + 2).$$

Thus the polynomial has the required form, with k representing the constant factor.

Problem 163.

The problem is exactly like the last one.

We can write it as:

$$P(x) = (x-1)(x-2)Q(x)$$

Given that $P(3) = 4$ we can make the following substitutions.

$$\begin{aligned}4 &= (3-1)(3-2)Q(x)\\ 4 &= 2Q(x)\\ Q(x) &= 2\end{aligned}$$

We could have reasoned out at the start itself that $Q(x)$ will actually be a constant because we are told that $P(x)$ cannot exceed degree 2.

Thus $P(x) = 2(x-1)(x-2)$ which is $2x^2 - 6x + 4$.

Problem 164.

This is solved in the book and uses proof by contradiction approach to solve. Summarizing the solution. If $P(x)$ and $Q(x)$ are two polynomials of degree not more than 2 and for three different numbers the two polynomials are same then both the polynomials are indeed equivalent.

Let a polynomial be defined as

$$R(x) = P(x) - Q(x)$$

Then for those three numbers we know that

$$R(x_1) = R(x_2) = R(x_3) = 0$$

But $R(x)$ cannot have a degree more than 2 and thus it cannot have three roots/solutions so $R(x)$ indeed must be equal to 0. From this we derive that $P(x) = Q(x)$.

Problem 165.

This is fairly simple in terms of looking at it from a point of view of simultaneous equations.

$$16a + 4b + c = 0$$
$$49a + 7b + c = 0$$
$$100a + 10b + c = 0$$

Solve the first two we get

$$11a + b = 0$$

From the second and third we get

$$17a + b = 0$$

Solving these two we get $a = 0$ and then we get $b = 0$ plugging these two back in one of the original equations we also getting $c = 0$.

Problem 166.

As guided we can use the same line of reasoning for the previous problem for degree 2.

Let the polynomial be $P(x)$ and $Q(x)$ be equal for numbers x_1 up to x_{n+1}. Now we have another polynomial which is given as $R(x) = P(x) - Q(x)$. From this we state that

$$R(x_1) = R(x_2) = ... = R(x_n) = R(x_{n+1}) = 0$$

So $R(x)$ has $(n+1)$ roots which is not possible since it can have only n roots since its degree is at the maximum n, thus $R(x) = 0$. Hence we get that $P(x) = Q(x)$ showing that such a polynomial is unique.

Problem 167.

(a) Given 2 roots already

$$P(x) = (x - 1)(x - 2)Q(x)$$
$$P(3) = 4 = (3 - 1)(3 - 2)Q(x)$$
$$Q(x) = 2$$

Thus $P(x) = 2(x - 1)(x - 2)$.

(b) Same as preceding problem

$$P(x) = (x-1)(x-3)Q(x)$$
$$P(2) = 2 = (2-1)(2-3)Q(x)$$
$$Q(x) = -2$$

Thus $P(x) = -2(x-1)(x-3)$.

(c) Same as preceding problem

$$P(x) = (x-2)(x-3)Q(x)$$
$$P(1) = 6 = (1-2)(1-3)Q(x)$$
$$Q(x) = 3$$

Thus $P(x) = 3(x-2)(x-3)$.

(d) Though the problem is solved in two ways in the book I find the easier method is to simply solve linear equations.

Let

$$P(x) = ax^2 + bx + c$$

We know that the degree cannot exceed 2 thus the above form. Now plug in given values we get the following set of linear equations.

$$a + b + c = 6$$
$$4a + 2b + c = 2$$
$$9a + 3b + c = 4$$

Solving for a, b, and c gives us.

$$a = 3$$
$$b = -13$$
$$c = 16$$

Hence the polynomial is $P(x) = 3x^2 - 13x + 16$.

Problem 168.

Again this could be solved as linear equations. Let the polynomial be $ax^3 + bx^2 + cx + d$. Plugging in the given values.

$$-a+b-c+d=2$$
$$d=1$$
$$a+b+c+d=2$$
$$8a+4b+2c+d=7$$

Substitute $d=1$

$$-a+b-c=1$$
$$d=1$$
$$a+b+c=1$$
$$8a+4b+2c=6$$

From the first and third equation we get $b=1$. Updating the equations again.

$$a=-c$$
$$b=1$$
$$d=1$$
$$a=-c$$
$$4a+c=1$$

From above we get

$$a=\frac{1}{3}$$
$$b=1$$
$$c=-\frac{1}{3}$$
$$d=1$$

Thus the polynomial can be stated as

$$P(x)=\frac{x^3}{3}+x^2-\frac{x}{3}+1$$

Problem 169.

This is similar to earlier problems in this chapter. We will use the same reasoning as in problem 164 and 166 but modified for this problem.

Given

$$
\begin{aligned}
P(x_1) &= y_1 \\
P(x_2) &= y_2 \\
&... \\
P(x_{10}) &= y_{10}
\end{aligned}
$$

Let another polynomial $Q(x)$ also have the same outputs with the given input arguments.

$$
\begin{aligned}
Q(x_1) &= y_1 \\
Q(x_2) &= y_2 \\
&... \\
Q(x_{10}) &= y_{10}
\end{aligned}
$$

Now we have another polynomial which is the difference of the above two, $R(x) = P(x) - Q(x)$. We can derive the following relationship from this for $i \in [1, 10]$

$$R(x_i) = P(x_i) - Q(x_i)$$

But we know that $P(x_i) = y_i$ and also $Q(x_i) = y_i$. Substituting this in the equation above.

$$R(x_i) = P(x_i) - Q(x_i) = y_i - y_i = 0$$

If $R(x_i) = 0$ then $P(x_i) = Q(x_i)$. Thus $P(x)$ is unique.

Problem 170.

This is a nice problem.

There are a few ways to do this. The most obvious is solve for the unknowns but the authors say do not do that. So let us keep that aside.

The other method is just visual inspection. We see the coefficients of a to be square numbers 10^2, 6^2, and 2^2. Also the bases of these are coefficients of b. So we have a generic structure like $ax^2 + bx + c = k$. We have boiled down the question to what we have been proving multiple times in this chapter. We have a polynomial of degree 2 $P(x)$ such that only one unique polynomial exists since we are given that $P(10) = 18.37$, $P(6) = 0.05$, and $P(2) = -3$. Thus a, b, and c will exist uniquely.

But this is the solution which did not come to my mind immediately. A very traditional way to show that there exists unique solutions to these equations is to use matrices. If we show that the determinant of the matrix of coefficients of these equations is non-zero then there is a solution to this.

$$A = \begin{pmatrix} 100 & 10 & 1 \\ 36 & 6 & 1 \\ 4 & 2 & 1 \end{pmatrix}$$

This coefficient matrix is not linearly dependent so determinant is non-zero. Anyways we can still check it. Doing some row operations. $R_1 - R_2$ and $R_2 - R_3$, we get

$$A = \begin{pmatrix} 64 & 4 & 0 \\ 32 & 4 & 0 \\ 4 & 2 & 1 \end{pmatrix}$$

Another row operation $R_1 - R_2$, this gives

$$A = \begin{pmatrix} 32 & 0 & 0 \\ 32 & 4 & 0 \\ 4 & 2 & 1 \end{pmatrix}$$

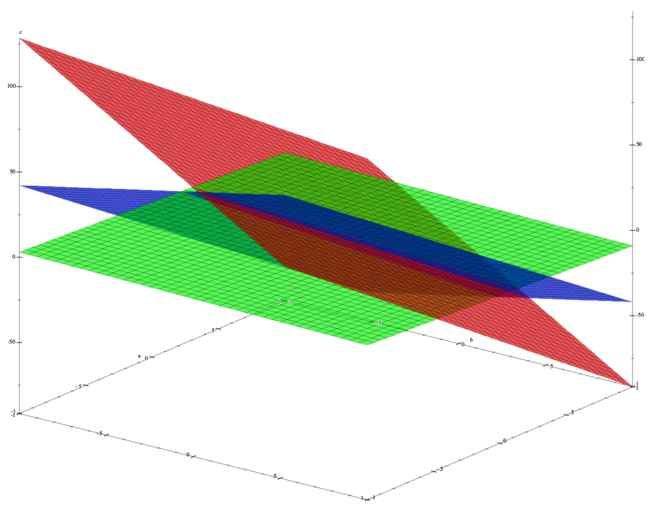
We get $\det(A) \neq 0$. Thus unique solution exists.

There is a third way to look at this problem. This is when we think of a 3-D co-ordinate system with axis a, b, and c. Each of the 3 given equations should represent a unique plane in this 3 dimensional co-ordinate system for there to be a unique solution. These 3 planes would then intersect at a single point. Assume (proof by contradiction) that if at least any of the two planes are either parallel to each other or are equal (basically the same). We immediately observe that given the coefficients none of these two conditions are possible because either the coefficients would have been same or reducible (after dividing by a suitable number) to a the same numbers. Thus these planes will always intersect at one point.

Let us try to see this using Lisp programming too.

```
#lang racket

(require plot)

(plot-new-window? #t)

;; 3 planes assume c to be 0 or we can use an offset constant either ways

(define (plane1 x y)
  (- 18.37 (* 100 x) (* 10 y)))

(define (plane2 x y)
  (- 0.05 (* 36 x) (* 6 y)))
```

```
(define (plane3 x y)
  (- -3 (* 4 x) (* 2 y)))

;; plot the planes

(plot3d
 (list
  (surface3d plane1 -1 1 -1 1
             #:color "red"
             #:alpha 0.6)
  (surface3d plane2 -1 1 -1 1
             #:color "blue"
             #:alpha 0.6)
  (surface3d plane3 -1 1 -1 1
             #:color "green"
             #:alpha 0.6))
 #:x-label "a"
 #:y-label "b"
 #:z-label "c")
```



The intersection point is not very clear but the actual co-ordinate for the planes used in the Lisp code returns $(0.4771875, -3.05625, -3.265)$.

Problem 171.

The answer is given but the solution isn't.

We can say that this polynomial has 10 roots and they are numbers from 1 to 10, there could be more roots but we know these 10 for sure. Since the highest coefficient of $P(x)$ is 1 it means that no other coefficient will be greater than 1.

The smallest degree polynomial which we can form is

$P(x) = (x-1)(x-2)(x-3)(x-4)(x-5)(x-6)(x-7)(x-8)(x-9)(x-10)$

If we substitute $x = 11$ then we basically get 10! which computes to 3628800.

$P(11) = (11-1)(11-2)(11-3)(11-4)(11-5)(11-6)(11-7)(11-8)(11-9)(11-10)$

39 Arithmetic progressions

Problem 172.

First $+2$, second -1.

Problem 173.

The difference is -7. Thus $-2-7 = -9$. So -9.

Problem 174.

This is a subset of natural numbers starting from 2. If it were starting from 1 the 1000_{th} number would be 1000. Since we are starting from 2 we move a number forward so the answer is 1001. Note that the set of natural numbers is also an arithmetic progression where the common difference between two successive elements is 1.

Problem 175.

Without churning out formulas we can derive them ourselves.

The 2_{nd} term is first term plus the common difference multiplied by the common difference. We can generalize that by

$$T_n = a + (n-1) \times d$$

Where T_n is the n_{th} term, a is the first term and d is the constant difference in this arithmetic progression.

Thus the 1000_{th} term will be $(2 + (1000-1) \times 2)$ which is 2000.

Problem 176.

We apply the same logic as the earlier problem and arrive at $1 + (1000-1) \times 2$ which is 1999.

Problem 177.

We just derived the formula in problem number 175. The n_{th} is given by

$$T_n = a + (n-1) \times d$$

Where T_n is the n_{th} term, a is the first term and d is the constant difference in this arithmetic progression.

Problem 178.

Yes this too is an arithmetic progression. We can write it as say the last term being n, the second last term being $n - d$, the third last term being $n - 2d$ and so on. Thus the common difference can be denoted as $-d$ and the n_{th} can be stated as

$$T_n = last_{term} - (n-1) \times d$$

where n is counted from the end.

Problem 179.

Yes the resultant series is also an arithmetic progression. The common difference now doubles.

$$a, \cancel{a+d}, a+2d, \cancel{a+3d}, a+4d, \cancel{a+5d}, a+6d, \cancel{a+7d}, a+8d, \cancel{a+9d}, a+10d, ...$$
$$a, a+2d, a+4d, a+6d, a+8d, a+10d, ...$$

Problem 180.

No. Now its not an arithmetic progression. The common difference pattern will become after omission as following

$$0, d, 2d, d, 2d, d, 2d, ...$$

Problem 181.

The first term is 5 and suppose the common difference is d then the second term will be $5 + d$. The third term should be $5 + 2d$ but we are given that is 8.

Therefore

$$5 + 2d = 8$$
$$d = \frac{3}{2}$$

The second term will be

$$5 + d$$
$$= 5 + \frac{3}{2}$$
$$= \frac{13}{2}$$
$$= 6.5$$

Problem 182.

This is simply the average technically since the middle number should be equidistant from the first and third number. This is one way to look at this problem. The other way is again the usual way of having the numbers as a, $a + d$ and $a + 2d$. Then equating $a + 2d$ with b which gives d as $\frac{b-a}{2}$. The middle term then can be $a + \frac{(b-a)}{2}$ which is $\frac{(a+b)}{2}$.

Problem 183.

We can write the progression as
$a, a + d, a + 2d, a + 3d, ...$

We are given that $a + 3d = b$

This gives the difference d as $\frac{(b-a)}{3}$. The second and third term will be.

Second term

$$a + d$$
$$a + \frac{(b-a)}{3}$$
$$\frac{2a+b}{3}$$

Third term

$$a + 2d$$
$$a + \frac{2(b-a)}{3}$$
$$\frac{a+2b}{3}$$

Problem 184.

The first term a is 1. The difference d is 2. The last term is 999. If there are n terms then the last term can be written as $a + (n-1)d$. Let us equate.

$$a + (n-1)d = 999$$
$$1 + (n-1)2 = 999$$
$$n = 501$$

There are 501 terms in this progression.

40 The sum of arithmetic progression

Problem 185.

We should do this the way the child Gauss would do. Let the sum of the series be S

$$S = 1 + 3 + 5 + 7 + ... + 999$$
$$S = 999 + 997 + 995 + 993 + ... + 1$$

Adding these two we get

$$2S = 1000 + 1000 + 1000 + ... + 1000$$

The important thing is to know how many terms are there. We can easily deduce that since $999 = 1 + (n-1)2$. This gives n as 500. Back to the earlier equation.

$$2S = 1000 \times 500$$
$$S = 250000$$

Problem 186.

This is similar to the previous problem.

$$S = a + (a + 1.d) + (a + 2.d) + ... + (a + (n - 1)d)$$

Here $(a + (n - 1)d) = b$.

Adding these two

$$S = a + (a + 1.d) + (a + 2.d) + ... + (a + (n - 1)d)$$
$$S = (a + (n - 1)d) + ... + (a + d) + a$$

We get

$$2S = (a + a + (n - 1)d) + (a + a + (n - 1)d) + ... + (a + a + (n - 1)d)$$

Which essentially is

$$2S = (a + b) + (a + b) + ... + (a + b)$$

since there are n terms on the right hand side

$$2S = n(a + b)$$

$$S = \frac{n(a + b)}{2}$$

Problem 187.

The reasoning is given in the book but we did not make this mistake. The explanation withe figure is quite good.
Essentially if we notice we fold the series over each other once reversed. This is exactly the child Gauss method.

Problem 188.

The hint given is a good one especially visually speaking. But let us try and solve it with equations.

An odd number can be given as $2k + 1$ for every $k > 0$ where k is an integer. Therefore, the sum of the first n odd numbers will be.

$$S_{odd} = [(2k+1)]+[(2k+1)+(1\times 2)]+[(2k+1)+(2\times 2)]+...+[(2k+1)+((n-1)\times 2)]$$

Let us rearrange the terms.

$$S_{odd} = 2kn + n + 2(1 + 2 + 3 + ... + (n-1))$$

We are in familiar territory of an arithmetic progression.

$$S_{odd} = 2kn + n + \frac{2n(n-1)}{2}$$
$$S_{odd} = 2kn + n + n^2 - n$$
$$S_{odd} = 2kn + n^2$$

Please note the question asks us that "the sum of n first odd numbers" meaning that $2k + 1$ should have been 1, therefore $k = 0$ in this instance. We get:

$$S_{odd} = n^2$$

Hence proved.

41 Geometric progressions

Problem 189.

For the first one $\frac{6}{3}$ which is 2.

For the second one $\frac{2}{6}$ which is $\frac{1}{3}$.

Problem 190.

Common ratio is $\frac{3}{2}$. Thus third term will be

$$3 \times \frac{3}{2} = \frac{9}{2}$$

Problem 191.

The first term let us start labeling it as a and the common ratio as r. Here

$a = 3$

$r = \frac{6}{3} = 2$

So the terms are in the below sequence for n terms:

$$a, ar, ar^2, ar^3, ar^4, ..., ar^{n-1}$$

For this specific problem we have the 1000th term as:

$$ar^{1000-1} = 3 \times 2^{1000-1} = 3 \times 2^{999}$$

Problem 192.

We just solved this in the previous problem. The nth term is given by $a \times q^{n-1}$.

Problem 193.

In this problem $a = 1$ and $ar^3 = 4$. We compute r as ± 2. The second term will be ar which will be ± 2.

Problem 194.

One bacteria grows in the following sequence every 1 minute.

$$1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, ...$$

which is the powers of 2

$$2^0, 2^1, 2^2, 2^3, 2^4, ...2^{30}$$

At the 30th minute there will be 2^{30} bacteria. Now if there were 2 bacteria at the start then the sequence would look like.

$$2.2^0, 2.2^1, 2.2^2, 2.2^3, 2.2^4, ...2.2^{30}$$

$$2^1, 2^2, 2^3, 2^4, 2^5, ...2^{29}, 2^{30}, 2^{31}$$

Here we note that 2^{30} is the 29th term denoting the 29th minute.

Problem 195.

Yes, we get another geometric progression where the common ratio is now the reciprocal of the original series. Let us look at a simple example.

$$2^0, 2^1, 2^2, 2^3, 2^4, ..., 2^{30}$$

Common ratio is $\frac{2^2}{2^1}$ which is 2.

$$2^{30}, 2^{29}, 2^{28}, ..., 2^2, 2^1, 2^0$$

Common ratio is $\frac{2^{28}}{2^{29}}$ which is $\frac{1}{2}$.

Problem 196.

The series will look like below

$$a, aq, aq^2, aq^3, aq^4, aq^5, aq^6, aq^7, ..., aq^t$$

After deleting every second term we get the following

$$a, aq^2, aq^4, aq^6, aq^8, ..., aq^{2k}$$

Now the common ratio is $\frac{aq^2}{a}$ which is q^2. Thus it will remain a geometric progression but with a different common ratio.

Problem 197.

The series will look like below

$$a, aq, aq^2, aq^3, aq^4, aq^5, aq^6, aq^7, ..., aq^t$$

After deleting every third term we get the following

$$a, aq, aq^3, aq^4, aq^6, aq^7, aq^9 ...$$

If we look at the first two terms we get the common ratio as $\frac{aq}{a} = q$ but if we look at the new third term and second term then the common ratio is $\frac{aq^3}{aq} = q^2$ which is different. Thus this new series is not a geometric progression.

Problem 198.

First term is a assume the common ratio is r then the second term will be ar and the third term will be ar^2. We are given that the third term is b. Therefore we have

$$ar^2 = b$$
$$r = \pm\sqrt{\frac{b}{a}}$$

Therefore the second term will be $\pm a\sqrt{\frac{b}{a}}$.

Problem 199.

Let the terms be as below where the common ratio is r

$$1, 1 \cdot r, 1 \cdot r^2, 1 \cdot r^3, ...$$

We are given that $a = 1 \cdot r^3$ where $a > 0$.

$$a = r^3$$
$$r = \sqrt[3]{a}$$

Thus the second term will be $\sqrt[3]{a}$ and the third term will be $\sqrt[3]{a^2}$.

42 The sum of a geometric progression

Problem 200.

This is the just the sum of powers of 2, from 2^0 to 2^{10}. The authors already have given a solution for this. I like the second solution more. But in my solution first we will generate a formula for the sum of a geometric progression.

Suppose the first term of the geometric series of n terms is a and the common ratio is r, then:

$$S = a + ar + ar^2 + ar^3 + ... + ar^{n-1}$$

Multiplying both sides by r.

$$rS = ar + ar^2 + ar^3 + ... + ar^{n-1} + ar^n$$

Subtracting the two equations.

$$S - rS = a - ar^n$$

$$S = \frac{a(1-r^n)}{(1-r)}$$

if r is larger than 1 then we can also modify it aS

$$S = \frac{a(r^n-1)}{(r-1)}$$

In this specific example there are 11 terms, the first term is 1 and the common ratio is 2. Plugging in the numbers we get

$$S = \frac{1(2^{11}-1)}{(2-1)}$$

$$S = \frac{2^{11}-1}{1}$$

$$S = 2^{11} - 1 = 2047$$

Problem 201.

I am unsure if the student is supposed to fund sums of each of the expressions or add all the expressions and find a single sum. The question actually says 'sums' so probably it is individual expressions getting summed up. But I'll add those up too.

Each of the expressions are geometric progressions with the first number as 1 and the common ratio as $\frac{1}{2}$. The only thing which is different in each expression is the number of elements. It starts off from 2 and goes till 11.

We will use the formula we derived ion the previous question.

$$1 + \frac{1}{2} = \frac{1 - \frac{1}{2^2}}{1 - \frac{1}{2}} = 2 \cdot \big(1 - \frac{1}{2^2}\big)$$

We can simply state this as

$$1 + \frac{1}{2} = 2 \cdot \big(1 - \frac{1}{2^2}\big)$$

$$1 + \frac{1}{2} + \frac{1}{4} = 2 \cdot \big(1 - \frac{1}{2^3}\big)$$

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = 2 \cdot \big(1 - \frac{1}{2^4}\big)$$

...

$$1 + \frac{1}{2} + \frac{1}{4} + ... + \frac{1}{1024} = 2 \cdot \big(1 - \frac{1}{2^{11}}\big)$$

Summing up all the terms on the right hand side of the equations above we get.

$$2 \cdot 10 - 2 \cdot \big(\frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + ... + \frac{1}{2^{11}}\big)$$

$$2 \cdot 10 - \big(\frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} + ... + \frac{1}{2^{10}}\big)$$

$$2 \cdot 10 - \frac{1}{2} \cdot \frac{(1 - \frac{1}{2^{10}})}{(1 - \frac{1}{2})}$$

$$2 \cdot 10 - (1 - \frac{1}{2^{10}})$$

$$19 - \frac{1023}{1024} = \frac{18433}{1024}$$

This is just a tad over 18.

Problem 202.

We derived this in an earlier question. Stating the same below.

The first term of the geometric series of n terms is a and the common ratio is q, then:

$$S = a + aq + aq^2 + aq^3 + ... + aq^{n-1}$$

Multiplying both sides by q.

$$qS = aq + aq^2 + aq^3 + ... + aq^{n-1} + aq^n$$

Subtracting the two equations.

$$S - qS = a - aq^n$$

$$S = \frac{a(1 - q^n)}{(1 - q)}$$

if q is larger than 1 then we can also modify it as

$$S = \frac{a(q^n - 1)}{(q - 1)}$$

Problem 203.

The author clearly explains the solution. The denominator would become 0 and thus the fraction would not be defined.

They also refer to the definition of a derivative but we can skip that for this book.

43 Different problems about progressions

Problem 204.

The authors give a hint. But the other way to look at it is that the numbers in the denominator could have a greatest common divisor difference. Having done this we would start getting terms which would start falling into an arithmetic progression.

Let us try it.

The first term is $\frac{1}{2} = \frac{15}{30}$, this we get after multiplying numerator and denominator by $\frac{1}{15}$. So the terms are.

$$\frac{15}{30}, ..., \frac{10}{30}, ..., \frac{6}{30}, ...$$

We can now easily see the pattern in the numerator that they are decreasing by 1. Thus it is an arithmetic progression.

Problem 205.

This is a good problem. The authors have solved it via proof by contradiction. Common ratio is taken as q.

$$3 = 2 \cdot q^n$$
$$5 = 3 \cdot q^m$$

for some value of m and n. We can manipulate

$$q^n = \frac{3}{2}$$

$$q^m = \frac{5}{3}$$

$$\left(\frac{3}{2}\right)^m = q^{mn} = \left(\frac{5}{3}\right)^n$$

$$\frac{3^m}{2^m} = \frac{5^n}{3^n}$$

$$3^{m+n} = 2^m \cdot 5^n$$

Now the left hand side is an odd number whereas the right hand side is always an even number when $m \neq 0$. For $m = 0$ we basically equate it as 3 equals to 5 which is also not true.

Therefore, this is not possible. This is the proof by contradiction.

Problem 206.

This can be reasoned very differently than actually deriving any formulas. In the previous problem we got the first equation given below. But the two other ways we can get the relation is mentioned after the first equation below. We note that these combinations rather sequence will arise depending on which order the numbers 2, 3, 5 come in the progression.

$$3^{m+n} = 2^m \cdot 5^n$$
$$2^{m+n} = 3^m \cdot 5^n$$
$$5^{m+n} = 2^m \cdot 3^n$$

We notice that any combination of 2, 3, and 5 will never yield any of the three equality above other than when $m = 0$ and $n = 0$. Why? Because 2, 3, and 5 are prime numbers (thus co-prime/relatively prime to each other).

Thus using the exact same line of argument as the last problem we can say that no matter what 2, 3, and 5 will not be the terms of a geometric progression.

Problem 207.

If two subsequent terms in an arithmetic progression are integers then the common difference will be an integer. Thus every term will have to be an integer.

Here the problem states that the first two terms are integers in the arithmetic progression.

Problem 208.

The problem is solved in the book with an example from a series constructed with the powers of 2. But it can be generic

$$...\frac{a}{r^4}, \frac{a}{r^3}, \frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2, ar^3, ar^4...$$

Here r is the common ratio.

Problem 209.

Solved in the book. This is not possible because we go from a to $a + d$ and then go from $a + d$ to $a + 2d$, the difference will be positive if d is positive and negative if negative. Both ways which it cannot change sign from positive to negative. It has to be consistent.

Problem 210.

But the answer flips in a geometric progression. Since we can have an example such as 1, -1, 1. This is also solved in the book.

Problem 211.

This too is solved in the book and a nice instance is given where 0 is one of the terms and the common difference is an irrational number such as $\sqrt{2}$.

Problem 212.

This is solved in the book but we can prove this fact as follows. Let the arithmetic progression be as $a, a+d, a+2d, ...$ and there are two integer terms $a+md$ and $a + nd$ where $m \neq n$ and both m and n belong to $\mathbb{Z}$, Then we have

$$(a + md) - (a + nd) = (m - n)d$$

From a we can look at every term which is at an interval of $(m - n)d$.

$$a + md + k(m - n)d$$

The term $k(m - n)d$ is an integer. So we get multiple repeating terms which are integers. Note that k is an integer. Hence proved.

Problem 213.

We can generalize the progression as

$$a, 2a + 1, 2(2a + 1) + 1, 2(2(2a + 1) + 1) + 1, ...$$

$$a, 2a + 1, 4a + 3, 8a + 7, ...$$

$$a, 2a + 2 - 1, 4a + 4 - 1, 8a + 8 - 1, ...$$

$$a, 2(a + 1) - 1, 2^2(a + 1) - 1, 2^3(a + 1) - 1, ...$$

The 100th term will be $2^{99}(a + 1) - 1$. Here $a = 1$. Thus we have

$$2^{99}(1 + 1) - 1 = 2^{100} - 1$$

Problem 214.

Let the progression be as

$$..., \frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2, ...$$

We know that $ar^2 = ar + a$. Solving for the common ratio r now.

$$ar^2 = ar + a$$
$$ar^2 - ar - a = 0$$
$$r^2 - r - 1 = 0$$
$$r = \frac{1 \pm \sqrt{1 + 4}}{2}$$
$$r = \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}$$

Problem 215.

This is one of the challenging problems. In this we have to find out the general n^{th} term of a Fibonacci sequence. This is known as Binet's Formula, named after the 19th century French mathematician Jacques Binet. Though the derivation of this formula was known a century earlier by mathematicians such as Abraham de Moivre, Daniel Bernoulli, and the master Euler.

The Art of Problem Solving wiki has multiple solutions. We will use a proof which is algebraic in nature and is a continuation of the previous problem (same as the first AoPS proof). A shootout to AoPS for writing really good books on problem solving. These books must also be consumed by the curious student.

In the previous problem we had a geometric series where one term is the sum of the previous two terms. If r was the common ratio this series we got the quadratic as $r^2 - r - 1 = 0$ and the roots are $r = \dfrac{1+\sqrt{5}}{2}, \dfrac{1-\sqrt{5}}{2}$.

We can use this quadratic to build all the terms of Fibonacci sequence and also generalize a term. Given from the quadratic:

$$r^2 = r + 1$$

First term

$$r = r$$

Second term

$$r^2 = r + 1$$

Third term

$$\begin{aligned} r^3 &= r \cdot r^2 \\ &= r \cdot (r+1) \\ &= r^2 + r \\ &= r + 1 + r \\ &= 2r + 1 \end{aligned}$$

Fourth term

$$\begin{aligned} r^4 &= r \cdot r^3 \\ &= r \cdot (2r+1) \\ &= 2r^2 + r \\ &= 2r + 2 + r \\ &= 3r + 2 \end{aligned}$$

Similarly we get other terms
1st term: r
2nd term: $r + 1$
3rd term: $2r + 1$
4th term: $3r + 2$
5th term: $5r + 3$
6th term: $8r + 5$
7th term: $13r + 8$
8th term: $21r + 13$
9th term: $34r + 21$

We get the pattern that $r^n = F_n r + F_{n-1}$.

From the previous problem we know that the two roots will be $r = \dfrac{1+\sqrt{5}}{2}, \dfrac{1-\sqrt{5}}{2}$, thus we can form two equations.

Equation 1:

$$\left(\frac{1+\sqrt{5}}{2}\right)^n = F_n\frac{1+\sqrt{5}}{2} + F_{n-1}$$

Equation 2:

$$\left(\frac{1-\sqrt{5}}{2}\right)^n = F_n\frac{1-\sqrt{5}}{2} + F_{n-1}$$

Subtracting equation 2 from equation 1 we get:

$$\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n = F_n\left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2}\right)$$

Rearranging and simplifying the above equation

$$F_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$F_n = \left(\frac{1}{\sqrt{5}}\right)\left(\frac{1+\sqrt{5}}{2}\right)^n + \left(-\frac{1}{\sqrt{5}}\right)\left(\frac{1-\sqrt{5}}{2}\right)^n$$

Hence we have derived the value of A and B as $\dfrac{1}{\sqrt{5}}$ and $-\dfrac{1}{\sqrt{5}}$ respectively.

44 The well-tempered clavier

Problem 216.

Let the frequencies of tones be denoted as f_1, f_2, and f_3.
Given that

$$\frac{f_1}{f_2} = \frac{3}{2}$$

$$\frac{f_2}{f_3} = \frac{4}{3}$$

$$\frac{f_1}{f_3} = \frac{f_1}{f_2} \times \frac{f_2}{f_3} = \frac{3}{2} \times \frac{4}{3} = \frac{2}{1}$$

Thus we get an octave.

Problem 217.

We are given in the text the value of Major third as $\frac{5}{4}$. Major is denoted with a capital M thus a Major third can be written as $M_3 = \frac{5}{4}$. Minor is denoted with a small m. Given in the text is $m_3 = \frac{6}{5}$.

Also given in the problem

$$m_6 \times M_3 = \frac{2}{1}$$

$$m_6 = \frac{2}{1} \times \frac{4}{5} = \frac{8}{5}$$

$$M_6 \times m_3 = \frac{2}{1}$$

$$M_6 = \frac{2}{1} \times \frac{5}{6} = \frac{5}{3}$$

We have to find

$$\frac{m_6}{M_6} = \frac{\frac{8}{5}}{\frac{5}{3}} = \frac{24}{25}$$

Problem 218.

Not very clear what the question asks but I assume that the question asks what is the change in frequency when we play a $33\frac{1}{3}$ record in a 45 rpm player. Suppose the frequencies are $f_1, f_2, f_3, f_4, ...$ then the frequencies will simply get multiplied by the ratio of record player to the record which is $\frac{45}{33\frac{1}{3}}$, this is equal to $\frac{27}{20}$.

Thus the new transposed frequencies will be $\frac{27}{20}f_1, \frac{27}{20}f_2, \frac{27}{20}f_3, \frac{27}{20}f_4, ...$

Problem 219.

The solution is given by the authors. But essentially transposing by one semitone maps every key to its neighboring key (on the right). Every transposition multiplies every frequency by one constant. Therefore each neighboring tone is obtained by multiplying the previous one by the same constant, which essentially is a geometric progression.

Problem 220.

The passage before this problem is enlightening, something which I was not aware of despite knowing a little bit about music. Suppose the frequencies are in a geometric progression $a, aq, aq^2, aq^3, aq^4, ...$ where q is the common ratio of the tone progression.

Suppose there are n equal steps per octave, then $q^n = 2$ and now suppose a fifth occupies k of those steps. Thus we have $q^k = \frac{3}{2}$

Doing some manipulations for the n equal steps per octave and inserting the fifth ratio.

$$\begin{aligned} q^n &= 2 \\ (q^n)^k &= 2^k \\ (q^k)^n &= (\frac{3}{2})^n \\ 2^k &= \frac{3^n}{2^n} \\ 2^{k+n} &= 3^n \end{aligned}$$

The last expression arrived at is not possible since a power of 2 cannot be odd or decomposed into a prime multiple of 3. Thus by contradiction we can say that we can not get true fifths even if that n varies.

Problem 221.

A youtube video of Bach's Well-Tempered Clavier is available here.

In the preface of the book *Geometry* by I.M.Gelfand and Tatiana A. (Gelfand), Tatiana says in the last few lines the following: *Many times Gelfand himself said that, if he could have been born again, he would have become a composer. I would say he actually was one.*

45 The sum of an infinite geometric progressions

Problem 222.

The series is actually a divergent one meaning that successive terms are increasing (not really the language I should be using but it means that the successive terms of distances when added do not converge to an answer).

Suppose Achilles speed is $\frac{v}{10}$ where v is the speed of the turtle. At the start of catch up race Achilles is at say at $x = 0$ on the X axis and the turtle is at $x = 1$ on the X axis. Intuitively we know as time t increases the distance between Achilles and the turtle will keep increasing so we must go backwards in time to see at what point they were together. Since at that t the x coordinate of them both will be same we can say that

$$\frac{v}{10} \times t = 1 + vt$$

vt is the distance covered by the turtle in time t, let's call it S_t.

$$\frac{S_t}{10} = 1 + S_t$$

$$S_t = -\frac{1}{9}$$

This means that Achilles and the turtle would have been together at $x = -\frac{1}{9}$ on the X axis, that is in the past time wise.

46 Equations

No problems in this chapter.

47 A short glossary

No problems in this chapter.

48 Quadratic equations

Problem 223.

The problem is solved. But intuitively we can see the roots will be some positive or negative 1 and 2.

$$x^2 - 3x + 2 = 0$$
$$(x - 1)(x - 2) = 0$$

Thus the roots are $x = 1$ and $x = 2$.

Problem 224.

(a)

$$
\begin{aligned}
x^2 - 4 &= 0 \\
x^2 - 2^2 &= 0 \\
(x-2)(x+2) &= 0 \\
x &= 2, -2
\end{aligned}
$$

(b)

$$
\begin{aligned}
x^2 + 2 &= 0 \\
x^2 &= -2 \\
x &= \sqrt{-2} \\
x &= \sqrt{2}i
\end{aligned}
$$

(c)

$$
\begin{aligned}
x^2 - 2x + 1 &= 0 \\
(x-1)^2 &= 0 \\
x &= 1
\end{aligned}
$$

(d)

$$
\begin{aligned}
x^2 - 2x + 1 &= 9 \\
x^2 - 2x - 8 &= 0 \\
(x-4)(x+2) &= 0 \\
x &= 4, -2
\end{aligned}
$$

(e)

$$
\begin{aligned}
x^2 - 2x - 8 &= 0 \\
(x-4)(x+2) &= 0 \\
x &= 4, -2
\end{aligned}
$$

(f)

$$
\begin{aligned}
x^2 - 2x - 3 &= 0 \\
(x-3)(x+1) &= 0 \\
x &= 3, -1
\end{aligned}
$$

(g)

$$
\begin{aligned}
x^2 - 5x + 6 &= 0 \\
(x-3)(x-2) &= 0 \\
x &= 3, 2
\end{aligned}
$$

(h)

$$x^2 - x - 2 = 0$$
$$(x-2)(x+1) = 0$$
$$x = 2, -1$$

Problem 225.

When $bx + c = 0$ and $b = 0$, it implies that $c = 0$. In this specific case all a, b, and c turn out to be 0. To correct this error we need to have a constraint that $x = -\frac{c}{b}$ only when $b \neq 0$.

49 The case $p = 0$. Square roots

Problem 226.

This is solved in the book elegantly looking at various sub cases. The attempt below is a smaller proof which uses the same proof by contradiction approach but also uses the fundamental theory of arithmetic. This proof is fairly similar to what the authors have written but this was given by the Greeks.

Let $\sqrt{2} = \frac{m}{n}$ where m and n are integers and n is not 0 and also they have no common factors.

Squaring

$$m^2 = 2n^2$$

So m^2 is even

Now let k be an integer

$$m = 2k$$

Substituting

$$4k^2 = 2n^2$$

So,

$$n^2 = 2k^2$$

Here now n is also even. But here both m and n are even and divisible by 2, contradicting the assumption that m and n have no common factors. Hence proved.

Problem 227.

We can use the same approach as our last problem. Let $\sqrt{3} = \frac{m}{n}$. Here m and n are integers with no common factors and also n is not 0.

Squaring

$$m^2 = 3n^2$$

So m^2 is divisible by 3 and we can write $3|m$

Now let k be some integer

$$m = 3k$$

Substituting

$$9k^2 = 3n^2$$

So,

$$n^2 = 3k^2$$

Here now n is also divisible by 3 which is $3|n$. But we had m and n as numbers with no common factors at the start and now we have ended with 3 as the common factor. This contradicts the assumption that m and n have no common factors. Hence proved.

We can generally prove that $\sqrt{p}$ for any prime p is irrational with the same logic.

50 Rules for square roots

Problem 228.

Solved in the book. We need to show that $\sqrt{a} \cdot \sqrt{b}$ when squared is non negative.

$$(\sqrt{a} \cdot \sqrt{b})^2 = (\sqrt{a})^2 \cdot (\sqrt{b})^2 = a \cdot b$$

Problem 229.

Similar approach to the previous problem

$$\left(\frac{\sqrt{a}}{\sqrt{b}}\right)^2 = \frac{a}{b}$$

We note that $b \neq 0$.Thus $\frac{a}{b}$ is non negative and defined given the properties of a and b.

Problem 230.

No, it can be positive or negative a. As rightly mentioned in the book a will be:

$$|a| = \begin{cases} +a, & \text{if } a \geq 0, \\ -a, & \text{if } a < 0. \end{cases}$$

Problem 231.

(a)

$$\frac{1}{2+\sqrt{3}} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= \frac{2-\sqrt{3}}{2^2-(\sqrt{3})^2}$$

$$= \frac{2-\sqrt{3}}{4-3}$$

$$= 2-\sqrt{3}$$

(b)

$$\frac{1}{\sqrt{7}-\sqrt{5}} = \frac{1}{\sqrt{7}-\sqrt{5}} \times \frac{\sqrt{5}+\sqrt{7}}{\sqrt{5}+\sqrt{7}}$$

$$= \frac{\sqrt{5}+\sqrt{7}}{(\sqrt{7})^2-(\sqrt{5})^2}$$

$$= \frac{\sqrt{5}+\sqrt{7}}{2}$$

Problem 232.

Simplify the expression $\sqrt{1001} - \sqrt{1000}$.

$$= \sqrt{1001} - \sqrt{1000} \times \frac{\sqrt{1001} + \sqrt{1000}}{\sqrt{1001} + \sqrt{1000}}$$

$$= \frac{1001 - 1000}{\sqrt{1001} + \sqrt{1000}}$$

$$= \frac{1}{\sqrt{1001} + \sqrt{1000}}$$

Whereas the other term is $\frac{1}{10} = \frac{1}{\sqrt{100}}$. This clearly shows that

$$\frac{1}{\sqrt{1001} + \sqrt{1000}} < \frac{1}{10}$$

Implying

$$\sqrt{1001} - \sqrt{1000} < \frac{1}{10}$$

Problem 233.

Now this is a problem which requires some intuition. Additionally, we are unsure what is meant by simplification but since this is solved in the book we understand what the authors want us to do.

$$\begin{aligned} 3 + 2\sqrt{2} &= 1 + 2 + 2\sqrt{2} \\ &= (1 + \sqrt{2})^2 \\ &\text{thus plugging this into the original expression} \\ &= \sqrt{(1 + \sqrt{2})^2} \\ &= 1 + \sqrt{2} \end{aligned}$$

Problem 234.

The solution says that we have to look at the positive expression only i.e. $\sqrt{2}-1$.

51 The equation $x^2 + px + q = 0$

Problem 235.

The next set of questions can easily be solved using discriminants but the authors have not introduced the concept so we will stick to how they want us to think.

$$
\begin{gathered}
x^2 + 2x - 6 = 0 \\
x^2 + 2x + 1 - 7 = 0 \\
(x+1)^2 = 7 \\
x + 1 = \pm\sqrt{7} \\
x = \pm\sqrt{7} - 1
\end{gathered}
$$

Problem 236.

$$
\begin{gathered}
x^2 + 2x - 8 = 0 \\
x^2 + 2x + 1 - 9 = 0 \\
(x+1)^2 = 9 \\
x + 1 = \pm 3 \\
x = 2, -4
\end{gathered}
$$

Problem 237.

$$
\begin{aligned}
x^2 + 3x + 1 &= 0 \\
x^2 + 2 \times \frac{3}{2} \times 1 \times x + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 + 1 &= 0 \\
\left(x + \frac{3}{2}\right)^2 - \frac{9}{4} + 1 &= 0 \\
\left(x + \frac{3}{2}\right)^2 &= \frac{5}{4} \\
x + \frac{3}{2} &= \pm\sqrt{\frac{5}{4}} \\
x &= \pm\sqrt{\frac{5}{4}} - \frac{3}{2}
\end{aligned}
$$

Problem 238.

Now this problem is solved in the book. In the real number set this equation will have no solution but in the complex or imaginary world it does have a solution.

$$
\begin{aligned}
x^2 - 2x + 2 &= 0 \\
(x-1)^2 + 1 &= 0 \\
(x-1)^2 &= -1 \\
x - 1 = \sqrt{-1} &= \pm i \\
x &= 1 \pm i
\end{aligned}
$$

In the end of this chapter the authors do talk about imaginary roots while explaining the sub cases of completing the square method.

52 Vieta's theorem

Problem 239.

In case of an equation whose both roots are the same then we can say that the two root α and β both are the same. Thus for the equation $x^2 + px + q = 0$ we have

$$\alpha + \beta = -p$$
$$\alpha + \alpha = 2\alpha = -p$$
$$\alpha = -\frac{p}{2}$$
$$\alpha \times \alpha = q$$
$$\alpha = \sqrt{q}$$

We can write the equation as

$$x^2 + px + \frac{p}{4} = 0$$

or

$$x^2 - 2\sqrt{q}x + q = 0$$

Yes, both proofs are still valid.

Problem 240.

Given a cubic equation $x^3 + px^2 + qx + r$ and let the roots be α, β, and γ, we have

$$x^3 + px^2 + qx + r = (x - \alpha)(x - \beta)(x - \gamma)R(x)$$

$R(x)$ is some polynomial but we know that the left hand side is a 3 degree equation then $R(x)$ has to be a constant. Also that the coefficient of x^3 is 1 so that the constant $R(x) = 1$. Therefore we now have

$$x^3 + px^2 + qx + r = (x - \alpha)(x - \beta)(x - \gamma) \times 1$$
$$x^3 + px^2 + qx + r = (x - \alpha)(x - \beta)(x - \gamma)$$
$$x^3 + px^2 + qx + r = x^3 - \gamma x^2 - \beta x^2 + \beta\gamma x - \alpha x^2 + \alpha\gamma x + \alpha\beta x - \alpha\beta\gamma$$
$$x^3 + px^2 + qx + r = x^3 - \alpha x^2 - \beta x^2 - \gamma x^2 + \alpha\beta x + \beta\gamma x + \gamma\alpha x - \alpha\beta\gamma$$
$$x^3 + px^2 + qx + r = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$$

Comparing the coefficients we get

$$\alpha + \beta + \gamma = -p$$
$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$
$$\alpha\beta\gamma = -r$$

Hence proved.

Problem 241.

Using Vieta's theorem

$$
\begin{gathered}
(x_1 + x_2)^2 = x_1^2 + x_2^2 + 2x_1x_2 \\
(-p)^2 = x_1^2 + x_2^2 + 2q \\
x_1^2 + x_2^2 = p^2 - 2q
\end{gathered}
$$

Problem 242.

Using Vieta's theorem

$$(x_1 - x_2)^2 = x_1^2 + x_2^2 - 2x_1x_2 = p^2 - 2q - 2q = p^2 - 4q$$

Problem 243.

This problem and the next one are challenging. Also at the end of the next problem I have collated the history about solving these equations in generality. It is very interesting.

Before jumping onto the solutions and history there was a young brilliant French mathematician named Évariste Galois. His theory essentially leads us to

> *Can we extract information about the roots of a polynomial without actually solving for the roots?*

In this solution I will use the simplest differentiation method from calculus possible i.e. differentiation of a polynomial. I will leave the explanation for it but just give the Wikipedia reference. Also for differentiating a polynomial ax^n with respect to x the derivative is nax^{n-1}.

Consider the cubic equation

$$f(x) = x^3 + px^2 + qx + r = 0$$

with three distinct roots x_1, x_2, x_3. We wish to express

$$(x_1 - x_2)^2(x_2 - x_3)^2(x_1 - x_3)^2$$

entirely in terms of the coefficients p, q, r.

By Vieta's formulas,

$$x_1 + x_2 + x_3 = -p$$

$$x_1x_2 + x_1x_3 + x_2x_3 = q$$

$$x_1x_2x_3 = -r$$

Let

$$\Delta = (x_1 - x_2)^2(x_2 - x_3)^2(x_1 - x_3)^2$$

Notice first why the differences are squared. The expression

$$(x_1 - x_2)(x_2 - x_3)(x_1 - x_3)$$

changes sign whenever two roots are interchanged, whereas its square is unchanged by every permutation of the roots. Thus Δ is a symmetric polynomial in the roots and can therefore be expressed in terms of the coefficients of the cubic.

Since

$$f(x) = (x - x_1)(x - x_2)(x - x_3)$$

we have

$$f'(x) = 3x^2 + 2px + q$$

Evaluating the derivative at each root gives

$$f'(x_1) = (x_1 - x_2)(x_1 - x_3)$$

$$f'(x_2) = (x_2 - x_1)(x_2 - x_3)$$

$$f'(x_3) = (x_3 - x_1)(x_3 - x_2)$$

Multiplying,

$$\begin{aligned} f'(x_1)f'(x_2)f'(x_3) &= (x_1 - x_2)(x_1 - x_3)(x_2 - x_1)(x_2 - x_3)(x_3 - x_1)(x_3 - x_2) \\ &= -(x_1 - x_2)^2(x_2 - x_3)^2(x_1 - x_3)^2 \end{aligned}$$

Therefore

$$\Delta = -f'(x_1)f'(x_2)f'(x_3)$$

Since

$$f'(x_i) = 3x_i^2 + 2px_i + q$$

we obtain

$$\Delta = -(3x_1^2 + 2px_1 + q)(3x_2^2 + 2px_2 + q)(3x_3^2 + 2px_3 + q)$$

Expanding this symmetric expression and using Vieta's formulas gives

$$f'(x_1)f'(x_2)f'(x_3) = 4p^3r - p^2q^2 - 18pqr + 4q^3 + 27r^2$$

Hence

$$\Delta = p^2q^2 - 4q^3 - 4p^3r - 27r^2 + 18pqr$$

Thus the discriminant of

$$x^3 + px^2 + qx + r$$

is

$$p^2q^2 - 4q^3 - 4p^3r - 27r^2 + 18pqr$$

The meaning of the discriminant is especially clear from its definition:

$$\Delta = (x_1 - x_2)^2(x_2 - x_3)^2(x_1 - x_3)^2$$

Thus

$$\Delta = 0$$

if and only if at least two roots coincide.

For a cubic with real coefficients,

$$\Delta > 0$$

means that the cubic has three distinct real roots,

$$\Delta < 0$$

means that it has one real root and two non-real complex-conjugate roots, and

$$\Delta = 0$$

means that it has a repeated root.

Problem 244.

Here we continue the story from the previous problem. Also give the amazing history behind this.

Suppose

$$f(x) = x^2 + px + q$$

has roots x_1, x_2, and

$$g(y) = y^2 + ry + s$$

has roots y_1, y_2

We wish to express

$$(y_1 - x_1)(y_2 - x_1)(y_1 - x_2)(y_2 - x_2)$$

as a polynomial in p, q, r, s

Since y_1, y_2 are the roots of g,

$$g(t) = (t - y_1)(t - y_2)$$

Therefore

$$g(x_1) = (x_1 - y_1)(x_1 - y_2) = (y_1 - x_1)(y_2 - x_1)$$

because the two minus signs cancel.

Similarly,

$$g(x_2) = (y_1 - x_2)(y_2 - x_2)$$

Hence the required expression is

$$R = g(x_1)g(x_2)$$

Now

$$g(x_i) = x_i^2 + rx_i + s$$

But x_i satisfies

$$x_i^2 + px_i + q = 0$$

so

$$x_i^2 = -px_i - q$$

Thus

$$\begin{aligned} g(x_i) &= -px_i - q + rx_i + s \\ &= (r-p)x_i + (s-q) \end{aligned}$$

Therefore

$$\begin{aligned} R &= ((r-p)x_1 + (s-q))\,((r-p)x_2 + (s-q)) \\ &= (r-p)^2 x_1 x_2 + (r-p)(s-q)(x_1 + x_2) + (s-q)^2 \end{aligned}$$

By Vieta's formulas,

$$x_1 + x_2 = -p, \qquad x_1 x_2 = q$$

Hence

$$R = q(r-p)^2 - p(r-p)(s-q) + (s-q)^2$$

Expanding,

$$R = qr^2 - pqr - prs + p^2 s + s^2 - 2qs + q^2$$

This polynomial is called the *resultant* of the two quadratics.

Its meaning follows immediately from

$$R = (y_1 - x_1)(y_2 - x_1)(y_1 - x_2)(y_2 - x_2).$$

We have

$$R = 0$$

if and only if one of these four factors is zero. Thus

$$R = 0 \iff \text{the two polynomials have a common root.}$$

There is a close connection between the resultant and the discriminant. A polynomial has a repeated root exactly when the polynomial and its derivative have a common root. Thus the discriminant of a polynomial is essentially the resultant of the polynomial and its derivative.

Historical Note — From Cubics to Galois Theory

For centuries mathematicians knew how to solve quadratic equations, but a general method for cubic equations remained unknown.

In the early sixteenth century the Italian mathematician Scipione del Ferro discovered a method for solving an important class of cubic equations, but he did not publish it widely. His method eventually became known to his student Antonio Fior.

In 1535 Fior challenged Niccolò Tartaglia to a mathematical contest. Tartaglia had independently discovered methods for solving cubic equations and defeated Fior decisively.

Girolamo Cardano later persuaded Tartaglia to reveal his method under a promise of secrecy. Cardano and his student Lodovico Ferrari developed the ideas further, and Ferrari discovered a method for solving quartic equations. Cardano later found evidence that del Ferro's discovery predated Tartaglia's. In 1545 he published the methods for the cubic and quartic in his famous book *Ars Magna*, crediting del Ferro, Tartaglia, and Ferrari. The publication nevertheless led to a bitter dispute between Tartaglia and Cardano.

Thus, by the middle of the sixteenth century, general equations of degrees could all be solved using radicals.

$$2, \qquad 3, \qquad 4$$

This naturally raised the question:

Is there a similar formula for equations of degree 5*?*

For nearly three hundred years mathematicians searched for one.

In 1824 the Norwegian mathematician Niels Henrik Abel (after whom the Abel prize is named) proved that there is no formula using radicals that solves every general quintic equation.

But this produced a subtler question. Some particular quintic equations *can* be solved by radicals. How can we decide which ones?

This question was answered by the young French mathematician Évariste Galois. Instead of searching directly for formulas, Galois studied the ways in which the roots of an equation could be *permuted* while preserving the algebraic relations between them. These permutations form what we now call the *Galois group* of the polynomial.

Galois discovered that the possibility of solving an equation by radicals is controlled by the structure of this group.

Problem 243 contains a small glimpse of this idea. Consider

$$V = (x_1 - x_2)(x_2 - x_3)(x_1 - x_3)$$

If two roots are interchanged, V changes sign. However,

$$V^2 = (x_1 - x_2)^2(x_2 - x_3)^2(x_1 - x_3)^2$$

is unchanged by every permutation of the roots.

Because it is unchanged by every permutation, it can be expressed entirely in terms of the coefficients of the polynomial. This quantity is the discriminant.

Thus the discriminant and resultant illustrate an important theme that eventually became central to modern algebra:

Learn about the roots without explicitly solving for them.

They also hint at an even deeper question:

What information remains unchanged when the roots are permuted?

It was Galois's extraordinary insight to make that question the foundation of a theory.

Tragically both Galois and Abel died at a very young age, 20 and 26 respectively. We will never know what else could they have discovered and proved to the world.

Problem 245.

In a quadratic equation if one root is irrational the other is a conjugate of the first root. In this case we are given that the first root is $4-\sqrt{7}$ then the conjugate of that is $4+\sqrt{7}$. The derivation in the previous chapter is given

$$x_{1,2} = -\frac{p}{2} \pm \sqrt{\frac{p^2}{4} - q}$$

Now using Vieta's theorem to get the polynomial.

$$4 - \sqrt{7} + \beta = -p$$
$$4 - \sqrt{7} + 4 + \sqrt{7} = -p$$
$$p = -8$$

and

$$(4 - \sqrt{7})\beta = q$$
$$(4 - \sqrt{7})(4 + \sqrt{7}) = q$$
$$q = 16 - 7 = 9$$

Therefore the quadratic equation is $x^2 - 8x + 9 = 0$.

Problem 246.

Given the roots are α and β then by Vieta's theorem we have

$$\alpha + \beta = -p$$

$$\alpha\beta = q$$

We also know that p and q are integers.

(a)

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$
$$\alpha^2 + \beta^2 = (-p)^2 + 2q = p^2 + 2q$$

Since p and q are integers so will $p^2 + 2q$ will be an integer.

(b)

$$(\alpha + \beta)^3 = \alpha^3 + \beta^3 + 3\alpha\beta(\alpha + \beta)$$
$$\alpha^3 + \beta^3 = -p^3 + 3q(-p)$$
$$\alpha^3 + \beta^3 = -p^3 - 3pq$$

Since p and q are integers $-p^3 - 3pq$ will also be an integer.

(c) This is an interesting problem and can be easily deduced using binomial expansion of $(\alpha + \beta)^n$ or even looking at Pascal's triangle (on the cover page of the book itself).

We saw from (a) and (b) above that $\alpha^2 + \beta^2$ and $\alpha^3 + \beta^3$ are both integers. If we can show that for powers of 4, 5, 6, 7, ... and then for n case and then $(n+1)$ cas this still holds true we would have proved our assertion that $\alpha^n + \beta^n$ is also an integer.

Let us look at $\alpha^4+\beta^4$ and expand using binomial expansion (taking an easy way to just use the numbers from Pascal's triangle since we have derived it earlier in the book).

$$(\alpha+\beta)^4 = \alpha^4 + 4\alpha^3\beta + 6\alpha^2\beta^2 + 4\alpha\beta^3 + \beta^4$$
$$(\alpha+\beta)^4 = \alpha^4 + \beta^4 + 4\alpha^3\beta + 4\alpha\beta^3 + 6\alpha^2\beta^2$$
$$(\alpha+\beta)^4 = \alpha^4 + \beta^4 + 4\alpha\beta(\alpha^2+\beta^2) + 6(\alpha\beta)^2$$
$$\alpha^4+\beta^4 = (\alpha+\beta)^4 - 4\alpha\beta(\alpha^2+\beta^2) - 6(\alpha\beta)^2$$
$$\alpha^4+\beta^4 = (-p)^4 - 4q(\alpha^2+\beta^2) - 6q^2$$

We notice that the only term $\alpha^2+\beta^2$ needs to be shown to be an integer. This we have already shown in (a) of this problem. Note that this was an even power i.e. 4. Now lets look at power 5.

$$(\alpha+\beta)^5 = \alpha^5 + 5\alpha^4\beta + 10\alpha^3\beta^2 + 10\alpha^2\beta^3 + 5\alpha\beta^4 + \beta^5$$
$$\alpha^5+\beta^5 = (\alpha+\beta)^5 - 5\alpha^4\beta - 5\alpha\beta^4 - 10\alpha^3\beta^2 - 10\alpha^2\beta^3$$
$$\alpha^5+\beta^5 = (\alpha+\beta)^5 - 5\alpha\beta(\alpha^3+\beta^3) - 10(\alpha\beta)^2(\alpha+\beta)$$
$$\alpha^5+\beta^5 = (-p)^5 - 5q(\alpha^3+\beta^3) - 10(q)^2(-p)$$

Again we notice every term on the right hand side is an integer and the term $\alpha^3+\beta^3$ has been proven to be an integer in (b) part of this problem.

So we get a pattern here: If we look at the even powers of $(\alpha^{2k}+\beta^{2k})$ then the term we need to be sure is an integer is $(\alpha^{2k-2}+\beta^{2k-2})$ and similarly for odd powers we need to prove for $(\alpha^{2m-1}+\beta^{2m-1})$. This will always be true since that follows from prior proofs. This is a nice intuitive proof. But here is a generalized induction based proof without using binomial expansion and Pascal's triangle (though this requires some deductive manipulations).

Let α and β are the two roots of the equation $x^2+px+q=0$. Substituting the root in the equation.

$$\alpha^2 + p\alpha + q = 0$$
$$\alpha^2 = -p\alpha - q$$

Similarly

$$\beta^2 = -p\beta - q$$

Multiplying $\alpha^2 = -p\alpha - q$ with α^{n-2} and for $\beta^2 = -p\beta - q$ with β^{n-2}. We get

$$\alpha^n = -p\alpha^{n-1} - q\alpha^{n-2}$$
$$\beta^n = -p\beta^{n-1} - q\beta^{n-2}$$

Now sum these two equations above, we get

$$\alpha^n + \beta^n = -p(\alpha^{n-1} + \beta^{n-1}) - q(\alpha^{n-2} + \beta^{n-2})$$
$$S_n = -pS_{n-1} - qS_{n-2}$$

Here S_n sum of squares of the roots of n degree equation. We can see that $S_0 = \alpha^0 + \beta^0 = 2$ and $S_1 = \alpha + \beta = -p$ and so on and so forth. So now S_{n-1} and S_{n-2} both will be integers and hence S_n will also be an integer.

The logic used is the same in both the proofs.

Problem 247.

(a) Computing square

$$(a + b\sqrt{2})^2$$
$$a^2 + 2b^2 + 2\sqrt{2}ab$$

Here the expression is of the form

$$a^2 + 2b^2 + 2\sqrt{2}ab = k + l\sqrt{2}$$
$$k = a^2 + 2b^2$$
$$l = 2ab$$

k and l will be integers since a and b are integers.

(b) We can expand the given expression using binomial expansion and then we will be able to prove what we are asked to do so.

$$(a + b\sqrt{2})^n = a^n + \binom{n}{1}a^{n-1}(b\sqrt{2}) + \binom{n}{2}a^{n-2}(b\sqrt{2})^2 + \binom{n}{3}a^{n-3}(b\sqrt{2})^3 ... + \binom{n}{n-1}a(b\sqrt{2})^{n-1} + (b\sqrt{2})^n$$

Let us look at the right hand side. Every term barring the odd powers of $\sqrt{2}$ will be an integer. The odd powers of $\sqrt{2}$ will end up with just a single $\sqrt{2}$ term. So actually we will get exactly what we want here. Rearranging the terms in the equation below.

$$\left[a^n + \binom{n}{2}a^{n-2}(b\sqrt{2})^2 ...\right] + \left[\binom{n}{1}a^{n-1}(b\sqrt{2}) + \binom{n}{3}a^{n-3}(b\sqrt{2})^3 ...\right]$$
$$\left[a^n + \binom{n}{2}a^{n-2}(2b^2) ...\right] + \left[\binom{n}{1}a^{n-1}(b) + \binom{n}{3}a^{n-3}(2b^3) ...\right]\sqrt{2}$$
$$k + l\sqrt{2}$$

$$k = \left[a^n + \binom{n}{2}a^{n-2}(2b^2)...\right]$$

$$l = \left[\binom{n}{1}a^{n-1}(b) + \binom{n}{3}a^{n-3}(2b^3)...\right]$$

Both k and l will be integers as shown. Hence proved.

(c) The number $(a - b\sqrt{2})^n$ will also be of the form $k + l\sqrt{2}$ because when we do the binomial expansion just like the previous (b) case we will get the same expansion only with an alternating minus sign for the odd powers of the $b\sqrt{2}$ term. A simple way to compute this is to substitute $-b$ instead of b in the previous problem. We will end up with the following.

$$\left[a^n + \binom{n}{2}a^{n-2}(-b\sqrt{2})^2...\right] + \left[\binom{n}{1}a^{n-1}(-b\sqrt{2}) + \binom{n}{3}a^{n-3}(-b\sqrt{2})^3...\right]$$
$$\left[a^n + \binom{n}{2}a^{n-2}(2b^2)...\right] - \left[\binom{n}{1}a^{n-1}(b) + \binom{n}{3}a^{n-3}(2b^3)...\right]\sqrt{2}$$
$$k - l\sqrt{2}$$

$$k = \left[a^n + \binom{n}{2}a^{n-2}(2b^2)...\right]$$

$$l = -\left[\binom{n}{1}a^{n-1}(b) + \binom{n}{3}a^{n-3}(2b^3)...\right]$$

Hence we get the form of $k - l\sqrt{2}$ which again shows that k and l are integers.

(d) This is solved in the book. This is a typical Gelfand - Shen set up. They asked us in the first three parts of the problem to show that $a + b\sqrt{2}$ has the form $k + l\sqrt{2}$ and the same proof for $(a + b\sqrt{2})^n$. In this current problem we can convert the given term into $a + b\sqrt{2}$.

$$a^2 - 2b^2 = 1$$
$$(a + b\sqrt{2})(a - b\sqrt{2}) = 1$$

Now if we take n^{th} power on both sides.

$$(a + b\sqrt{2})^n(a - b\sqrt{2})^n = 1^n$$
$$(k + l\sqrt{2})(k - l\sqrt{2}) = 1$$

By trial and error we know that one solution is $a = 3$ and $b = 2$ which gives $3^2 - 2 \cdot 2^2 = 1$. So in the form $(k + l\sqrt{2})(k - l\sqrt{2}) = 1$ we can take this one solution of $(3, 2)$ and get any power of it. Let us do that. For different ns.

$$n = 1 \Rrightarrow (3 + 2\sqrt{2})^1 \to (3, 2)$$
$$n = 2 \Rrightarrow (3 + 2\sqrt{2})^2 \to (17, 12)$$
$$n = 3 \Rrightarrow (3 + 2\sqrt{2})^3 \to (99, 70)$$
$$n = 4 \Rrightarrow (3 + 2\sqrt{2})^4 \to (577, 408)$$
$$...$$

Thus we will get infinitely many integer solutions of the form $(3 + 2\sqrt{2})^n$ for this equation.

This is the famous Pell's equation in Elementary Number Theory which is of the form $x^2 - Ny^2$. A little bit of history around this equation. The great Euler by mistake associated the name of the English mathematician John Pell to this equation. Pell had never worked on it surprisingly! But the name stuck ever since. This equation was first studied in the 7th century around 628 AD by the Indian mathematician Brahmagupta and then 500 years later by another Indian mathematician Bhaskara II. Bhaskara developed an algorithm to solve these equations. Then in the 17^{th} century the French genius mathematician Fermat worked on these equations.

Ideally the equation of the form $x^2 - Ny^2$ should have been called Brahmagupta - Bhaskara Equation. But I think real mathematicians do not care about names but rather ideas and realizations and proofs.

Problem 248.

By Vieta's theorem we know that if α and β were the roots then.

$$\alpha\beta = q$$

We can reason it out as that if q is negative then α and β have to be of opposite signs. But the roots must exist. The authors also show a visual graphical reasoning in the solution.

53 Factoring $ax^2 + bx + c$

Problem 249.

In this set of problems the formula from the next chapter is being used indirectly so easier and quicker to solve that way.

$$2x^2 + 5x - 3$$

$$x_{1,2} = \frac{-5 \pm \sqrt{5^2 - 4 \cdot 2 \cdot -3}}{2 \cdot 2}$$

$$x_{1,2} = \frac{-5 \pm \sqrt{25 + 24}}{4}$$

$$x_{1,2} = \frac{-5 \pm 7}{4}$$

$$x_{1,2} = -3, \frac{1}{2}$$

Therefore the factorization is

$$(x+3)(x - \frac{1}{2})$$
$$(x+3)(2x-1)$$

Problem 250.

Finding Roots

$$2x^2 + 2x + \frac{1}{2} = 0$$

$$4x^2 + 4x + 1 = 0$$

$$x_{1,2} = \frac{-4 \pm \sqrt{4^2 - 4 \cdot 4 \cdot 1}}{2 \cdot 4}$$

$$x_{1,2} = \frac{-4 \pm \sqrt{0}}{8}$$

$$x_{1,2} = -\frac{1}{2}$$

So the factorization will be $(x + \frac{1}{2})^2$

Problem 251.

Finding Roots assuming the variable is a

$$2a^2 + 5ab - 3b^2 = 0$$

$$a_{1,2} = \frac{-5b \pm \sqrt{25b^2 - 4 \cdot 2 \cdot (-3b^2)}}{2 \cdot 2}$$

$$a_{1,2} = \frac{-5b \pm \sqrt{25b^2 + 24b^2}}{4}$$

$$a_{1,2} = \frac{-5b \pm 7b}{4}$$

$$a_{1,2} = -3b, -\frac{b}{2}$$

So the factor equation will be

$$(a+3b)(a-\frac{b}{2}) = (a+3b)(2a-b)$$

54 A formula for $ax^2 + bx + c = 0$ (where $a \neq 0$)

Problem 252.

The roots of equation $ax^2 + bx + c = 0$ are given as

$$x_{1,2} = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}}$$

Now we know that $a \neq 0$ but we do not if its positive or negative. Let us assume the positive case $a > 0$. Then the roots are given by

$$x_{1,2} = \left(-\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a}\right), \left(-\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}\right)$$

Now if $a < 0$ then the roots are

$$x_{1,2} = \left(-\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{-2a}\right), \left(-\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{-2a}\right)$$

This can be rewritten as

$$x_{1,2} = \left(-\frac{b}{2a} - \frac{\sqrt{b^2-4ac}}{2a}\right), \left(-\frac{b}{2a} + \frac{\sqrt{b^2-4ac}}{2a}\right)$$

Basically the order of the roots does not matter.

Problem 253.

This is solved in the book smartly. The implicit substitution is $y = \dfrac{1}{x}$.

$$\begin{gathered} ax^2 + bx + c = 0 \\ a + \frac{b}{x} + \frac{c}{x^2} = 0 \\ a + by + cy^2 = 0 \end{gathered}$$

Now $\dfrac{1}{x_1}$ and $\dfrac{1}{x_2}$ will satisfy the equation $cy^2 + by + a = 0$.

55 One more formula concerning quadratic equatons

Problem 254.

This is the exact derivation we did in problem 253. For the equation $ax^2 + bx + c = 0$ if the roots are given by

$$x_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

then the equation $cy^2 + by + a = 0$ has the roots as $\dfrac{1}{x_1}$ and $\dfrac{1}{x_2}$.

Now if we equate $x_{1,2}$ and $\dfrac{1}{y_{1,2}}$ we should get the result as true.

$$
\begin{aligned}
\frac{-b \pm \sqrt{b^2-4ac}}{2a} &= \frac{2c}{-b \mp \sqrt{b^2-4ac}} \\
&= \frac{2c}{-b \mp \sqrt{b^2-4ac}} \times \frac{-b \pm \sqrt{b^2-4ac}}{-b \pm \sqrt{b^2-4ac}} \\
&= \frac{-2bc \pm 2c\sqrt{b^2-4ac}}{b^2-b^2-4ac} \\
&= \frac{-2c(b \pm \sqrt{b^2-4ac})}{-4ac} \\
&= \frac{-b \pm \sqrt{b^2-4ac}}{2a}
\end{aligned}
$$

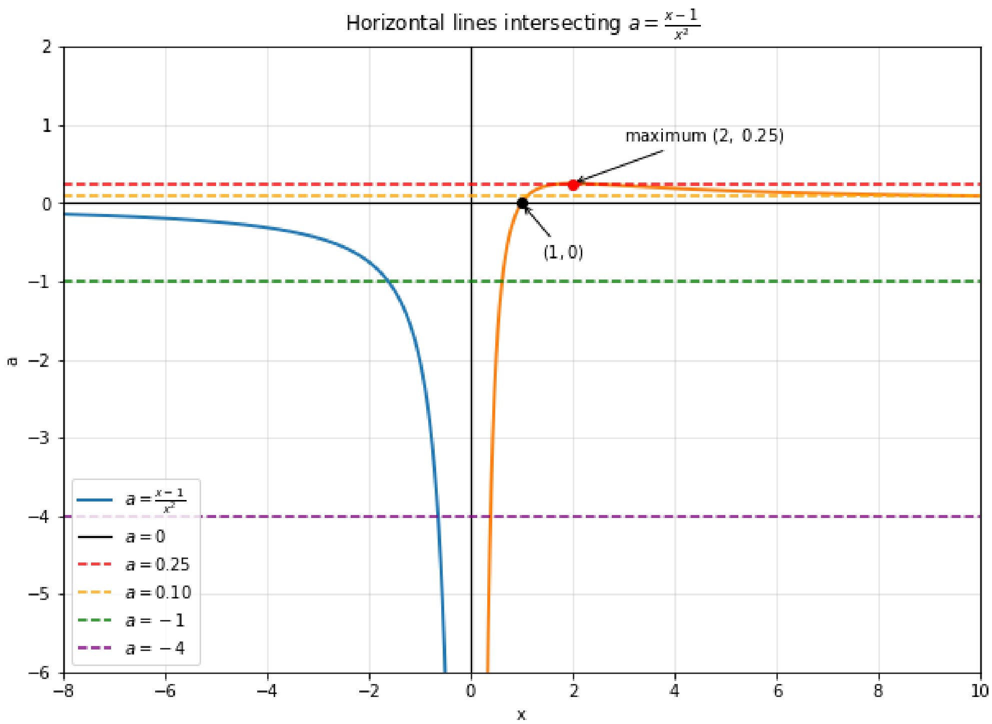
Hence we were able to prove this.

56 A quadratic equation becomes linear

Problem 255.

We go into the side of functions and graphs here. Imagine the $a = c$ for some c constant line going down from top to bottom. The authors explain it beautifully in the given solution.

Here is a graph plotted in Python. The dark continuous blue and orange curves is the graph of the equation $a = \frac{(x-1)}{x^2}$



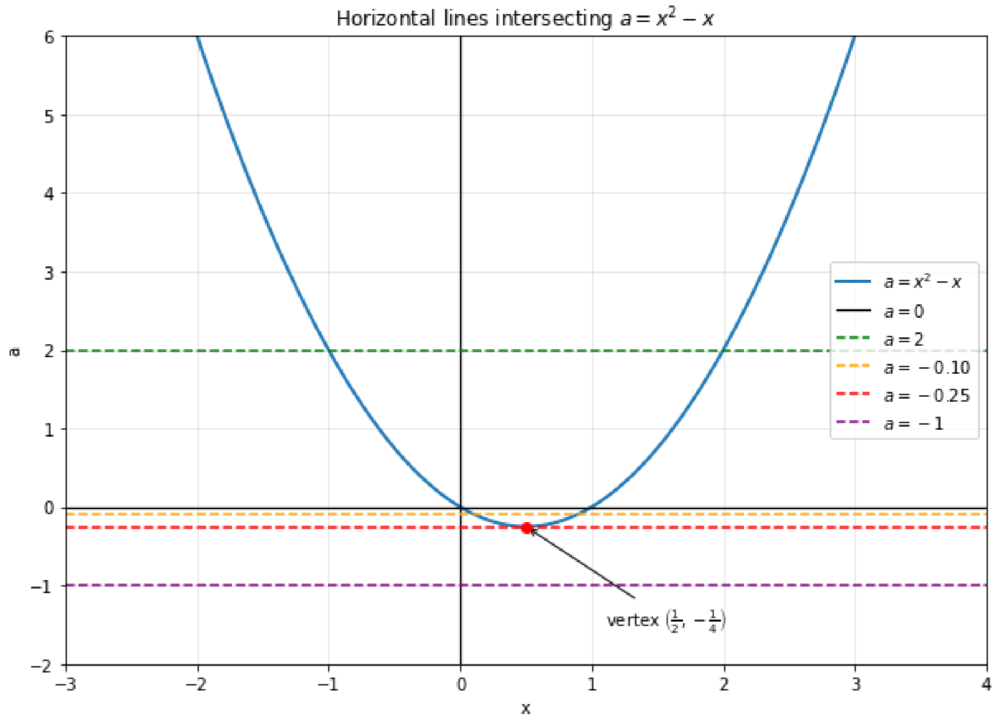
Problem 256.

(a) This is essentially a parabola. We can see the following 3 cases:

For $a > -\dfrac{1}{4}$ there are 2 real roots

For $a = -\dfrac{1}{4}$ there is 1 root

For $a < -\dfrac{1}{4}$ there are no real roots



(b) This is now essentially visually a hyperbola. We can see the following 5 cases:

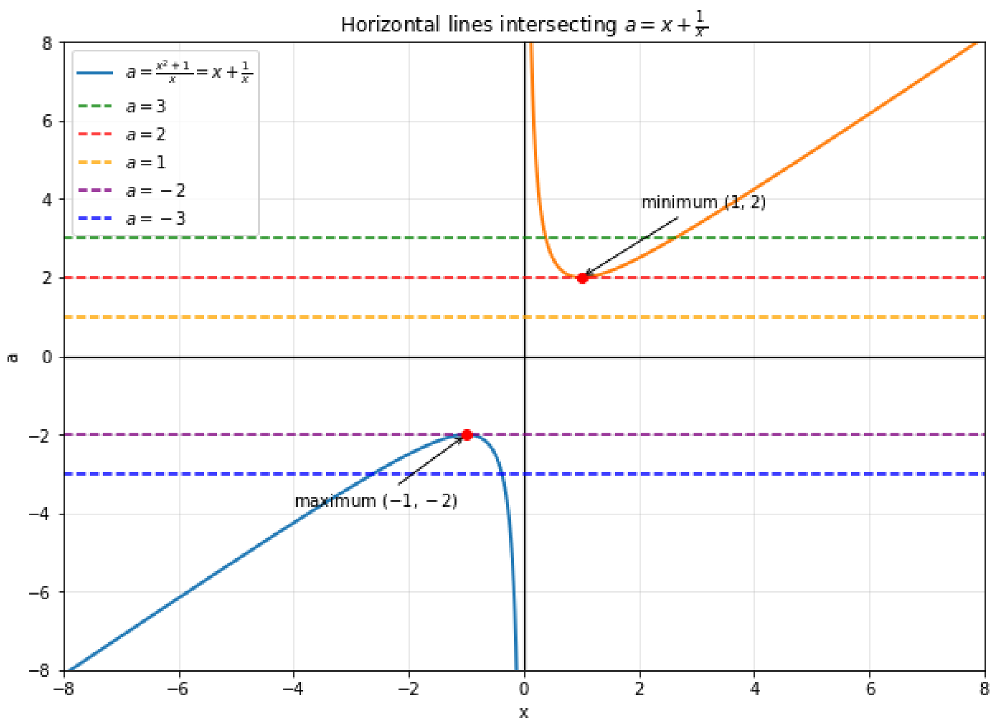
For $a > 2$ there are 2 real positive roots

For $a = -2$ there is 1 root which is $x = 1$

For $-2 < a < 2$ there are no real roots

For $a = -2$ there is again 1 root which is $x = -1$

For $a < -2$ there are 2 real negative roots



57 The graph of a quadratic polynomial

Problem 257.

This is an important chapter especially when we move into subjects like calculus. For this specific question we can just translate the X and Y axis back by shifting the origin. This is basically the same technique the authors have used in this chapter.

What we do is that if we have something like $y + k$ we need to get new y_{new} by shifting it to $y_{new} + k = 0$. Similarly for x axis.

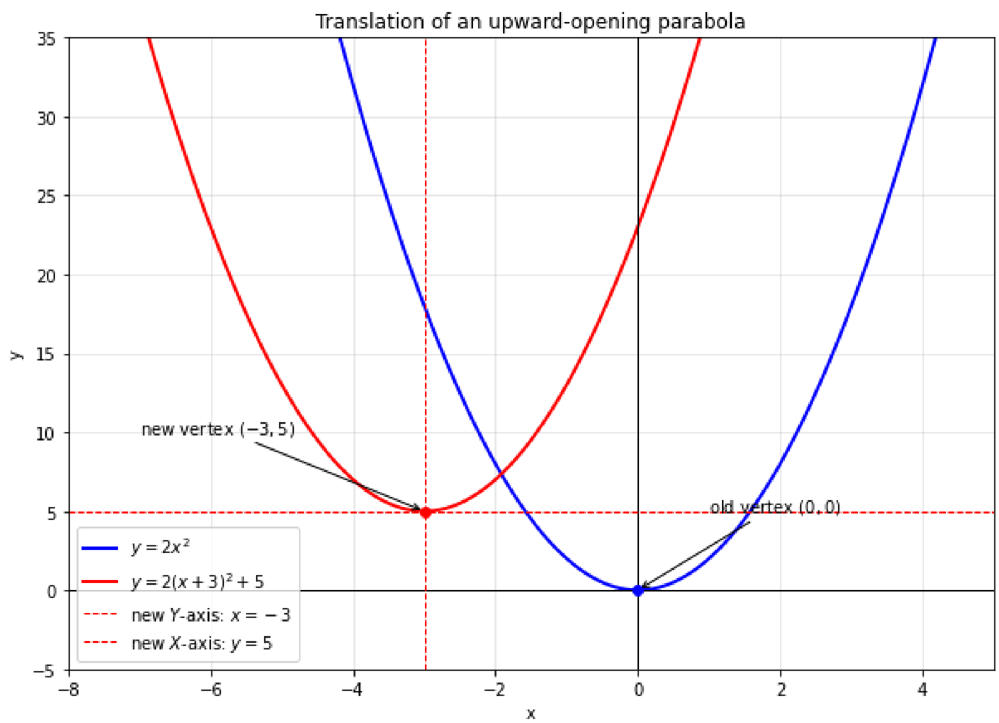
$$y = a(x+m)^2 + n$$
$$y - n = a(x+m)^2$$

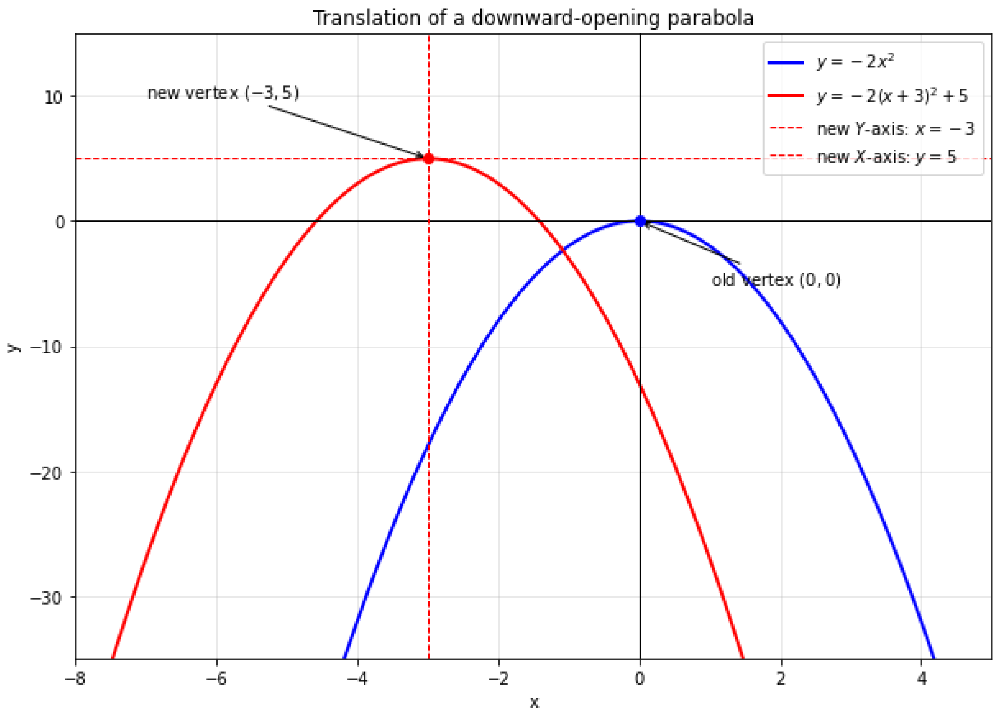
We can see how to get new Origin after translation.

$$y_{new} - n = 0$$
$$y_{new} = n$$
$$x_{new} + m = 0$$
$$x_{new} = -m$$

Now we have the equation as $y_{new} = ax_{new}^2$, the transposed origin is at $(-m, n)$ which is the bottom point of the parabola if a were positive and top of parabola if a were negative.

See the plots of the shifting below for $a > 0$ and $a < 0$ respectively.





Problem 258.

This is solved in the book and yes ordering is certainly important. Let me try and visually represent it with the example given $y = x^2$.

Let us take a concrete example for a, m, and n to keep things simple such as $a = 2$, $m = 3$, and $n = 5$.

So we initially have $y = x^2$ and we will do the steps in order first and then in the sequence the problem asks us.

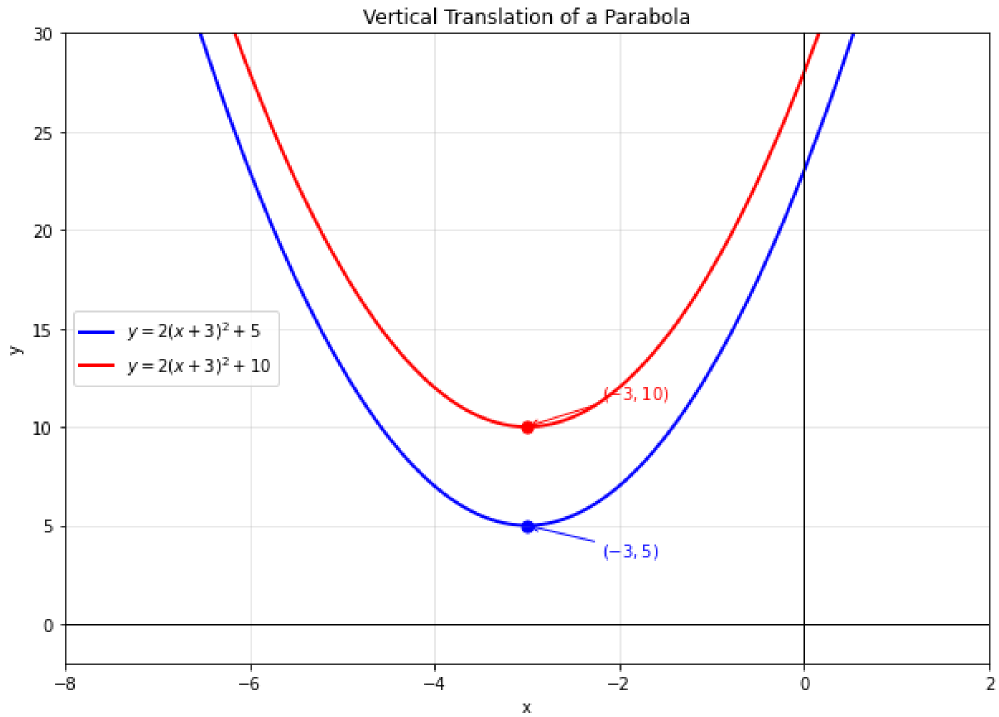
Ordered Steps:

$$y = x^2$$
$$y = 2x^2$$
$$y = 2(x+3)^2$$
$$y = 2(x+3)^2 + 5$$

Now random ordering as per the problem given:

$$y = x^2$$
$$y = x^2 + n$$
$$y = (x+3)^2 + 5$$
$$y = 2\left((x+3)^2 + 5\right)$$
$$y = 2(x+3)^2 + 10$$

We can see both the plots now are different. The plot of them is below and in this instance shows the vertical translation of the curves.



Problem 259.

We can go through all the 6 cases and see what graph we end up with (6 cases because of 3!).

$abc \Rrightarrow y = a(x+m)^2 + n$

$acb \Rrightarrow y = a(x+m)^2 + n$

$bac \Rrightarrow y = a(x+m)^2 + n$

$bca \Rrightarrow y = a[(x+m)^2 + n] = a(x+m)^2 + an$

$cab \Rrightarrow y = a[(x+m)^2 + n] = a(x+m)^2 + an$

$cba \Rrightarrow y = a[(x+m)^2 + n] = a(x+m)^2 + an$

3 of them end up with the same and 3 others with a different curve. If we notice the only difference is in the Y axis as such.

Problem 260.

This is solved in the book but the solution via the means of calculus is pretty elegant for this one. Anyways let us try and avoid calculus.

<u>a term</u>
If $a > 0$ then the parabola opens upward, for $a < 0$ it opens downwards, and for $a = 0$ the curve basically becomes a straight line $y = bx + c$.

<u>b term</u>
This is actually the curvature of the curve and without calculus its unintuitive. But as the authors say the sign of $\dfrac{b}{a}$ is determined by the vertex's x co-ordinate. The left half corresponds to the positive value of this term and the right half to the negative value.

<u>c term</u>
c is fairly simple as we can imagine the transposed Y axis (see problem 257 for plots and explanation).

58 Quadratic inequalities

Problem 261.

This is a very interesting topic but the book covers the very basics.

$$x^2 - 3x + 2 < 0$$
$$(x-1)(x-2) < 0$$

First imagine the graph of this curve $x^2 - 3x + 2$. The points where it intersects on the X axis are the roots which we already know since we have factored the polynomial. The roots i.e. the points where it touches the X axis are $x = 1$ and $x = 2$. We also know that the coefficient of the x^2 term is positive so the curve is opening upwards.

We used to call these critical points. Critical points include the points where the curve intersects or touches the axes, and the maximum and the minimum points also. Anyways we notice patterns when we transition from one critical point to another. So let us start at $+\infty$ and move towards $+2$, we notice that both $(x-1)$ and $(x-2)$ will be positive. The moment the the x value changes to 2 the curve intersects the X axis and then $(x-2)$ term becomes negative whereas the $(x-1)$ remains positive meaning the product is negative, so y is negative and below the X axis. Then the x value keep reducing towards $-\infty$ and when it touches 1 it again intersects the X axis and after that both $(x-1)$ and $(x-2)$ become negative and their product is positive. This is clearly visible from the curve at the end of this solution.

A beautiful way is to change signs on the number line and the authors have demonstrated that in the solution.

I will take a slight diversion and explain the calculus way of things too. Given $f(x) = x^2 - 3x + 2$. At $x = 0$ the function $f(x) = 2$. The maxima or minima is got by differentiating the function.

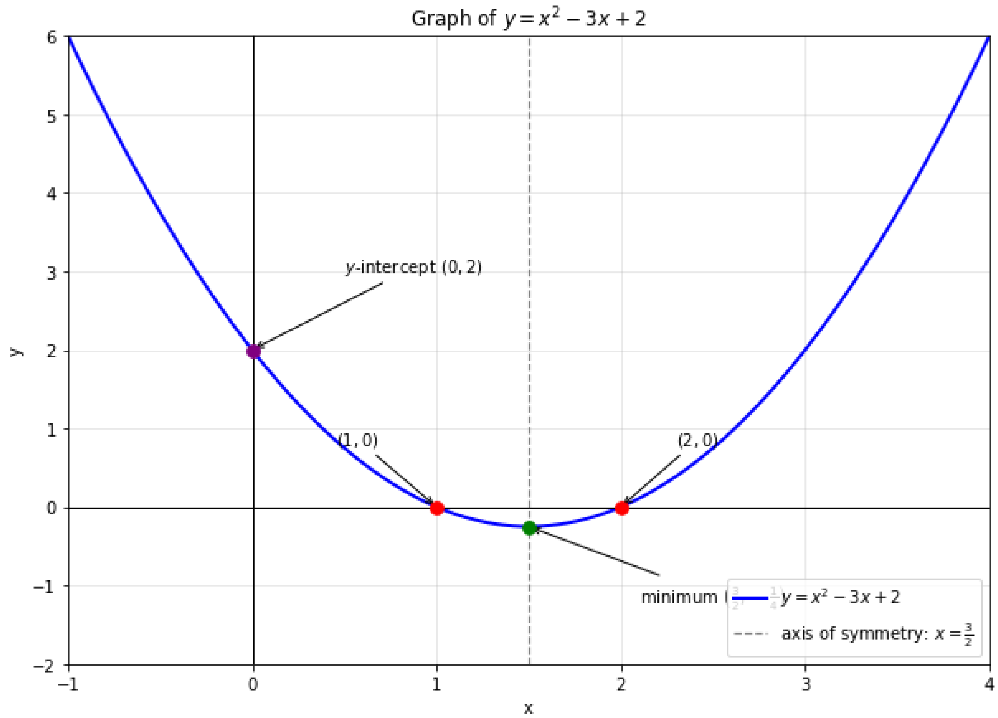
$$f'(x) = \frac{dy}{dx} = 2x - 3$$

$$2x - 3 = 0$$

$$x = \frac{3}{2}$$

So we have 4 critical points now the two roots and the intersection on Y axis as well as the maxima or minima. This greatly helps plot the curve. Double differentiation will also help get the curvature.

Here is the curve below.



59 Maximum and minimum values of a quadratic polynomial

Problem 262.

This is again a calculus problem. Rather it is very easy to solve using differentiation. Let the number be x so the product is $x \cdot (x-1)$. This is a polynomial curve in this case a parabola again with roots at $x = 0$ and $x = 1$, also notice that the coefficient of x^2 is negative so that the parabola opens downwards. So we know the maximum value is at the vertex which is in between the roots and that is $x = \dfrac{1}{2}$ Therefore the maximum product is $\dfrac{1}{2} \times \dfrac{1}{2} = \dfrac{1}{4}$.

The other way is to again differentiate the function and find maxima/minima.

Problem 263.

This is a fact I just taught my daughter today morning. It was mentioned in her workbook that for a given perimeter the rectilinear shape which has the largest area is the square. We used a piece of rope to try and construct a few examples. The challenge initially was to explain what rectilinear shapes are. *Recti* comes from Latin meaning straight or upright and *linear* again comes from Latin meaning simply line. So any shape made of straight lines be it a square, rectangle, some kind of L shape, a hexagon or a dodecagon etc. are all rectilinear shapes. With this aside let us prove this.

Let there be a rectangle with length l and breadth or width b.

$Perimeter = p = 2(l + b)$

$Area = a = lb$

Let us substitute value of l from the perimeter equation into the area since we know p is a constant here. We get:

$$a = \left(\frac{p}{2} - b\right) b$$
$$a = \frac{pb - 2b^2}{2}$$
$$a = -b^2 + \frac{p}{2}b$$

We have a quadratic equation above. We can see the the parabola formed opens downwards. We also see at $a = 0$ the parabola touches the horizontal axis at $b = 0$ and $b = \frac{p}{2}$. The middle point from 0 to $\frac{p}{2}$ will be the vertex and the highest point of the parabola. The middle point is $\frac{p}{4}$. So we have $l = \frac{p}{4}$ for the figure with maximum area.

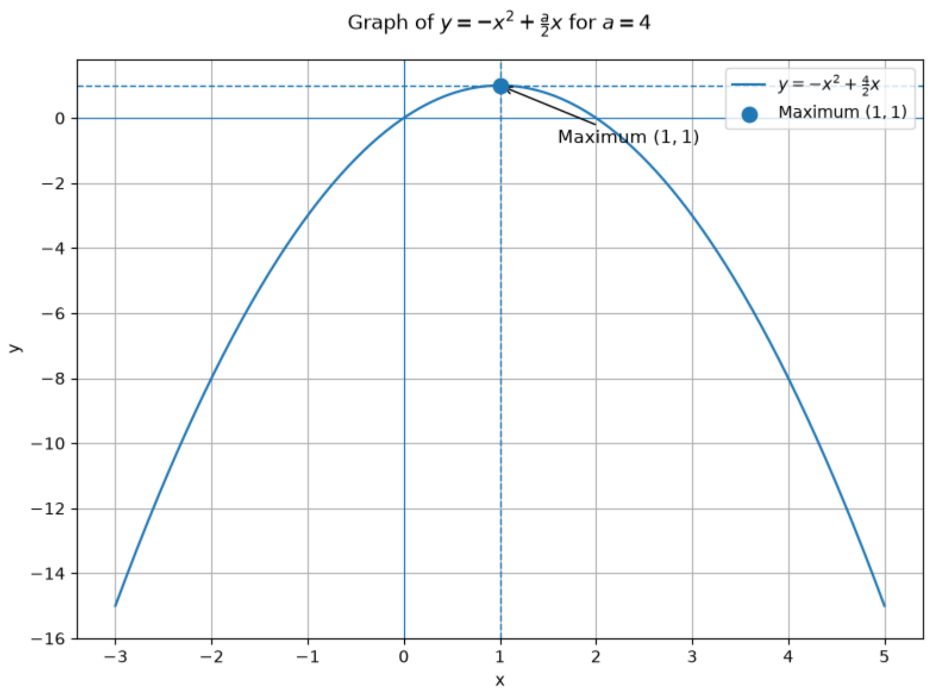
$$p = 2(l + b)$$
$$p = 2(\frac{p}{4} + b)$$
$$b = \frac{p}{2} - \frac{p}{4}$$
$$b = \frac{p}{4}$$

As we can see both length and breadth of this rectangle is equal to $\frac{p}{4}$, hence it is a square.

If we were to use calculus then we would find the maxima of this curve easily.

$$a = -b^2 + \frac{p}{2}b$$
$$\frac{da}{db} = -2b + \frac{p}{2} = 0$$
$$2b = \frac{p}{2}$$
$$b = \frac{p}{4}$$

From here on we can we do the same substitution to arrive at the proof. The graph for an instance of this problem looks like this



Problem 264.

This should be similar to the proof above but I would derive the proof without using the result of the preceding problem.

Let the rectangle have its length as l and breadth/width as b. Since we are given the area let us call it a and the perimeter which we have to minimize as p.

$$
\begin{aligned}
p &= 2(l + b) \\
a &= lb \\
l &= \frac{a}{b}
\end{aligned}
$$

$$
\begin{aligned}
p &= 2\left(\frac{a}{b} + b\right) \\
2b^2 - pb + 2a &= 0
\end{aligned}
$$

Now we have a quadratic equation with the variable b and we have to minimize p here. Let us look at the discriminant of this quadratic.

$$
D = \sqrt{p^2 - 16a}
$$

For the equation to have real roots (the number under the square root in the discriminant should not be a negative number) we need to have $D >= 0$.

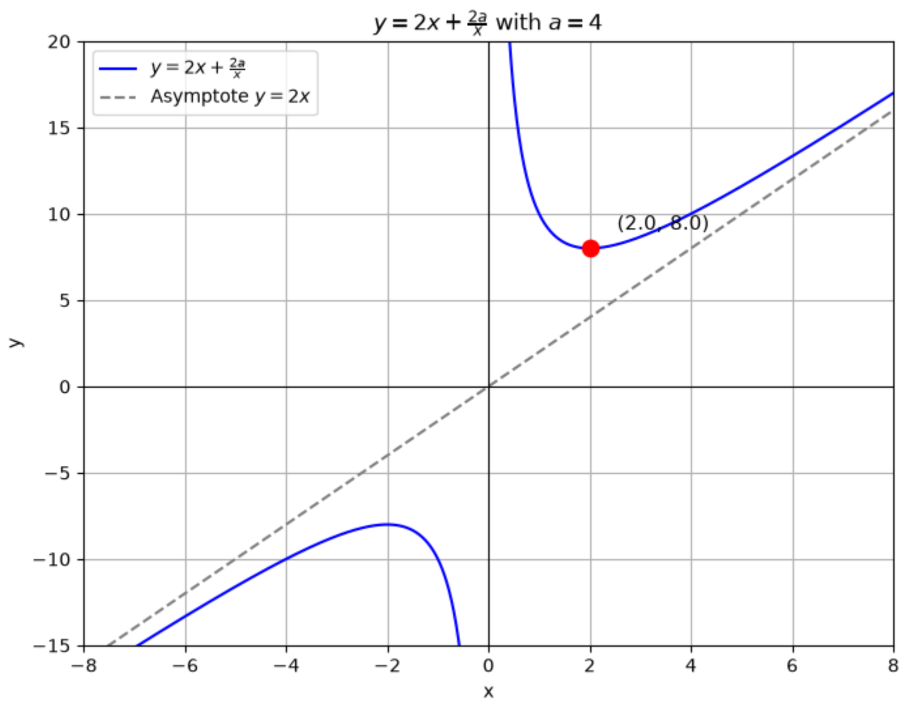
$$
\begin{gathered}
\sqrt{p^2 - 16a} \geq 0 \\
p^2 \geq 16a \\
p \geq 4\sqrt{a} \\
\text{or} \\
p \leq -4\sqrt{a}
\end{gathered}
$$

Now p cannot be negative so we have the condition as $p \geq 4\sqrt{a}$. So the minimum perimeter will be $p = 4\sqrt{a}$. Now we can put this back in the quadratic equation.

$$
\begin{aligned}
2b^2 - 4\sqrt{a}b + 2a &= 0 \\
b^2 - 2\sqrt{a}b + a &= 0 \\
b_{1,2} &= \frac{2\sqrt{a} \pm \sqrt{4a - 4a}}{2} \\
b &= \sqrt{a}
\end{aligned}
$$

If $b = \sqrt{a}$ then $l = \dfrac{a}{b} = \dfrac{a}{\sqrt{a}} = \sqrt{a}$. Thus $l = b = \sqrt{a}$ meaning the shape is a square. Hence proved.

This again can be solved using calculus. I will leave that here. But plot the curve with an example since the plot generated is that of a hyperbola and we can observe the curvature, positive solution and the minimum points on the curve.



Problem 265.

This is exactly the problem we solved previously, so will directly go to the calculations.

Let c be the value of the given expression then we have

$$x + \frac{2}{x} = c$$
$$x^2 - cx + 2 = 0$$

Looking at the discriminant now which should not be negative

$$D = \sqrt{c^2 - 8}$$

Now this discriminant has to non negative

$$D \geq 0$$
$$\sqrt{c^2 - 8} \geq 0$$
$$c \geq 2\sqrt{2}$$
or
$$c \leq -2\sqrt{2}$$

We look at the minimum value where x is positive which is at $c = 2\sqrt{2}$

60 Biquadratic equations

Problem 266.

Since both the x terms are even powers we can substitute $x^2 = y$ and then reduce this biquadratic equation to a quadratic equation.

$$\begin{gathered}
x^4 - 3x^2 + 2 = 0 \\
y^2 - 3y + 2 = 0 \\
(y-1)(y-2) = 0
\end{gathered}$$

Now let us put back the value of x.

$$\begin{gathered}
x^2 - 1 = 0 \\
x = \pm 1 \\
\text{and} \\
x^2 - 2 = 0 \\
x = \pm\sqrt{2}
\end{gathered}$$

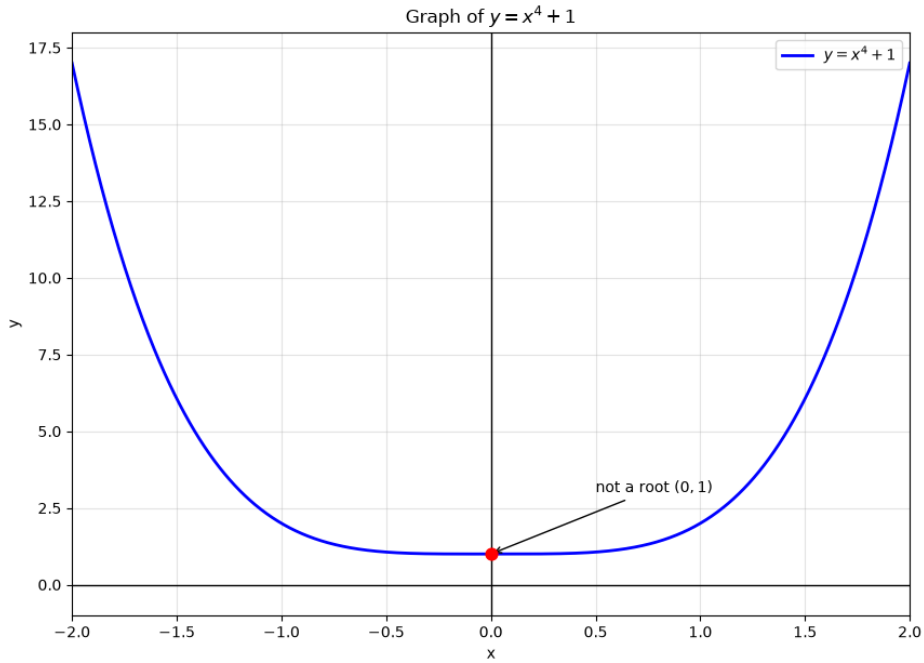
Therefore $x = \pm 1, \pm\sqrt{2}$.

Problem 267.

This is a good set of problems and I will draw the plots of every example.

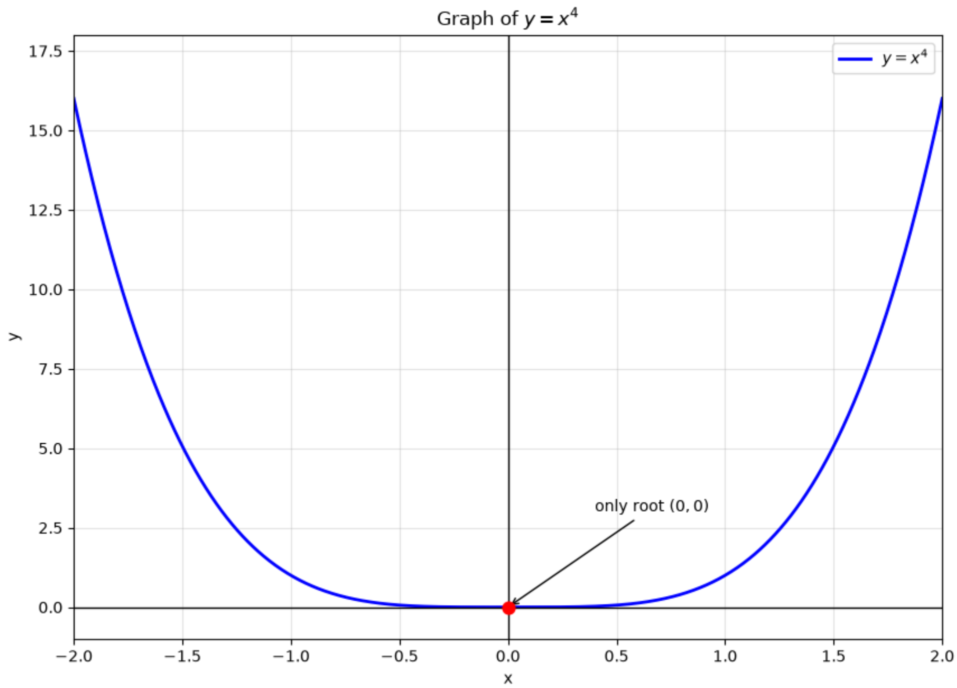
(a) Having no solution the authors imply no real solutions. This has only complex roots.

$$x^4 + 1 = 0$$



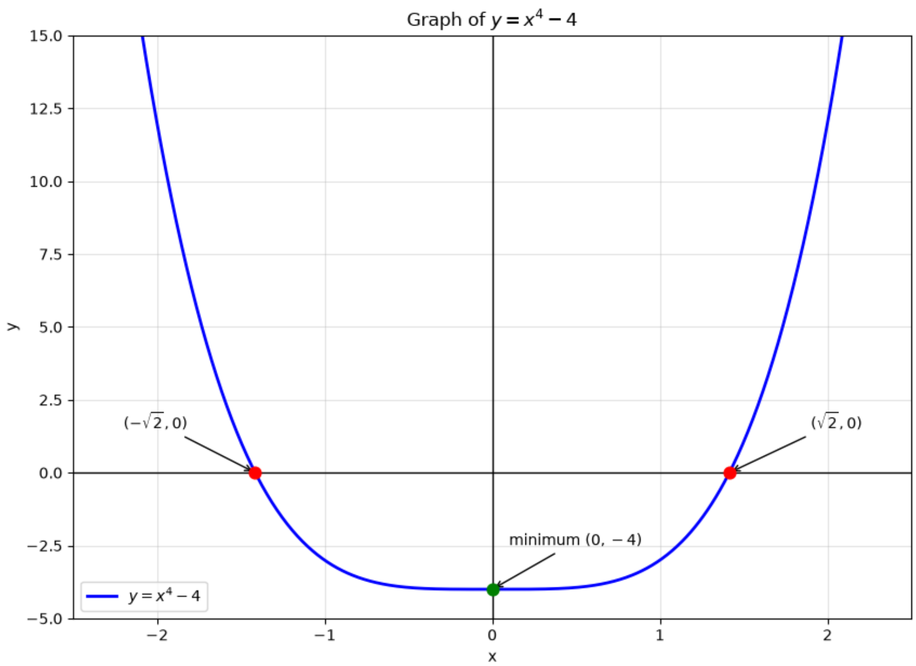
(b) So all roots are same. The only root is 0.

$$x^4 = 0$$



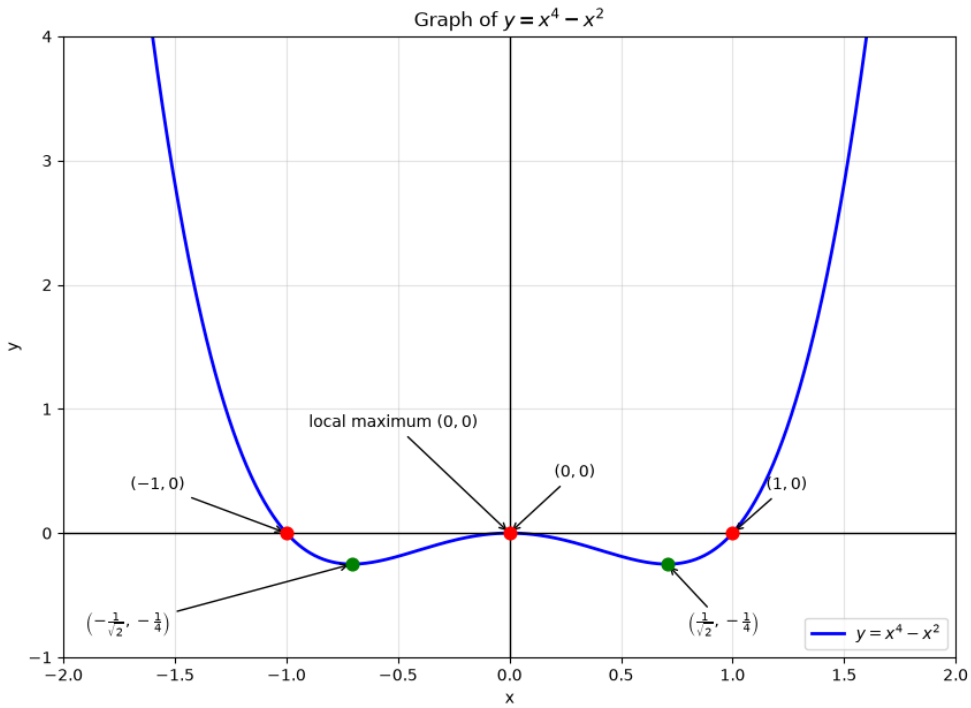
(c) Only two roots which are $\pm\sqrt{2}$ in this example.

$$x^4 - 4 = 0$$



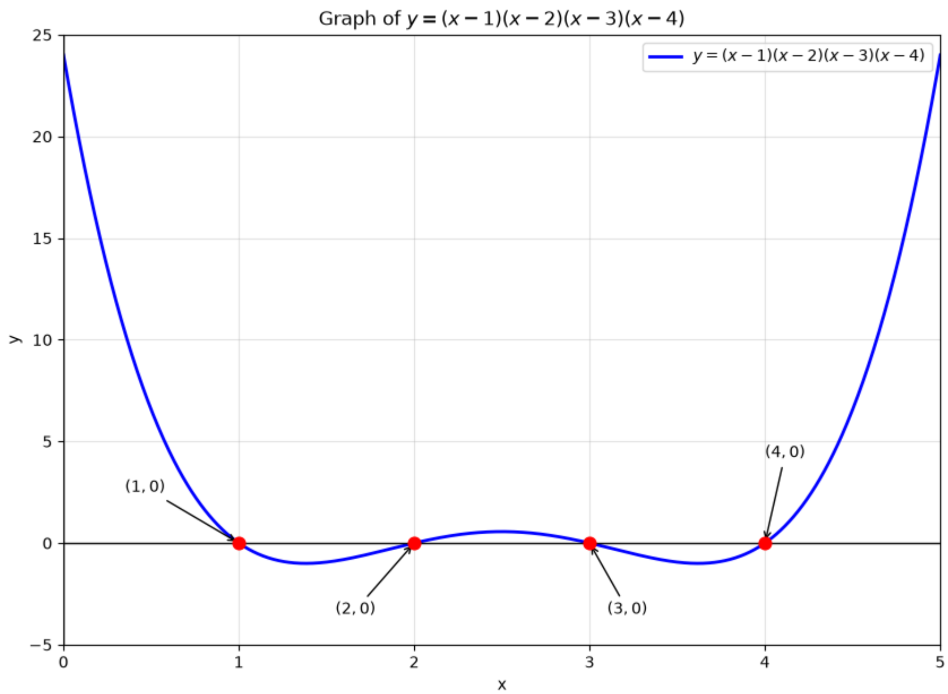
(d) In this case one root should be 0 and the other two will be conjugates, in this example ± 1.

$$x^4 - x^2 = 0$$



(e) Four roots is fairly easy to construct which is of the form $(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)$. In this example the roots are 1, 2, 3, and 4.

$$(x-1)(x-2)(x-3)(x-4) = 0$$



(f) It is not possible to have 5 roots for an equation which has a power of 4.

Problem 268.

If we substitute $t = x^3$ we will get a quadratic equation which will give us three solutions (0 roots, 1 or 2 roots). This is so because for $t = 0$ x will also be 0, so that's one solution. Now $x = \sqrt[3]{t}$ so for only positive values of t will we get a solution, thus two positive real roots of t will give two x roots. The last case is when there are no real roots itself. The authors probably mean that if $a = b = c = 0$ then any value of x can satisfy this 6 degree equation, therefore they say infinitely many solutions are possible.

Problem 269.

Same as the last question we can substitute $t = x^4$ we again get a quadratic but in this case we end up with $x = \sqrt[4]{t}$ and hence we can have the possibility of 0, 1, 2, 3, and 4 roots. Also if $a = b = c = 0$ then they are infinitely many solutions like the earlier problem.

61 Symmetric equations

Problem 270.

This problem is solved in the book. It is a challenging problem.

First set of observations, $x = 0$ or $x = -1$ will not be solutions to this equation. All the coefficients are positive meaning plugging in a positive integer will not give us a solution. We need some other way to think about it.

But there is one pattern we see that coefficients are "2 - 7 - 4 - 7 - 2", they probably can be regrouped. The book now has a typo when it says dividing by x whereas it should be dividing by x^2.

$$
\begin{gathered}
2x^4 + 7x^3 + 4x^2 + 7x + 2 = 0 \\
2x^2 + 7x + 4 + \frac{7}{x} + \frac{2}{x^2} = 0 \\
2x^2 + \frac{2}{x^2} + 7x + \frac{7}{x} + 4 = 0 \\
2\left(x^2 + \frac{1}{x^2}\right) + 7\left(x + \frac{1}{x}\right) + 4 = 0
\end{gathered}
$$

We also know that

$$
\begin{gathered}
\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 \\
x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2
\end{gathered}
$$

Let $y = \left(x + \frac{1}{x}\right)$ and putting this back in the rearranged equation earlier.

$$
\begin{gathered}
2\left(x^2 + \frac{1}{x^2}\right) + 7\left(x + \frac{1}{x}\right) + 4 = 0 \\
2(y^2 - 2) + 7y + 4 = 0 \\
2y^2 + 7y = 0 \\
y(2y + 7) = 0 \\
y_{1,2} = 0, -\frac{7}{2}
\end{gathered}
$$

Now solving

$$\left(x + \frac{1}{x}\right) = 0$$
$$x^2 + 1 = 0$$
$$x = \sqrt{-1}$$

This does not give any real roots for x. Solving for the other value of y.

$$\left(x + \frac{1}{x}\right) = -\frac{7}{2}$$
$$2x^2 + 7x + 2 = 0$$
$$x_{1,2} = \frac{-7 \pm \sqrt{33}}{4}$$

62 How to confuse students on an exam

No problems in this chapter.

63 Roots

Problem 271.

We can make comparisons of the 10^{th} power like this

$$x = \sqrt[10]{2}$$
$$x^{10} = 2$$

And for 1.2

$$y = 1.2$$
$$y^{10} = (1.2)^{10}$$

Thus $y^{10} > x^{10}$ hence $y > x$ and $1.2 > \sqrt[10]{2}$.

We confirm this with python code below.

```
# Problem 271

import math

x = math.pow(2, 1 / 10)
y = 1.2

print(x)

>> 1.0717734625362931
```

Problem 272.

This is a nice question. We can think of it as

$$x = \sqrt[7]{0.999}$$
$$x = \sqrt[7]{(1-x)^7}$$

Now going back to the binomial expansion of $(1-x)^7$ and using Pascal's triangle we have.

$$0.999 = (1-x)^7 = 1 - 7x + 21x^2 - 35x^3 + 35x^4 - 21x^5 + 7x^6 - x^7$$
$$0.001 = 7x - 21x^2 + 35x^3 - 35x^4 + 21x^5 - 7x^6 + x^7$$

Now we can observe that at a certain value of x the $7x$ term will be almost equal to 0.001 but above that the rest of the terms as well as $7x$ itself will make the total value above 0.001. Thus the value can be computed to be $x \approx \dfrac{0.001}{7} \approx 0.000143$

The first 3 decimals will be 0.999 itself. Rest of the terms of the binomial expansion will contribute terms which will have a decimal form of something like 0.00000525 and smaller. Thus no impact from these terms.

We confirm this with python code.

```
# Problem 272

import math

x = math.pow(0.999, 1 / 7)

print(x)

>> 0.9998570815947251
```

Problem 273.

Again a nice question. Let us do some manipulations.

$$
\begin{aligned}
x &= \sqrt{2} \\
x^2 &= 2 \\
y &= \sqrt[3]{3} \\
y^3 &= 3
\end{aligned}
$$

We observe that x is raised to 2 and y is raised to 3. To compare them let us raise both to such a power that they are comparable. So the minimum power in which both will be same is the Least Common Multiple of 2 and 3 which is 6. So squaring for y and cubing for x now.

$$
\begin{aligned}
x^6 &= 2^3 \\
y^6 &= 3^2
\end{aligned}
$$

We can see now that $y > x$ that is $\sqrt[3]{3} > \sqrt{2}$. Let us verify with some python code.

```
# Problem 273

import math

x = math.pow(2, 1 / 2)
y = math.pow(3, 1 / 3)

print(f"x is {x} and y is {y}")

>> x is 1.4142135623730951 and y is 1.4422495703074083
```

Problem 274.

Let

$$
\begin{aligned}
x &= \sqrt[3]{3} \\
x^3 &= 3 \\
y &= \sqrt[4]{4} \\
y^4 &= 4
\end{aligned}
$$

Now the least common multiple of 3 and 4 is 12, so let us try to raise both x and y to the power of 12.

$$x^{12} = 3^{12}$$
$$y^{12} = 4^3 = 2^6$$

Thus $x > y$ which is $\sqrt[3]{3} > \sqrt[4]{4}$. Verifying this with python.

```
# Problem 274

import math

x = math.pow(3, 1 / 3)
y = math.pow(4, 1 / 4)

print(f"x is {x} and y is {y}")

>> x is 1.4422495703074083 and y is 1.4142135623730951
```

Problem 275.

Let

$$x = \sqrt{\sqrt{2}}$$
$$x^2 = \sqrt{2}$$
$$x^4 = 2$$
$$y = \sqrt[4]{2}$$
$$y^4 = 2$$

We see that $x = y$ hence $\sqrt{\sqrt{2}} = \sqrt[4]{2}$. Verifying this in python

```
# Problem 275

import math

x =math.pow(math.pow(2, 1 / 2), 1 / 2)
y = math.pow(2, 1 / 4)

print(f"x is {x} and y is {y}")

>> x is 1.189207115002721 and y is 1.189207115002721
```

Problem 276.

The definition given in this chapter is that a root of a number x is the number whose value is the power raised to that root and equal to x. Example a square root of 4 is the number whose square is equal to 4. The cube root of 9 is the number whose value when raised to 3 is equal to 9.

Therefore a one root of a number a is the value of the number equal to when it is raised by 1 equal to a in this case it is a itself. Here a is non negative. Answer is given in the book.

Problem 277.

The proof is given in the book and simply uses the multiplication of $(xy)^n = x^n \cdot y^n$ as the proof in a way. Let $x = \sqrt[n]{a}$ and $y = \sqrt[n]{b}$ then putting it back in the earlier equation we get.

$$(xy)^n = x^n \cdot y^n$$
$$(\sqrt[n]{a} \cdot \sqrt[n]{b})^n = (\sqrt[n]{a})^n \cdot (\sqrt[n]{b})^n = ab$$

Hence shown as required.

Problem 278.

Simply use the derivation from the last problem, here the numbers are a and $\frac{1}{b}$. But the authors mention that a and b are non negative. I think we should be saying a is non negative and b is positive (as b cannot be 0).

It simply becomes exactly like the previous problem and thus proved.

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$$
$$\left(\frac{\sqrt[n]{a}}{\sqrt[n]{b}}\right)^n = \frac{(\sqrt[n]{a})^n}{(\sqrt[n]{b})^n} = \frac{a}{b}$$

Problem 279.

Same as the last problem again. Here the only change is that $\sqrt[n]{1} = 1$ always. This is by the definition of the root given in the chapter. Using this we prove that

$$\sqrt[n]{\frac{1}{a}} = \frac{\sqrt[n]{1}}{\sqrt[n]{a}} = \frac{1}{\sqrt[n]{a}}$$

Problem 280.

This is solved in the book and is exactly a generalization of problem 277.

$$\sqrt[n]{abc} = \sqrt[n]{(ab)c} = \sqrt[n]{ab} \cdot \sqrt[n]{c} = \sqrt[n]{a} \cdot \sqrt[n]{b} \cdot \sqrt[n]{c}$$

And again we expand the ab n^{th} root. This can be generalized to any number of terms under the root, it does not matter.

Problem 281.

This gets used a lot in problem solving therefore nice to prove this. Use the proof from the last problem.

$$\sqrt[n]{a^m} = \sqrt[n]{\underbrace{a \cdot a \cdot a ... a}_{\text{m times}}}$$

Now using the relationship $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$

$$\sqrt[n]{a^m} = \sqrt[n]{\underbrace{a \cdot a \cdot a ... a}_{\text{m times}}}$$

$$\sqrt[n]{a^m} = \underbrace{\sqrt[n]{a} \cdot \sqrt[n]{a} \cdot \sqrt[n]{a} ... \sqrt[n]{a}}_{\text{m times}}$$

$$\sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Problem 282.

The book says there is a flaw. Not really. The solution is getting more generalized in this problem over the last one. The authors prove it for a negative exponent and it holds true. Assume third root of a square of a fraction.

$$\sqrt[3]{a^{-2}} = \sqrt[3]{\frac{1}{a^2}} = \frac{1}{\sqrt[3]{a^2}} = \frac{1}{(\sqrt[3]{a})^2} = (\sqrt[3]{a})^{-2}$$

Problem 283.

This is solved in the book. As per the definition of roots in the chapter we have

$$\left(\sqrt[m]{\sqrt[n]{a}}\right)^{mn} = a$$

Now raising to the power of mn on both sides of the given expression.

$$\sqrt[mn]{a} = \sqrt[m]{\sqrt[n]{a}}$$

$$\left(\sqrt[mn]{a}\right)^{mn} = \left(\left(\sqrt[m]{\sqrt[n]{a}}\right)^{m}\right)^{n} = \left(\sqrt[n]{a}\right)^{n} = a$$

Problem 284.

Using the identity proven from the last problem we can prove this.

$$\sqrt[mn]{a^n} = \sqrt[m]{a}$$

$$\sqrt[mn]{a^n} = \sqrt[m]{(\sqrt[n]{a})^n} = \left(\sqrt[m]{a}\right)$$

Problem 285.

Simple proof given all that we know about powers right now.

$$\sqrt[n]{a^n b} = a\sqrt[n]{b}$$

$$\sqrt[n]{a^n} \cdot \sqrt[n]{b} = a \cdot \sqrt[n]{b} = a\sqrt[n]{b}$$

64 Non-integer inequalities

Problem 286.

This is a broad question but I will try and cover every property (7 of them) from the last chapter. Listing them down

Property 1:

$$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

Property 2:

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

Property 3:

$$\sqrt[n]{abc} = \sqrt[n]{a} \cdot \sqrt[n]{b} \cdot \sqrt[n]{c}$$

Property 4:

$$\sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

Property 5:

$$\sqrt[mn]{a} = \sqrt[m]{\sqrt[n]{a}}$$

Property 6:

$$\sqrt[mn]{a^n} = \sqrt[m]{a}$$

Property 7:

$$\sqrt[n]{a^n b} = a\sqrt[n]{b}$$

Now we will substitute $\sqrt[m]{\ldots}$ with the power raised to $\frac{1}{m}$ and $\sqrt[n]{\ldots}$ with the power raised to $\frac{1}{n}$ and check all the 7 properties. The preceding text in this chapter has already shown the basic ones so not covering that. Also we will equate $p = \frac{1}{m}$ and $q = \frac{1}{n}$.

Property 1: Holds true

$$\begin{aligned}\sqrt[n]{ab} &= a^{\frac{1}{n}} b^{\frac{1}{n}}\\ &= a^p b^p\\ &= (ab)^p\end{aligned}$$

Property 2: Holds true

$$\begin{aligned}\sqrt[n]{\frac{a}{b}} &= \frac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}}\\ &= \frac{a^p}{b^p}\\ &= \left(\frac{a}{b}\right)^p\end{aligned}$$

Property 3: Holds true

$$\begin{aligned}\sqrt[n]{abc} &= a^p \cdot b^p \cdot c^p\\ &= (abc)^p\end{aligned}$$

Property 4: Holds true

$$\begin{aligned}\sqrt[n]{a^m} &= (a^m)^p \\ &= (a^p)^m\end{aligned}$$

Property 5: Holds true

$$\begin{aligned}\sqrt[mn]{a} &= a^{pq} \\ &= (a^p)^q\end{aligned}$$

Property 6: Holds true

$$\begin{aligned}\sqrt[mn]{a^n} &= ((a^n)^{\frac{1}{n}})^q \\ &= a^q\end{aligned}$$

Property 7: Holds true

$$\begin{aligned}\sqrt[n]{a^n b} &= (a^n)^{\frac{1}{n}} \cdot b^p \\ &= ab^p\end{aligned}$$

Problem 287.

The solution is given in the book. The printing looks a bit tricky in the book.

$$\begin{aligned}(\sqrt[15]{a})^{10} &= (\sqrt[3\times 5]{a})^{2\times 5} \\ &= \left(\left(\sqrt[5]{\sqrt[3]{a}}\right)^5\right)^2 \\ &= \left(\sqrt[3]{a}\right)^2\end{aligned}$$

Problem 288.

We just proved this in the previous problem. But we can generalize the proof a bit.

Let m be a number which can be decomposed into a product of two integers $m = kp$ and n be a number which is decomposed into $n = kq$. Here k is the greatest common divisor of m and n, therefore p and q are relatively prime (co-prime).

$$
\begin{aligned}
(\sqrt[n]{a})^m &= a^{\frac{m}{n}} \\
&= (\sqrt[k\times q]{a})^{k\times p} \\
&= \left(\left(\sqrt[k]{\sqrt[q]{a}}\right)^k\right)^p \\
&= \left(\sqrt[q]{a}\right)^p \\
&= a^{\frac{p}{q}}
\end{aligned}
$$

Thus we show $a^{\frac{m}{n}} = a^{\frac{p}{q}}$.

Problem 289.

Not sure why the authors have proved it the way they have done.

$$a^p \cdot a^q = \underbrace{a \cdot a \cdot a ...}_{\text{p times}} \times \underbrace{a \cdot a \cdot a ...}_{\text{q times}}$$

$$= \underbrace{a \cdot a \cdot a ...}_{\text{p + q times}}$$

$$= a^{p+q}$$

Problem 290.

We can do the following using the previous proven properties.

$$(ab)^{\frac{m}{n}} = (a^m \times b^m)^{\frac{1}{n}} = (a^m)^{\frac{1}{n}} \times (b^m)^{\frac{1}{n}} = a^{\frac{m}{n}} \cdot b^{\frac{m}{n}}$$

Problem 291.

This is solved in detail in the book.

Without replicating the solution here the broad idea is to show p as a rational number of form $\frac{m}{n}$ as one case and then do the same with q. The proof therefore elegantly follows from what we have done earlier in this very chapter.

Problem 292.

Let us first consider the simple case of $a = 1$ although that is not asked in the question. For the this the solution will always be $1^p = 1$. Now for $a > 1$ the value of $p \, \epsilon \, (-\infty, +\infty)$ meaning p can be negative, zero or positive. In fact this holds true irrespective of the value of a.

Now let

$$p = \frac{m}{k}, q = \frac{n}{k},$$

where $k > 0$ and $m, n \in \mathbb{Z}$. Since $p < q$, we have

$$m < n,$$

so that

$$n - m > 0.$$

Now,

$$a^q = a^{n/k} = a^{m/k} a^{(n-m)/k} = a^p a^{(n-m)/k}.$$

Case 1. Suppose $a > 1$.

Since $n - m > 0$,

$$a^{(n-m)/k} > 1.$$

Hence,

$$a^q = a^p a^{(n-m)/k} > a^p.$$

Therefore,

$$p < q \implies a^p < a^q,$$

so a^p is an increasing function of p.

Case 2. Suppose $0 < a < 1$.

Since $n - m > 0$,

$$a^{(n-m)/k} < 1.$$

Hence,

$$a^q = a^p a^{(n-m)/k} < a^p.$$

Therefore,

$$p < q \implies a^p > a^q,$$

so a^p is a decreasing function of p.

Problem 293.

Nice problem.

$$\sqrt[\frac{1}{2}]{a} = (a)^{\frac{1}{\left(\frac{1}{2}\right)}} = a^2$$

Similarly,

$$\sqrt[-\frac{1}{2}]{a} = (a)^{\frac{1}{\left(-\frac{1}{2}\right)}} = a^{-2}$$

65 Proving inequalities

Problem 294.

Again a smart problem which can be proven verbally. The authors say that for the minimum value of the term in the series which is $\frac{1}{200}$ the sum of the series would be half if this was the only value. Similarly, for the maximum value the sum would be 1. Therefore, the the inequality will always hold true.

Problem 295.

Again a verbal solution.

To show that the series is greater than $\frac{1}{2}$ the terms are arranged as

$$\left(1-\frac{1}{2}\right)+\left(\frac{1}{3}-\frac{1}{4}\right)+...+\left(\frac{1}{199}-\frac{1}{200}\right)$$

The first term in the brackets is $\frac{1}{2}$ and rest of the terms are positive, so certainly this series is greater than $\frac{1}{2}$.

Again rearranging but slightly differently.

$$1-\left(\frac{1}{2}-\frac{1}{3}\right)-\left(\frac{1}{4}-\frac{1}{5}\right)-...-\left(\frac{1}{198}-\frac{1}{199}\right)-\frac{1}{200}$$

We subtract positive values from 1 and thus the series will smaller than 1.

Hence inequality is proven.

Problem 296.

Again nicely proven in the book. These kind of solutions require inductive intelligence and probably will come with a lot, I mean really a lot of practice.

$$= \frac{1}{101} + ... + \frac{1}{200}$$
$$= \left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{200}\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{100}\right)$$

The step here is to cancel out the common terms. So in a way the student has to be able to think one step backwards.

Then,

$$\left(1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{200}\right) - 2 \cdot \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + ... + \frac{1}{200}\right)$$

In this second step now we are building up to the fractional subtraction we need to do for the common denominators.

$$\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + ... + \frac{1}{199} - \frac{1}{200}\right)$$

Very good problem and requires practice to do such manipulations and be able to see forward and backward and mould the transformations.

Problem 297.

This is solved in the book.

To reason this out is fairly simple. We can imagine the kind of products we would get if we were to hypothetically multiply all the terms.

To see that play out let us consider a simpler version of this problem $(1.01)^2$ as Polya has taught us. When we expand this we get

$$1.01^2 = (1 + 0.01)(1 + 0.01)$$
$$1.01^2 = 1 + 0.01 + 0.01 + 0.01 \cdot 0.01$$

We observe that all the 1s multiply with each other to give 1. All the 0.01 multiply as 0.01^2. Rest of the terms also contribute to the sum.

Now let us look at the original term $(1.01)^{100}$. This can be written as a product of $(1 + 0.01)$ 100 times.

$$1.01^{100} = (1 + 0.01)(1 + 0.01)...(1 + 0.01)$$

Like our simplified problem of power 2 we can get 1s here by multiplying 100 1s. Similarly we can multiply all the 0.01s which is also 100 in number. Ignoring all the other terms for the moment which indeed will contribute to the value of this expression. We get

$$1.01^{100} = 1 \times 100 + 0.01 \times 100 + ...$$

$$1.01^{100} = 1 + 1 + ... = 2 + ...$$

Therefore we have proved just by reasoning that $(1.01)^{100} \geq 2$.

Problem 298.

This is solved elegantly in the book. Again this requires insight to manipulate given expressions.

The mental approach could be that if we can prove that 1 plus the remaining terms is less than 2 when the 'remaining' terms is less than 1. Also we notice that the denominators in the 'remaining' terms are squares. Next thing we can think of is terms cancelling out. Here is the crux.

$$\frac{1}{4} = \frac{1}{2^2} < \frac{1}{1 \cdot 2} = \frac{1}{1} - \frac{1}{2}$$

$$\frac{1}{9} = \frac{1}{3^2} < \frac{1}{2 \cdot 3} = \frac{1}{2} - \frac{1}{3}$$

$$\frac{1}{16} = \frac{1}{4^2} < \frac{1}{3 \cdot 4} = \frac{1}{3} - \frac{1}{4}$$

$$...$$

All the terms begin to cancel out in the expansion we have built here.

$$1 + \frac{1}{4} + \frac{1}{9} + ... + \frac{1}{100^2} < 1 + \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + ... + \left(\frac{1}{99} - \frac{1}{100}\right)$$

$$= 1 + 1 - \frac{1}{100} < 2$$

Problem 299.

This is a nice problem. My first instinct is binomial expansion. But there is a much shorter way to figure this out. Let us do binomial expansion method first.

$$1000^{2000} = 1000^{1000} \times 1000^{1000}$$

$$2000^{1000} = (1000 + 1000)^{1000}$$

Now we can think of $(1000+1000)^{1000}$ as $(a+b)^n$ and then expand. We get

$$\binom{1000}{0}1000^{1000}\cdot 1000^0+\binom{1000}{1}1000^{999}\cdot 1000+\binom{1000}{2}1000^{998}\cdot 1000^2+...+\binom{1000}{1000}1000^0\cdot 1000^{1000}$$

$$1000^{1000}\left(1+\binom{1000}{1}+\binom{1000}{2}+...+\binom{1000}{999}+\binom{1000}{1000}\right)$$

This term $\left(1+\binom{1000}{1}+\binom{1000}{2}+...+\binom{1000}{999}+\binom{1000}{1000}\right)$ is actually the same as $(1+1)^{1000}$ since we will be multiplying by 1 to all the coefficients. Therefore we know that

$$\left(1+\binom{1000}{1}+\binom{1000}{2}+...+\binom{1000}{999}+\binom{1000}{1000}\right)=2^{1000}$$

Therefore in the original expression we now have

$$1000^{1000}\left(1+\binom{1000}{1}+\binom{1000}{2}+...+\binom{1000}{999}+\binom{1000}{1000}\right)=1000^{1000}\times 2^{1000}$$

We can now say that $1000^{2000}>2000^{1000}$

The other way to do this is to simply take the 1000^{th} root of both these expressions. So we have

$$\sqrt[1000]{1000^{2000}}=\sqrt[1000]{1000^{1000}}\times\sqrt[1000]{1000^{1000}}=1000\times 1000=1000000=10^6$$

$$\sqrt[1000]{2000^{1000}}=2000$$

We can again say that $1000^{2000}>2000^{1000}$ because a million is much greater than 2000.

Problem 300.

We are given that

$$1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{1,000,000}<20$$

Grouping the terms as follows (basis the hint):

$$1+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+...$$

The groups are formed between consecutive powers of 2. Thus the k^{th} group has 2^{k-1} terms:

$$\frac{1}{2^{k-1}+1} + \frac{1}{2^{k-1}+2} + ... + \frac{1}{2^k}.$$

Every term in this group is less than

$$\frac{1}{2^{k-1}}$$

and hence the sum of the group is less than

$$2^{k-1} \cdot \frac{1}{2^{k-1}} = 1$$

Now

$$1,000,000 < 2^{20} = 1,048,576$$

Therefore all the terms up to $\dfrac{1}{1,000,000}$ are contained in the first 20 groups after the initial term 1.

$$1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{1,000,000} < 1 + 19 = 20$$

Thus

$$1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{1,000,000} < 20$$

For the second part, consider the same grouping:

$$1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + ...$$

In the k^{th} group, every term is at least

$$\frac{1}{2^k}$$

and there are 2^{k-1} terms. Therefore each group is at least

$$2^{k-1} \cdot \frac{1}{2^k} = \frac{1}{2}$$

In fact, except for the first group, the inequality is strict. Hence, after 40 such groups,

$$1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{2^{40}} > 1 + 40 \cdot \frac{1}{2} = 21$$

Therefore, for example, taking

$$n = 2^{40}$$

we obtain

$$1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{n} > 20$$

Hence such an n exists.

66 Arithmetic and geometric means

Problem 301.

This is nicely proved in the book.

Consider a point a and b on the X axis and also $a < b$.

Distance between the two points is $b - a$

Half of this distance is $\frac{b-a}{2}$

Midway from a to b will be $a + \frac{b-a}{2}$. This is equal to $\frac{a+b}{2}$.

Hence proved.

Problem 302.

An easy problem.

$$7 = \frac{1+a}{2}$$
$$a = 13$$

Problem 303.

Another easy one. You can now suspect Gelfand and Shen are setting you up for some interesting concept.

$$7 = \sqrt{1 \times a}$$
$$a = 49$$

Problem 304.

Here we have the meaning of the previous two problems now come up.

(a) Let side of the square be s then

$$4s = 2(a + b)$$
$$s = \frac{a+b}{2}$$

The side of the square is the average of the two sides of a rectangle given that their perimeter is the same. Average is the arithmetic mean.

(b) Again side of the square is s

$$s^2 = ab$$
$$s = \sqrt{ab}$$

The side of the square is the geometric mean of the two sides of a rectangle given that their area is the same.

Problem 305.

This is explained concisely in the book.

For an arithmetic progression:

$$a, a + d, a + 2d, a + 3d, a + 4d, ...$$

Every term in the arithmetic progression is the mean of its two adjacent numbers, thus it is called an arithmetic mean. Let us try an example.

The numbers $a + 2d$ and $a + 4d$, their average is

$$= \frac{a + 2d + a + 4d}{2}$$
$$= \frac{2a + 6d}{2}$$
$$= a + 3d$$

$a + 3d$ is the term between $a + 2d$ and $a + 4d$.

Similarly, for a geometric progression

$$a, ar, ar^2, ar^3, ar^4, ar^5, ...$$

Let us look at ar^3 and ar^5.

$$= \sqrt{ar^3 \times ar^5}$$
$$= \sqrt{a^2r^8} = ar^4$$

Thus ar^4 is the geometric mean of ar^3 and ar^5

67 The geometric mean does not exceed the arithmetic mean

Problem 306.

This is an easy proof in the book but this is an inequality which is super useful and used a lot in problem solving generally speaking.

We have to prove

$$\sqrt{ab} \leq \frac{a+b}{2}$$

Meaning Arithmetic Mean is greater than or equal to the Geometric Mean. Later we should come across Harmonic Mean too but let us look at that in the future.

Let us square both sides of the inequality.

$$ab \leq \left(\frac{a+b}{2}\right)^2$$

$$ab \leq \frac{(a+b)^2}{4}$$
$$4ab \leq a^2 + b^2 + 2ab$$
$$a^2 + b^2 - 2ab \geq 0$$
$$(a-b)^2 \geq 0$$

Since we know that a and b are non negative therefore $(a - b)^2$ is always non-negative. Hence we proved it.

Problem 307.

Let the numbers be a and b.

$$\sqrt{ab} = \frac{a+b}{2}$$
$$ab = \frac{(a+b)^2}{4}$$
$$(a-b)^2 = 0$$
$$a = b$$

This happens only when a and b are equal.

68 Problems about maximum and minimum

Problem 308.

(a) We use the inequality $A.M. \geq G.M.$ here the acronyms stand for arithmetic mean and geometric mean respectively. The A.M. for a and b is $\frac{c}{2}$. The maximum the G.M. can be is $\frac{c}{4}$. We have

$$\sqrt{ab} \leq \frac{a+b}{2} = \frac{c}{2}$$
$$ab \leq \frac{c^2}{4}$$

Thus the maximum value is $\dfrac{c^2}{4}$ and this happens when $a = b$.

(b) The minimum value is 0 since the numbers a and b are non negative even if one of them is 0 then the product is 0.

Problem 309.

Let the non negative numbers be a and b, we are given that

$$\sqrt{ab} = \sqrt{c}$$
$$ab = c$$

We are given that $c > 0$. We know that the minimum value of A.M. will be the maximum value of G.M. can be which is $\dfrac{(a+b)}{2}$. Therefore, this is the minimum value of $a + b$, this is when both the numbers are equal.

$$\sqrt{c} = \frac{a+b}{2}$$

$$a + b = \sqrt{2}c$$

The maximum value can be anything since it is not upper bound (while the other number can be close to 0).

Problem 310.

Let the length and breadth be l and b respectively. We know that $2(l+b) = 120$ meaning $l + b = 60$.

$$GM \leq AM$$

$$lb \leq \frac{l+b}{2}$$

$$lb \leq \frac{60}{2}$$

$$lb \leq 30$$

Therefore the maximum area can be $30m^2$.

Problem 311.

The solution with reasoning is elegant in the book. We imagine the quadrilateral to extend with a mirror image into the sea. So the 120 perimeter is essentially 240. Now it becomes exactly like the previous problem. To maximize the area we know it has to be a square. So the side of the square will be 60. Now back to the original 3 sides square (since we will not have a fence along the ocean). The longer side will be 60 and the other two sides will 30 each (half of 60 since we cut the structure into half).

The area is $60 \times 30 = 1800m^2$.

The calculus way of doing this is to consider the sides of a rectangle say x and y, so the given fence length is $2x + y = 120$. We have to maximize xy.

$$A = xy = x \times (120 - 2x)$$
$$A = 120x - 2x^2$$

Differentiating with respect to x and finding the solution by equating to 0.

$$\frac{dA}{dx} = 120 - 4x = 0$$
$$x = 30$$

So we have it again $x = 30$ and $y = 120 - 2x = 60$ and area is $xy = 30 \times 60 = 1800m^2$.

Problem 312.

This is exactly the same question as above. I would prefer to use the calculus way to do it but the authors outline the solution in the book. Let us call $ab = A$

$$ab = A = (3 - 2b) \times b$$
$$A = 3b - 2b^2$$

Now maximizing A.

$$\frac{dA}{db} = 3 - 4b$$
$$3 - 4b = 0$$
$$b = \frac{3}{4}$$

$$a = 3 - 2b = 3 - 2 \times \frac{3}{4}$$
$$a = \frac{3}{2}$$

So the product is $\frac{3}{2} \times \frac{3}{4} = \frac{9}{8}$.

69 Geometric illustrations

Problem 313.

Two ways to look at this problem.

First way:

We can see that each side of the leftover square is $a + b - b - b = a - b$, thus the leftover area is $(a - b)^2$.

Second way:

The total area is $(a + b)^2$. The area of the 4 smaller rectangles is $4ab$. The leftover area is

$$(a + b)^2 - 4ab$$
$$a^2 + b^2 + 2ab - 4ab$$
$$(a - b)^2$$

Both ways we get the same answer as $(a - b)^2$.

70 The arithmetic and geometric means of several numbers

Problem 314.

This solution can be easily derived with the relationship between AM and GM for several numbers. We know That

$$\sqrt[n]{a_1...a_n} \leq \frac{a_1 + ... + a_n}{n}$$

Given the condition that $a_1 + ... + a_n$ is less than equal to n then we have that

$$\frac{a_1 + ... + a_n}{n} \leq \frac{n}{n}$$

$$\frac{a_1 + ... + a_n}{n} \leq 1$$

So we have from the AM GM relationship that

$$\sqrt[n]{a_1...a_n} \leq 1$$
$$(\sqrt[n]{a_1...a_n})^n \leq 1^n$$
$$a_1...a_n \leq 1$$

Hence shown.

Problem 315.

We will use the same AM GM Inequality to solve this. We know that when the numbers are equal (as in the previous problem) the product gets maximized. Let

$$a_1 + a_2 + ... + a_n = S$$

for all a_i which is non negative

We know that

$$\frac{a_1 + a_2 + ... + a_n}{n} \geq \sqrt[n]{a_1 a_2 ... a_n}$$

We can substitute the value of $a_1 + a_2 + ... + a_n = S$

$$\frac{S}{n} \geq \sqrt[n]{a_1 a_2 ... a_n}$$

$$a_1 a_2 ... a_n \leq \left(\frac{S}{n}\right)^n$$

Equality in AM GM occurs if and only if

$$a_1 = a_2 = ... = a_n$$

Since their sum is S, this means

$$a_1 = a_2 = ... = a_n = \frac{S}{n}$$

Thus the product is maximized when all the numbers are equal, and the maximum product is

$$\left(\frac{S}{n}\right)^n$$

Problem 316.

This is very similar to the previous problem.

Let

$$a_1 a_2 ... a_n = P$$

where $a_i \geq 0$

By AM GM inequality we get,

$$\frac{a_1 + a_2 + ... + a_n}{n} \geq \sqrt[n]{a_1 a_2 ... a_n}$$

Since

$$a_1 a_2 ... a_n = P$$

we have

$$\frac{a_1 + a_2 + ... + a_n}{n} \geq \sqrt[n]{P}$$

Multiplying both sides by n

$$a_1 + a_2 + ... + a_n \geq n\sqrt[n]{P}$$

Equality in AM GM occurs if and only if

$$a_1 = a_2 = ... = a_n$$

Since their product is P, in the equality case,

$$a_1^n = P$$

and hence

$$a_1 = a_2 = ... = a_n = \sqrt[n]{P}$$

Therefore, for n nonnegative numbers with fixed product P

$$a_1 + a_2 + ... + a_n \geq n\sqrt[n]{P}$$

with equality if and only if

$$a_1 = a_2 = ... = a_n = \sqrt[n]{P}$$

Thus the sum is minimized when all the numbers are equal.

From the last two problems using AM GM inequality relationship we learn that:
1. Fixed Sum $\implies$ maximum Product at Equality
2. fixed Product $\implies$ minimum Sum at Equality

Problem 317.

This is solved algebraically in the book. The solution is on the lines of taking two numbers then finding out the arithmetic mean of theirs. By doing so the sum remains the same but the product gets maximized. Repeating this process for the other two numbers and then finally repeating this for a different pair again. It eventually lets us maintain a constant sum but keep maximizing the product till all the numbers are the same and is equal to the arithmetic mean of all four.

I will give the math transformations just after this paragraph but the authors allude to a proof by calculus. I am more interested to try that out.

Here is the algebraic proof.

Let the numbers be a, b, c, d. Taking the averages (arithmetic mean):

$$a, b, c, d$$

$$\frac{a+b}{2}, \frac{a+b}{2}, c, d$$

$$\frac{a+b}{2}, \frac{a+b}{2}, \frac{c+d}{2}. \frac{c+d}{2}$$

$$\frac{a+b+c+d}{4}, \frac{a+b}{2}, \frac{a+b+c+d}{4}, \frac{c+d}{2}$$

$$\frac{a+b+c+d}{4}, \frac{a+b+c+d}{4}, \frac{a+b+c+d}{4}, \frac{a+b+c+d}{4}$$

Let

$$S = \frac{a+b+c+d}{4}$$

Therefore we have

$$a \cdot b \cdot c \cdot d \leq S \cdot S \cdot S \cdot S$$

or

$$\sqrt[4]{abcd} \leq S$$

Now for the calculus based proof.

Let $a + b + c + d = S$, then $d = S - a - b - c$

So

$$f(a, b, c) = abc(S - a - b - c)$$

At an interior maximum,

$$\frac{\partial f}{\partial a} = 0, \frac{\partial f}{\partial b} = 0, \frac{\partial f}{\partial c} = 0$$

Now,

$$\frac{\partial f}{\partial a} = bc(S - 2a - b - c)$$

$$\frac{\partial f}{\partial b} = ac(S - a - 2b - c)$$

$$\frac{\partial f}{\partial c} = ab(S - a - b - 2c)$$

Hence,

$$\begin{aligned} S - 2a - b - c = 0 \\ S - a - 2b - c = 0 \\ S - a - b - 2c = 0 \end{aligned}$$

Solving these three linear equations we get,

$$a = b = c$$

Also,

$$d = S - a - b - c = a$$

Therefore,

$$a = b = c = d = \frac{S}{4}$$

Hence,

$$abcd \leq \left(\frac{S}{4}\right)^4 = \left(\frac{a+b+c+d}{4}\right)^4$$

Therefore,

$$\sqrt[4]{abcd} \leq \frac{a+b+c+d}{4}$$

Problem 318.

This is simply equating the last problems final result.

$$\sqrt[4]{abcd} \leq \frac{a+b+c+d}{4}$$

We equate it now by raising to the power of 4.

$$(\sqrt[4]{abcd})^4 = \left(\frac{a+b+c+d}{4}\right)^4$$

$$abcd = \left(\frac{a+b+c+d}{4}\right)^4$$

Assume if $a = b = c = d = x$ then the equation would change into

$$x \cdot x \cdot x \cdot x = \left(\frac{x+x+x+x}{4}\right)^4$$

$$x^4 = \left(\frac{4x}{4}\right)^4$$

$$x^4 = x^4$$

Thus we can show it now.

Problem 319.

This is just a cumbersome extension of the prior proof we have done. But the method outlined in the book is better to do here.

Let the numbers be $a_1, a_2, ..., a_8$. Taking pairs and averaging out we arrive at the following

$$\frac{a_1+a_2}{2}, \frac{a_3+a_4}{2}, \frac{a_5+a_6}{2}, \frac{a_7+a_8}{2}$$

Averaging again

$$\frac{a_1+a_2+a_5+a_6}{4}, \frac{a_3+a_4+a_7+a_8}{4}$$

Averaging again

$$\frac{a_1+a_2+a_3+a_4+a_5+a_6+a_7+a_8}{8}$$

Basically all 8 numbers are this. We have averaged them out and maximized the product.

Let that be equal to S.

$$S = \frac{a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 + a_8}{8}$$

Now plugging these into the AM GM expression

$$a_1 \cdot a_2 \cdot a_3 \cdot a_4 \cdot a_5 \cdot a_6 \cdot a_7 \cdot a_8 \leq S \cdot S \cdot S \cdot S \cdot S \cdot S \cdot S \cdot S$$
$$a_1 \cdot a_2 \cdot a_3 \cdot a_4 \cdot a_5 \cdot a_6 \cdot a_7 \cdot a_8 \leq S^8$$
$$\sqrt[8]{a_1 \cdot a_2 \cdot a_3 \cdot a_4 \cdot a_5 \cdot a_6 \cdot a_7 \cdot a_8} \leq S$$

Problem 320.

This is solved in the book in a creative way.

For an odd number of numbers the technique is to reduce the case to an even number case. This is done by taking another number which is the GM of the 3 numbers. Summarizing the approach given the book below:

$$a, b, c, \sqrt[3]{abc}$$

$$\sqrt[4]{abc\sqrt[3]{abc}} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

We know that

$$\sqrt[4]{abc\sqrt[3]{abc}} = \sqrt[4]{(\sqrt[3]{abc})^3\sqrt[3]{abc}} = \sqrt[4]{(\sqrt[3]{abc})^4} = \sqrt[3]{abc}$$

Going back to the earlier derivation

$$\sqrt[4]{abc\sqrt[3]{abc}} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

$$\sqrt[3]{abc} \leq \frac{a + b + c + \sqrt[3]{abc}}{4}$$

$$3\sqrt[3]{abc} \leq a + b + c$$
$$\sqrt[3]{abc} \leq \frac{a + b + c}{3}$$

This is what we intended to arrive at.

Problem 321.

As per Problem 317 where we proved the inequality for $n = 4$ we could extend that for an even number $n = 8$. So the following would hold true

$$\sqrt[8]{a_1a_2\cdots a_8} \leq \frac{a_1 + a_2 + \cdots + a_8}{8}$$

To prove for $n = 7$ we will use exactly the same approach we used in the last problem. Let the numbers be $a_1, a_2, \cdots, a_7$. Then the 8^{th} number will be $\sqrt[7]{a_1a_2\cdots a_7}$. Now the equality will hold true ie AM GM

$$\sqrt[8]{a_1a_2a_3\cdots a_7\sqrt[7]{a_1a_2\cdots a_7}} \leq \frac{a_1 + a_2 + \cdots + a_7 + \sqrt[7]{a_1a_2\cdots a_7}}{8}$$

Using the same manipulation as the one in the earlier problem

$$\sqrt[8]{a_1a_2a_3\cdots a_7\sqrt[7]{a_1a_2\cdots a_7}} = \sqrt[8]{(\sqrt[7]{a_1a_2\cdots a_7})^7\sqrt[7]{a_1a_2\cdots a_7}} = \sqrt[8]{(\sqrt[7]{a_1a_2\cdots a_7})^8} = \sqrt[7]{a_1a_2\cdots a_7}$$

Plugging in this value to the earlier derivation now

$$\sqrt[8]{a_1a_2a_3\cdots a_7\sqrt[7]{a_1a_2\cdots a_7}} \leq \frac{a_1 + a_2 + \cdots + a_7 + \sqrt[7]{a_1a_2\cdots a_7}}{8}$$

$$\sqrt[7]{a_1a_2\cdots a_7} \leq \frac{a_1 + a_2 + \cdots + a_7 + \sqrt[7]{a_1a_2\cdots a_7}}{8}$$

$$7\sqrt[7]{a_1a_2\cdots a_7} \leq a_1 + a_2 + \cdots + a_7$$

$$\sqrt[7]{a_1a_2\cdots a_7} \leq \frac{a_1 + a_2 + \cdots + a_7}{7}$$

Hence we have proved it.

Problem 322.

We can solve this by taking the 6 numbers then putting the 7^{th} number as the geometric mean of the first 6 numbers and plugging it into the final expression in the previous problem as hinted by the authors.

$$\sqrt[7]{a_1a_2a_3\cdots a_6\sqrt[6]{a_1a_2\cdots a_6}} \leq \frac{a_1 + a_2 + \cdots + a_6 + \sqrt[6]{a_1a_2\cdots a_6}}{7}$$

Using the same manipulation as the one in the earlier problem

$$\sqrt[7]{a_1a_2a_3\cdots a_6\sqrt[6]{a_1a_2\cdots a_6}} = \sqrt[7]{(\sqrt[6]{a_1a_2\cdots a_6})^6\sqrt[6]{a_1a_2\cdots a_6}} = \sqrt[7]{(\sqrt[6]{a_1a_2\cdots a_6})^7} = \sqrt[6]{a_1a_2\cdots a_6}$$

Plugging in this value to the earlier derivation now

$$\sqrt[7]{a_1a_2a_3\cdots a_6\sqrt[6]{a_1a_2\cdots a_6}} \leq \frac{a_1+a_2+\cdots+a_6+\sqrt[6]{a_1a_2\cdots a_6}}{7}$$

$$\sqrt[6]{a_1a_2\cdots a_6} \leq \frac{a_1+a_2+\cdots+a_6+\sqrt[6]{a_1a_2\cdots a_6}}{7}$$

$$6\sqrt[6]{a_1a_2\cdots a_6} \leq a_1+a_2+\cdots+a_6$$

$$\sqrt[6]{a_1a_2\cdots a_6} \leq \frac{a_1+a_2+\cdots+a_6}{6}$$

Hence we are able to show the relationship as required using the previous problem.

Problem 323.

We have seen that the equality holds for every even number of terms. And from every even number of terms we can derive the equality for all odd number of terms. Therefore, the AM GM inequality will always hold true for all $n \geq 2$.

The more elegant way would be to prove it through mathematical induction.

Let $P(n)$ denote the statement

$$\sqrt[n]{a_1a_2\cdots a_n} \leq \frac{a_1+a_2+\cdots+a_n}{n}$$

for all positive real numbers $a_1, \cdots, a_n$

We know that $P(2)$ is true, since

$$\sqrt{ab} \leq \frac{a+b}{2}$$

First, suppose $P(n)$ is true. For $2n$ positive numbers, divide them into two groups of n numbers. By $P(n)$,

$$\sqrt[n]{a_1\cdots a_n} \leq \frac{a_1+\cdots+a_n}{n}$$

and

$$\sqrt[n]{a_{n+1}\cdots a_{2n}} \leq \frac{a_{n+1}+\cdots+a_{2n}}{n}$$

Using the two variable AM GM inequality on the two arithmetic means gives

$$\sqrt[2n]{a_1 a_2 \cdots a_{2n}} \leq \frac{a_1 + a_2 + \cdots + a_{2n}}{2n}$$

Hence

$$P(n) \implies P(2n)$$

Next, suppose $P(n)$ is true. Given positive numbers $a_1, \cdots, a_{n-1}$, let

$$A = \frac{a_1 + \cdots + a_{n-1}}{n-1}$$

Apply $P(n)$ to the n numbers

$$a_1, a_2, \cdots, a_{n-1}, A$$

Then

$$\sqrt[n]{a_1 a_2 \cdots a_{n-1} A} \leq \frac{a_1 + \cdots + a_{n-1} + A}{n} = A$$

Therefore

$$a_1 a_2 \cdots a_{n-1} A \leq A^n$$

and since $A > 0$,

$$a_1 a_2 \cdots a_{n-1} \leq A^{n-1}$$

Thus

$$\sqrt[n-1]{a_1 a_2 \cdots a_{n-1}} \leq A = \frac{a_1 + \cdots + a_{n-1}}{n-1}$$

Hence

$$P(n) \implies P(n-1)$$

Starting from $P(2)$, repeated doubling proves $P(2^k)$ for every $k \geq 1$. From each $P(2^k)$, repeated backward induction proves all cases below it. Therefore $P(n)$ holds for every integer $n \geq 2$.

Problem 324.

The next couple of pages is good reasoning by the authors. Just before problem 325 they actually answer this current problem.

We have proved that for $n = 2$ the product of two numbers is maximum when they are equal in problems 307 and 308.

Now via proof by contradiction.

Assume for n equal numbers the sum is S and the product is P. Now this P is not maximum but is maximum for $a_1 = a_2 = \cdots = a_n$ where some of these numbers are not equal. Now assume two numbers a_i and a_j are not equal. Now we replace these two by their arithmetic mean $\frac{a_i+a_j}{2}$ then the sum does not change but the product now increases. So this product could not have been maximum.

Therefore, the product P is maximum for a set of numbers only when they are equal.

Problem 325.

The argument assumes that among all nonnegative numbers $a_1, \cdots, a_n$ having a fixed sum S, there actually exists a choice for which the product $a_1 a_2 \cdots a_n$ attains its maximum. This existence has not yet been proved.

Using calculus, or more precisely the Extreme Value Theorem, the gap can be filled as follows.

Consider the set

$$D = \{(a_1, \cdots, a_n) \in \mathbb{R}^n : a_i \geq 0, \quad a_1 + \cdots + a_n = S\}$$

Since $0 \leq a_i \leq S$ the set D is bounded. It is also closed, and hence compact.

Now consider the function $f(a_1, \cdots, a_n) = a_1 a_2 \cdots a_n$. This function is continuous. Therefore, by the Extreme Value Theorem, f attains a maximum at some point of D. Suppose that at this maximum two numbers, say a_1 and a_2, are unequal. Replace them by

$$\frac{a_1 + a_2}{2}, \qquad \frac{a_1 + a_2}{2}$$

Their sum is unchanged, while

$$\left(\frac{a_1 + a_2}{2}\right)^2 - a_1 a_2 = \frac{(a_1 - a_2)^2}{4} > 0$$

Thus the product increases, contradicting the assumption that it was already maximal. Hence no two numbers can be unequal, and therefore

$$a_1 = a_2 = \cdots = a_n = \frac{S}{n}$$

Thus the product is maximal when all the numbers are equal.

Problem 326.

The plain vanilla way to do is to compute the fractions on the left hand side and then we can reason out the AM GM relationship.

$$\frac{a_1^2 \cdot \ a_3 \cdots a_n + a_1 \cdot \ a_2^2 \cdot a_4 \cdots a_n + \cdots + a_2 \cdot a_3 \cdots a_n^2}{a_1 \cdot a_2 \cdots a_n} \geq n$$

We can start simplifying

$$n \prod_{i=1}^{n} a_i \leq a_1^2 \cdot \ a_3 \cdots a_n + a_1 \cdot \ a_2^2 \cdot a_4 \cdots a_n + \cdots + a_2 \cdot a_3 \cdots a_n^2$$

Now using the AM GM relationship

$$\prod_{i=1}^{n} a_i \leq \frac{1}{n} \sum_{i=1}^{n} a_i \leq \frac{a_1^2 \cdot \ a_3 \cdots a_n + a_1 \cdot \ a_2^2 \cdot a_4 \cdots a_n + \cdots + a_2 \cdot a_3 \cdots a_n^2}{n}$$

Thus we can prove this way.

But there is a much more elegant way to solve this.

When we look at the fractions on the left side we observe that if they were all to get multiplied the denominator of the preceding fraction will cancel out the numerator of the next fraction resulting in $\frac{a_1}{a_1} = 1$. We can try this exploration of this idea.

Let the numbers be $\frac{a_1}{a_2}, \frac{a_2}{a_3}, \cdots, \frac{a_{n-1}}{a_n}, \frac{a_n}{a_1}$. We also know that $a_1, \cdots a_n$ are all positive so these fractions will be defined and positive.

Now using the AM GM inequality for these fractional numbers which are n in number.

$$\sqrt[n]{\frac{a_1}{a_2} \cdot \frac{a_2}{a_3} \cdot \dots \cdot \frac{a_{n-1}}{a_n} \cdot \frac{a_n}{a_1}} \leq \frac{\left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}\right)}{n}$$

$$\sqrt[n]{\frac{1}{1}} \leq \frac{\left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}\right)}{n}$$

$$1 \leq \frac{\left(\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1}\right)}{n}$$

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} \geq n$$

Hence we can prove this elegantly this way. The observation of taking the product of all these fractions and applying AM GM is the key to this solution.

Problem 327.

This looks like a tricky problem but it is very easy. The moment we see cube root on one side and a fraction where denominator is 3 we know it is related to AM GM inequality.

But the b^2 terms puts us off. It is just 3 numbers where two numbers are equal to b.

So the 3 numbers are a, b, b

$$\sqrt[3]{a \cdot b \cdot b} \leq \frac{a + b + b}{3}$$

$$\sqrt[3]{ab^2} \leq \frac{a + 2b}{3}$$

Hence proved.

Problem 328.

At this point we realize that the authors are essentially creating a series of numbers which can be whole, fractional, repetitive etc.

In this current problem we can take the numbers in multiple ways, as given below, but all of them lead to a deadlock in the proving the given inequality

First: a and b^2
Second: a, b, and b
Third: ab and b

We some how need a square for b ie b^2 in the GM and a single b with no coefficient in the AM as simply b. Here we realize that we can do

$$b = \frac{b}{2} + \frac{b}{2}$$

So the numbers become a, $\frac{b}{2}$, and another $\frac{b}{2}$

Applying the AM GM inequality for these 3 numbers.

$$\sqrt[3]{a \cdot \frac{b}{2} \cdot \frac{b}{2}} \leq \frac{a + \frac{b}{2} + \frac{b}{2}}{3}$$

$$\sqrt[3]{\frac{ab^2}{4}} \leq \frac{a+b}{3}$$

$$a + b \geq 3\sqrt[3]{\frac{1}{4}}$$

Therefore the minimum value of $a + b$ will be $\sqrt[3]{\frac{1}{4}}$.

Problem 329.

Again we need to try out different series of numbers. One approach is to consider dividing the numerator and denominator of the right hand side by 2 so that we can 3 numbers and their arithmetic mean. This leads us to a deadlock again.

The other way is to think that there are 6 numbers for the AM. How? Because there is 1 a, 2 b's and 3 c's. Let us apply AM GM to that

$$\sqrt[6]{a \cdot b \cdot b \cdot c \cdot c \cdot c} \leq \frac{a+b+b+c+c+c}{6}$$

$$\sqrt[6]{a} \cdot \sqrt[6]{b^2} \cdot \sqrt[6]{c^3} \leq \frac{a+2b+3c}{6}$$

$$\sqrt[6]{a} \cdot \sqrt[3]{b} \cdot \sqrt{c} \leq \frac{a+2b+3c}{6}$$

Hence we have shown the required relationship.

Problem 330.

This one is an easy one and this would be one of the options we would have stumbled upon in the previous problem itself.

The 3 numbers in the series are a, $2b$, and $3c$. Applying the AM GM inequality now.

$$\sqrt[3]{a \cdot 2b \cdot 3c} \leq \frac{a + 2b + 3c}{3}$$

$$\sqrt[3]{6abc} \leq \frac{a + 2b + 3c}{3}$$

$$\sqrt[3]{abc} \leq \frac{a + 2b + 3c}{3\sqrt[3]{6}}$$

Hence we have shown the relationship which we were asked for.

Problem 331.

Though this problem is solved in the book, its got a great explanation.

The principle we are using that if the sum of any two numbers are constant then the product is maximum when the numbers are equal. We have proved this earlier in this chapter for 2 numbers as well as for n numbers.

For the left hand side we think of 10 $\left(1 + \frac{1}{10}\right)$'s and a single 1, so there are 11 terms.

On adding these the 1's make 11 and the $\frac{1}{10}$ makes 1 again since there are 10 of them. So the total for the left hand side (the sum) is 12.

Similarly, for the right hand side, which is simpler, we have 11 terms of $\left(1 + \frac{1}{11}\right)$'s.

The total of 1's is 11 and the fractional part gives us another 1. So the total is again 12.

Now we have the same sums but in the right hand side case all numbers are equal but in the left hand side case the numbers are not equal so its product will be certainly lower than the right hand side.

Thus we have proved

$$\left(1+\frac{1}{10}\right)^{10} < \left(1+\frac{1}{11}\right)^{11}$$

Problem 332.

The hint given in the book is essentially the solution itself.

Let us look at the left hand side of the inequality. In this we have 11 terms each equal to

$$\left(1+\frac{1}{10}\right)$$

So we can say that using AM GM relationship that

$$\sqrt[11]{\left(1+\frac{1}{10}\right)^{11}} \leq \frac{11\cdot\left(1+\frac{1}{10}\right)}{11}$$

$$\left(1+\frac{1}{10}\right) \leq 1.01$$

$$\left(1+\frac{1}{10}\right)^{11} \leq 1.01^{11}$$

Ideally we should not be using $\leq$ here since we know the equality especially when all the terms are equal. So we have

$$\left(1+\frac{1}{10}\right)^{11} = 1.01^{11}$$

Now looking at the right hand side we have 12 terms but to make this comparison we need 11 terms so we can make one term as the product of two $\left(1+\frac{1}{11}\right)$.

Similarly the AM GM relationship turns out to be

$$\sqrt[11]{\left(1+\frac{1}{11}\right)^{10}\cdot\left(1+\frac{1}{11}\right)^{2}} \leq \frac{10\cdot\left(1+\frac{1}{11}\right)+\left(1+\frac{1}{11}\right)^{2}}{11}$$

Simplifying the right side of this numerically

$$\implies \frac{1}{11}\cdot\left(10+\frac{10}{11}+1+\frac{2}{11}+\frac{1}{11^2}\right)$$

$$\implies 1+\frac{12}{11^2}+\frac{1}{11^3} \implies 1.099925$$

So the inequality becomes

$$\left(1+\frac{1}{11}\right)^{10}\cdot\left(1+\frac{1}{11}\right)^{2} \leq 1.099925^{11}$$

$$\left(1+\frac{1}{11}\right)^{12} \leq 1.099925^{11}$$

So the maximum value is

$$\left(1+\frac{1}{11}\right)^{12} = 1.099925^{11}$$

We have shown that

$$\left(1+\frac{1}{10}\right)^{11} > \left(1+\frac{1}{11}\right)^{12}$$

Problem 333.

Let us try and compare similar numbers but first we know

$$\left(1+\frac{1}{10}\right)^{10} < \left(1+\frac{1}{11}\right)^{11}$$

and,

$$\left(1+\frac{1}{10}\right)^{11} > \left(1+\frac{1}{11}\right)^{12}$$

We know that the following since the 12^{th} power of a number greater than 1 will be greater than its 11^{th} power.

So this will be true

$$\left(1+\frac{1}{11}\right)^{12} > \left(1+\frac{1}{11}\right)^{11}$$

Now simply merging the two inequalities from the previous two problems given this fact, we have

$$\left(1+\frac{1}{10}\right)^{10} < \left(1+\frac{1}{11}\right)^{11} < \left(1+\frac{1}{11}\right)^{12} < \left(1+\frac{1}{10}\right)^{11}$$

Let us verify this in Scheme.

```
#lang sicp

(define compute
  (lambda (denom power)
    (cond ((= power 0 ) 1)
          (else (* (compute denom (- power 1))
                   (+ 1 (/ 1 denom)))))))

(let ((a (compute 10 11))
      (b (compute 11 12))
      (c (compute 11 11))
      (d (compute 10 10)))
  (> a b c d))
```

We get the answer from the Scheme interpreter as **#t**

71 The quadratic mean

Problem 334.

So if the non-negative numbers are a and b then the quadratic 'geometric' mean would be

$$\sqrt{a^2 \cdot b^2} = ab$$

So that would just be the product of those two numbers.

Problem 335.

We are given the following relationship and we need to prove it. Square both sides.

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

$$\frac{a^2+b^2}{2} \geq \left(\frac{a+b}{2}\right)^2$$

$$\begin{gathered} 2a^2+2b^2 \geq a^2+b^2+2ab \\ a^2+b^2-2ab \geq 0 \\ (a-b)^2 \geq 0 \end{gathered}$$

This will always be non-negative.

Problem 336.

From the prior problem this happens when both the numbers are equal. For instance let the number be x.

Quadratic mean is

$$\sqrt{\frac{a^2+b^2}{2}} = \sqrt{\frac{2x^2}{2}} = x$$

and the arithmetic mean is

$$\frac{a+b}{2} = \frac{x+x}{2} = x$$

Thus shown.

Problem 337.

We have to show for two non negative numbers a and b that

$$\sqrt{ab} \leq \sqrt{\frac{a^2+b^2}{2}}$$

Squaring both the sides

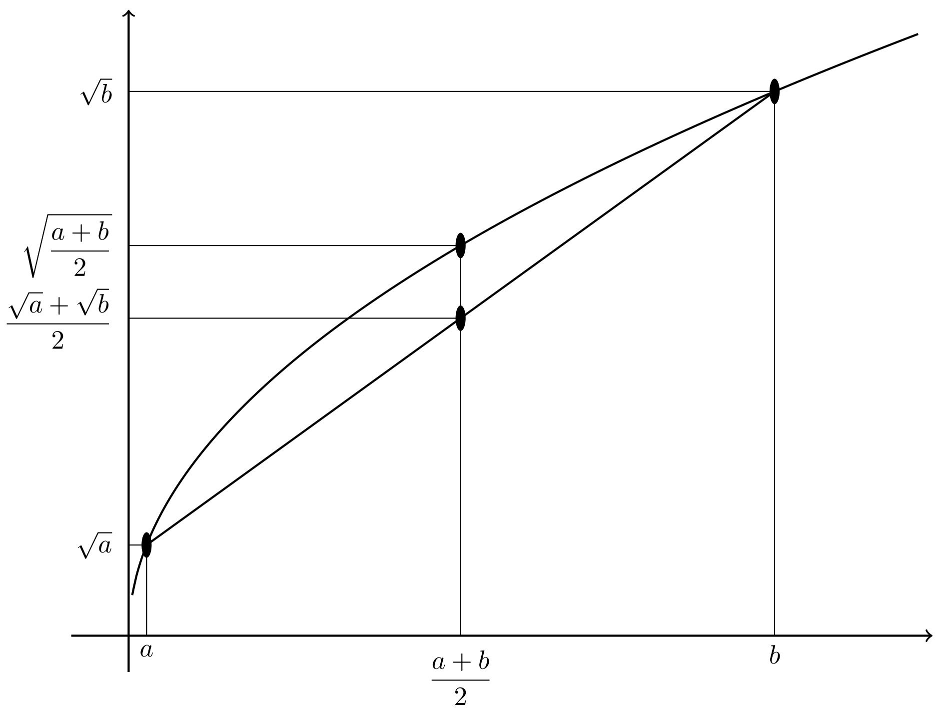
$$ab \leq \frac{a^2+b^2}{2}$$

$$\begin{gathered} a^2+b^2-2ab \geq 0 \\ (a-b)^2 \geq 0 \end{gathered}$$

This will always be true for non negative a and b, hence proved.

Problem 338.

Plotting the graph of this we can see the inequality.

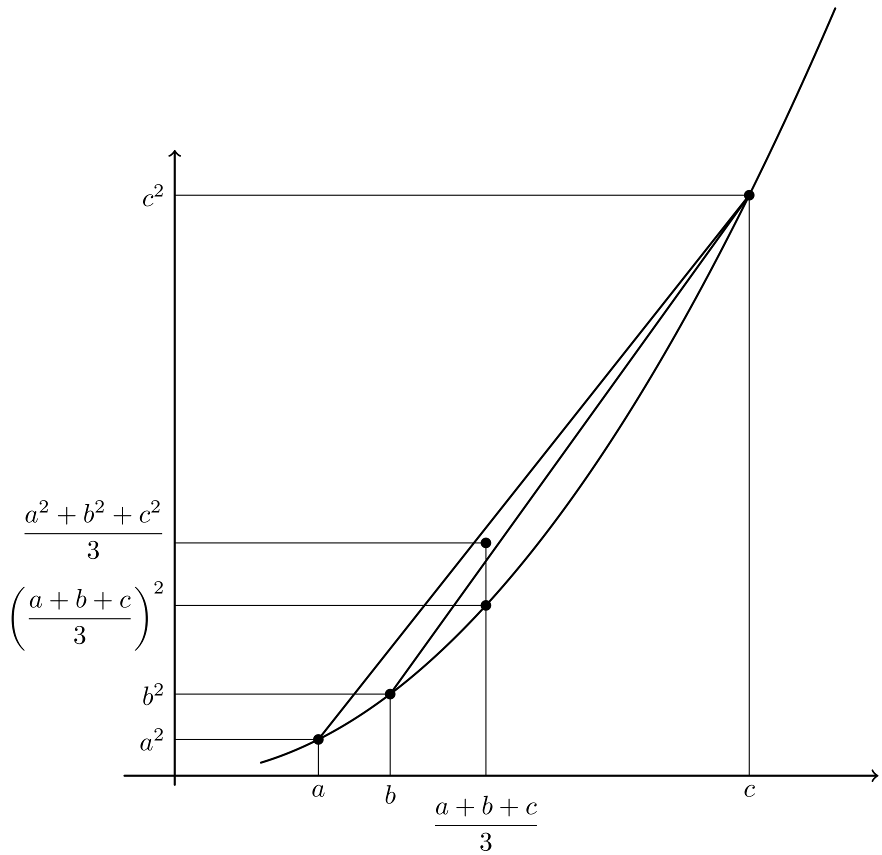


The proof visually can be seeing

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$$

Problem 339.

Now taking 3 points a, b, c on the curve $y = x^2$



We can see from the figure that

$$\left(\frac{a+b+c}{3}\right)^2 \leq \frac{a^2+b^2+c^2}{3}$$

$$\frac{a+b+c}{3} \leq \sqrt{\frac{a^2+b^2+c^2}{3}}$$

Problem 340.

(a) This is a simple application of the relationship we learnt of arithmetic mean and quadratic mean

$$\frac{a+b}{2} \leq \sqrt{\frac{a^2+b^2}{2}}$$

$$1 \leq \frac{a^2+b^2}{2}$$

$$a^2+b^2 \geq 2$$

So the minimum value of the sum of the squares is 2.

(b) Now we can use the same relationship but for 3 numbers.

$$\frac{a+b+c}{3} \leq \sqrt{\frac{a^2+b^2+c^2}{3}}$$

$$\left(\frac{2}{3}\right)^2 \leq \frac{a^2+b^2+c^2}{3}$$

$$a^2+b^2+c^2 \geq \frac{4}{3}$$

Therefore the minimum value is $\frac{4}{3}$.

72 The harmonic mean

Problem 341.

Let the two non negative numbers be a and b.

We have to prove That

$$\frac{1}{\left(\frac{\frac{1}{a}+\frac{1}{b}}{2}\right)} \leq \sqrt{ab}$$

Simplifying

$$\frac{2}{\left(\frac{1}{a}+\frac{1}{b}\right)} \le \sqrt{ab}$$

$$\frac{2ab}{a+b} \le \sqrt{ab}$$

Squaring both the sides

$$\begin{gathered}
4a^2b^2 \le ab(a+b)^2 \\
4a^2b^2 \le a^3b + ab^3 + 2a^2b^2 \\
a^3b + ab^3 - 2a^2b^2 \ge 0 \\
ab(a^2 + b^2 - 2ab) \ge 0 \\
ab(a-b)^2 \ge 0
\end{gathered}$$

The left hand side will always be non negative and thus we have shown that the harmonic mean does not exceed the geometric mean.

Problem 342.

The inequality may be rewritten as

$$\frac{a_1 + \cdots + a_n}{n} \ge \frac{1}{\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n}\right)/n}$$

Thus, we have to prove that the arithmetic mean of n positive numbers is greater than or equal to their harmonic mean.

By the AM GM inequality,

$$\frac{a_1 + \cdots + a_n}{n} \ge \sqrt[n]{a_1 a_2 \cdots a_n}$$

We get

$$\sqrt[n]{a_1 a_2 \cdots a_n} = \frac{1}{\sqrt[n]{\frac{1}{a_1}\frac{1}{a_2}\cdots\frac{1}{a_n}}}$$

Applying the AM GM inequality to the numbers

$$\frac{1}{a_1}, \frac{1}{a_2}, \cdots, \frac{1}{a_n}$$

we have

$$\frac{\frac{1}{a_1} + \cdots + \frac{1}{a_n}}{n} \geq \sqrt[n]{\frac{1}{a_1}\frac{1}{a_2}\cdots\frac{1}{a_n}}$$

All quantities are positive, rearranging the inequality.

$$\frac{1}{\sqrt[n]{\frac{1}{a_1}\frac{1}{a_2}\cdots\frac{1}{a_n}}} \geq \frac{1}{\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n}\right)/n}$$

Therefore,

$$\frac{a_1 + \cdots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \cdots a_n} \geq \frac{1}{\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n}\right)/n}$$

Hence,

$$(a_1 + \cdots + a_n)\left(\frac{1}{a_1} + \cdots + \frac{1}{a_n}\right) \geq n^2$$

This was an easy proof of the given inequality.

Another Solution given is the Cauchy-Schwarz inequality.

The Cauchy-Schwarz inequality states that

$$(p_1q_1 + \cdots + p_nq_n)^2 \leq (p_1^2 + \cdots + p_n^2)(q_1^2 + \cdots + q_n^2)$$

The authors choose

$$p_i = \sqrt{a_i}, \qquad q_i = \frac{1}{\sqrt{a_i}}$$

Then $p_iq_i = 1$ and therefore

$$(p_1q_1 + \cdots + p_nq_n)^2 = n^2$$

Also,

$$p_1^2 + \cdots + p_n^2 = a_1 + \cdots + a_n$$

and

$$q_1^2 + \cdots + q_n^2 = \frac{1}{a_1} + \cdots + \frac{1}{a_n}$$

Hence Cauchy-Schwarz gives

$$n^2 \leq (a_1 + \cdots + a_n) \left(\frac{1}{a_1} + \cdots + \frac{1}{a_n} \right)$$

which is precisely the desired inequality.

Proof of the Cauchy-Schwarz Inequality

Consider

$$(p_1 + q_1 x)^2 + (p_2 + q_2 x)^2 + \cdots + (p_n + q_n x)^2$$

Since this is a sum of squares, it is nonnegative for every real x. Expanding and collecting terms gives a quadratic polynomial $Ax^2 + Bx + C$ where

$$A = q_1^2 + q_2^2 + \cdots + q_n^2$$

$$B = 2(p_1 q_1 + p_2 q_2 + \cdots + p_n q_n)$$

$$C = p_1^2 + p_2^2 + \cdots + p_n^2$$

Since $Ax^2 + Bx + C \geq 0$ for every real x, its discriminant cannot be positive. Thus, $B^2 - 4AC \leq 0$

Therefore, $B^2 \leq 4AC$ so

$$\left(\frac{B}{2} \right)^2 \leq AC$$

Substituting the expressions for A, B, and C, we obtain

$$(p_1 q_1 + \cdots + p_n q_n)^2 \leq (p_1^2 + \cdots + p_n^2)(q_1^2 + \cdots + q_n^2)$$

This is the Cauchy-Schwarz inequality.

Postscript

With this problem, I come to the end of Gelfand and Shen's *Algebra*.

I began writing these solutions in November 2025, and finished the last of the 342 problems in end of August - early September 2026. The journey was not continuous. For several months, roughly from February through July I hardly looked at the book at all. Work, family, personal projects, and the ordinary pressures of life intervened. When I returned to it, my goal was deliberately modest: even one problem a day would be enough. Eventually, one problem at a time, I reached the end.

Working through the book has taught me several new methods and ways of thinking, but perhaps more importantly, it has refreshed mathematical ideas and methods that I had not seriously used in more than two decades. Some problems yielded immediately while others required considerable tinkering. A few stayed in my mind for days. That process of trying something, failing, changing the point of view, and trying again was often more valuable than the eventual solution.

I especially hope that my children will one day work through this book, beginning as early as is reasonably possible. If they do, I hope they *do not* use these pages as an ordinary solution manual.

When a problem refuses to yield, I would rather want that they leave it alone for a while and return to it the next day.

→ Draw something.

→ Rewrite it.

→ Try small cases.

→ Work backwards.

→ Look for a pattern.

→ Ask what changes if one of the assumptions is removed.

→ Try an idea that seems unlikely to work.

→ Let the problem remain unresolved for several days, perhaps even a week.

→ Only then should these solutions become useful.

The real purpose of a book such as this is not to arrive at the answers as quickly as possible. It is to learn how to explore a problem: to deconstruct it, play with it, notice opportunities, make mistakes, and slowly discover a path through it. I hope these solutions can eventually serve as hints, comparisons, or alternative approaches but never as a substitute for that exploration.

And I hope that other students, children, parents, and curious readers who happen upon these pages may find them useful in the same spirit.

These two hundred odd pages record the solutions. But the attempts, false starts, scraps of paper, forgotten ideas, rediscovered techniques, and hours spent thinking were the real project. I will continue editing, correcting mistakes and making explanations more clear.

This is not the end.

It's the beginning.

Deepak Venkatesh
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